MAP READING

4

CHAPTER WISE INDEX: MAP READING

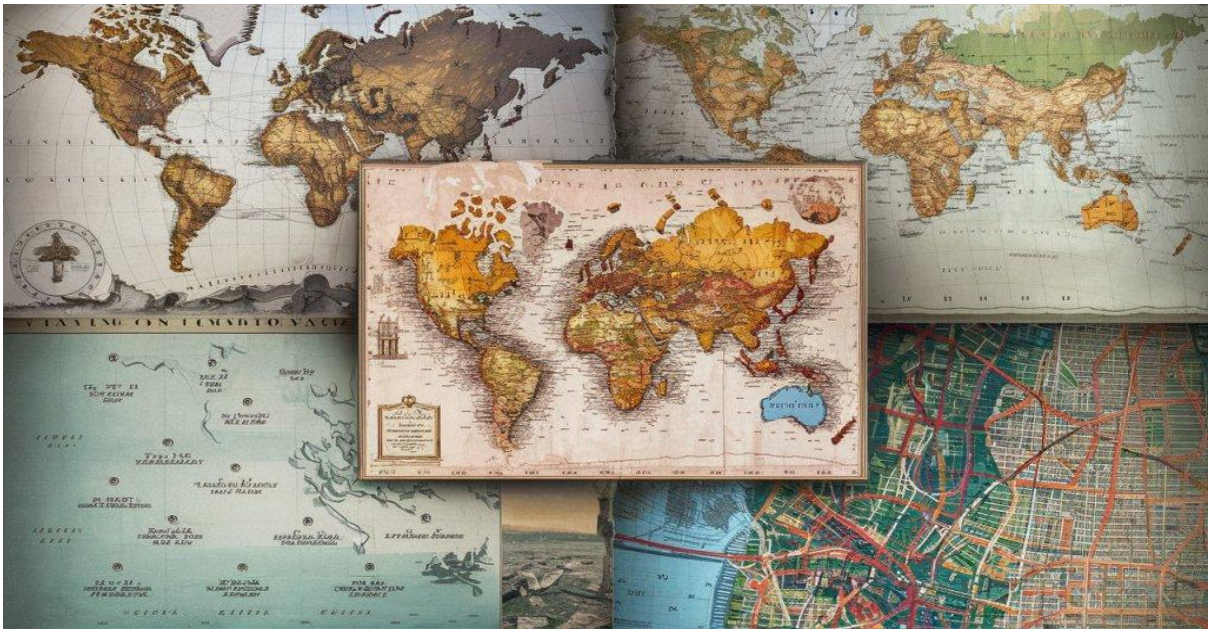
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MAP READING

CHAPTER MR-I: INTRO TO MAPS AND CONVENTIONAL SIGNS, SCALES & GRID SYSTEMS



TEACHING INSTRUCTIONS

Period : 01 (40 Mins)
Type : Lecture and practice
Year : 1st year SD/SW
Conducting officer : permanent instructor

Training Aids:
Classroom, Open Training area or Ground, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, Map Sheets, Scales and Map reading Charts

Time Plan:

- **Introduction : 05 Mins**
- **Part I : 10 Mins**
- **Part II : 05 Mins**
- **Part III : 05 Mins**
- **Part IV : 10 Mins**
- **Conclusion : 05 mins**

INTRODUCTION

1. Map Reading is an essential part of services training; it not only assists youth with methods of understanding surroundings but also paves the way for planning and executing any task given to a leader in the future. The study of maps helps in navigation and other daily activities in all civil affairs. Knowledge of grids and finding GR helps in finding the location of objects accurately. A good sense of directions, conventional signs and scales helps in finding out own location and location of objects. Map is true friend of soldier in battle field.

"We must navigate the stars, not the light of every passing ship."- Gen Omar Bradley

PREVIEW

The lecture will be covered in the following parts:-

- **Part I: Definition and Types of Maps & Conventional signs**.
- **Part II: Division of Indian sub-continent to map zones**.
- **Part III: Definition and methods of expressing scale and scale line Scale and grid system**.
- **Part IV: Definition of Grid and method of finding grid reference**.

LEARNING OBJECTIVES

- **To learn the basics of Map reading and conventional signs**.
- **Understand Division of Indian sub-continent to map zones**.
- **Comprehend scale and grid system Understanding Grid reference**.

PART: I DEFINITION AND TYPES OF MAPS & CONVENTIONAL SIGNS

Definition of Map:

2. A map is a representation of the features of the ground on a piece of paper or plain surface with the help of conventional signs on a particular scale. A map, however, has the following limitations:-

(a) Map details are correct only up to the given date of recording, as features on ground keep changing or getting added.

(b) As the map is two-dimensional, it cannot depict all features on the Earth's surface exactly.

3. Types of Maps

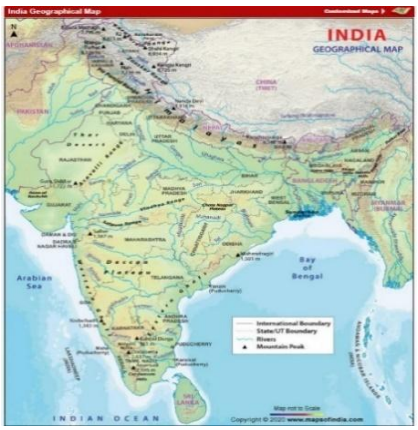
(a) **Physical Map**. This is a map that shows the natural features of a specific place, such as mountains and deserts.



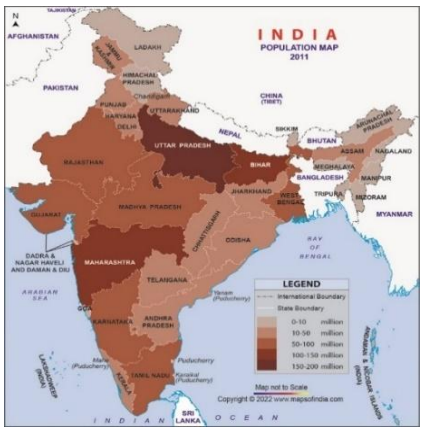
(b) **Political Map**. These maps represent the borders of different states and countries.



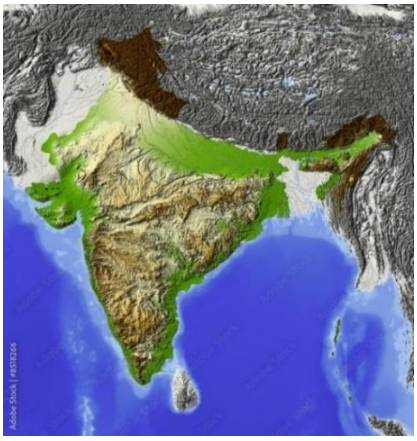
(c) **Geographical Map**. It shows the geographical features of a specific area like rivers, mountains and plains along with boundaries of states and countries.



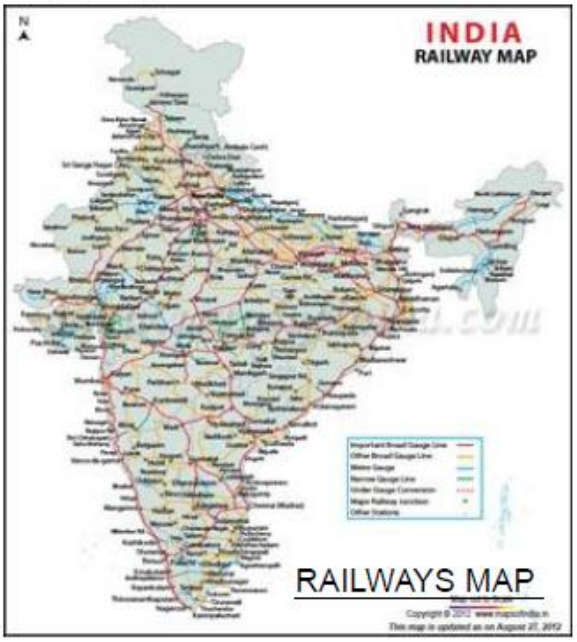
(d) **Statistical Map**. This map shows Statistical classifications such as population and crops, in an area.

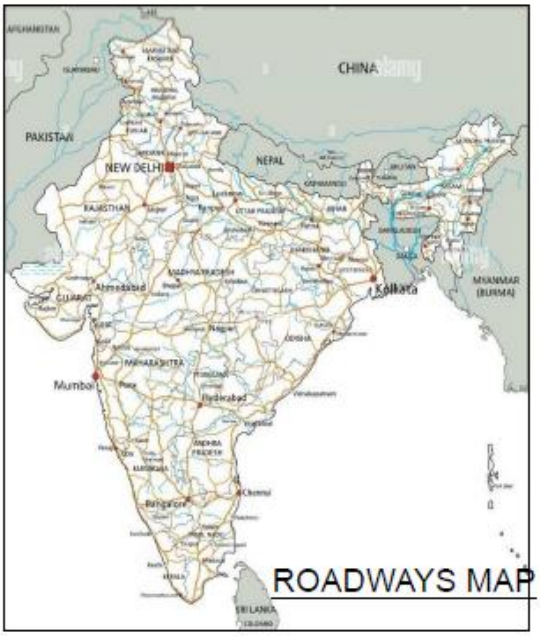


(e) **Relief Map**. This map shows the height and elevation of the land and use shading, tints, or contour lines to depict the land's physical terrain.



(f) **Rail/Road Map**. A road or railway map is a map of a rail/road network. Rail/road maps can show the relationship between train/road lines connecting various places.

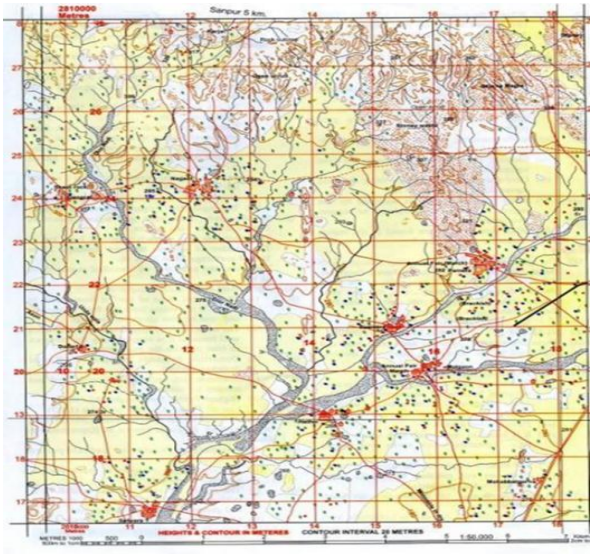




DID YOU KNOW?

- The word "map" comes from the medieval Latin mappa mundi, where mappa means "napkin" or "cloth" and mundi means "of the world".
- The topographical maps are prepared by the **Survey of India** in India for the entire country.
- The science of making maps is called as **Cartography**.

(g) **Topographic map**. This map is a detailed representation of shape of land area showing natural and man - made features and other information. These are the types of map which we shall be using for map reading Topographical features and others form are explained in next chapter.



DID YOU KNOW?

- **Atlas Maps** are small scale maps showing countries, continents, oceans etc.
- **Topographical Maps** are used in Army map reading. One of the main features that sets apart topographical maps from other maps are the **Grid Lines**.
- **Longitudes and Latitudes (Lat-Long)** are shown with black coloured curved lines on the maps.

4. **Conventional Signs**. These are the symbols approved by Survey of India used to represent certain features/objects of the ground on the map. The advantage of these signs are that any person belonging to any country who speaks a different language can understand the map with the help of conventional symbols.

INTERESTING FACT

- The use of conventional signs and symbols help us to quickly get an idea of the layout and characteristics of an area just by looking at a topographical map.

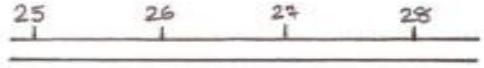
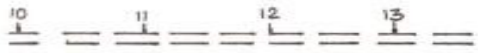
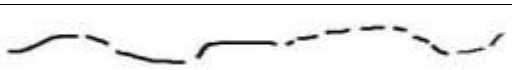
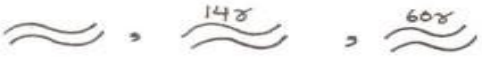
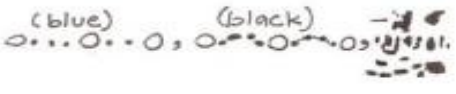
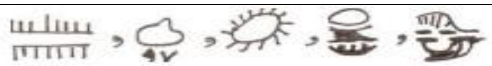
5. The Conventional Signs colour code is as follows:-

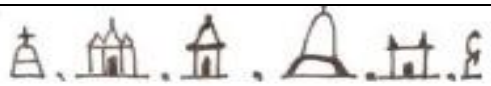
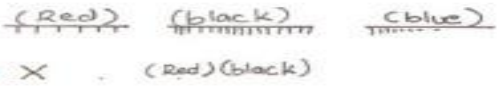
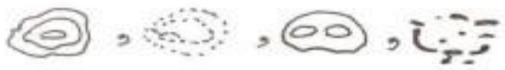
COLOUR CODES IN TOPOGRAPHICAL MAPS

Blue Perennial water bodies
Yellow Cultivated land
Green Vegetation
White Barren Lands
Brown Contours & sand features
Black Names, Lat. & Long, dry water bodies
Red Human-made features

Some Conventional signs are given below.

6. Some of the important conventional signs used in topographical maps are given below:

Ser No.	Description	Nomenclature
(a)	Roads-metalled with Km-stone.	
(b)	Roads-unmetalled with Km-stone.	
(c)	Cart track, camel track, mule path.	
(d)	Foot path, road in bed of stream, Level crossing.	
(e)	Bridges with piers and without, Causeway, Ford.	
(f)	Stream-Approx water course, canal.	
(g)	Riverbanks, shelving, steep 10 to 20 feet.	
(h)	Riverbeds-dry, with stream, with island and rocks.	
(i)	Tidal river-shoal-sub merged rocks.	
(j)	Wells-lined and unlined, spring, tanks-perennial and dry.	
(k)	Kaeaz-in, flow and dry, swamp, Reeds.	
(l)	Embankments, road or rail, tank cutting tunnel.	
(m)	Broken ground, camping ground, vine on trellis.	
(n)	Spaced names, Administrative, Locality, tribal.	

Ser No.	Description	Nomenclature
(o)	Villages: open, walled, ruined, deserted antiquities.	
(p)	Huts, permanent and temporary, Fort, Tower chhatvi.	
(q)	Church, Mosque, Temple, pagoda, Eidgah, tomb.	
(r)	Dams, masonry and earth work-weir (anicut in madras).	
(s)	Lighthouse-Lightship-Buoys Bamboo-plantation.	
(t)	Grass high and low cane, Bamboo-plantation.	
(u)	Palms, Areca, palmyra, other conifer, other trees, scrub.	
(v)	Contours, Form lines, Rocky slopes.	
(w)	Cliffs-sand features.	
(x)	Moraine, Glacier, Scree.	
(y)	Boundary demarcated; international.	
(z)	Boundary demarcated, province or state.	

PART II: DIVISION OF INDIAN SUBCONTINENT INTO MAP ZONES

7. In the context of military operations, topography, or mapping, the Indian subcontinent is divided into nine grid zones to minimise errors in scale. A unique set of parameters are there for each of these zones. Map zone of Indian Subcontinent are:-

(a) Zone 0 - India and Pakistan North of 35°35'N.

(b) One IA - India and Pakistan 28°N-35°35'N.

(c)Zone IB - Unknown parts of China (Tibet).

(d) Zone IIA - India 21°N-28°N & West of 82°E and Pakistan South of 28°N.

(e) Zone IIB - Complete Bangladesh, India North of 21°N & East of 82°E and Myanmar (Burma) North of 21°N.

(f) Zone IIIA - India 15°N-21°N.

(g) Zone IIIB - Myanmar (Burma) 15°N-21°N.

(h) Zone IVA - India South of 15°N.

(j) Zone IVB - Myanmar (Burma) South of 15°N.

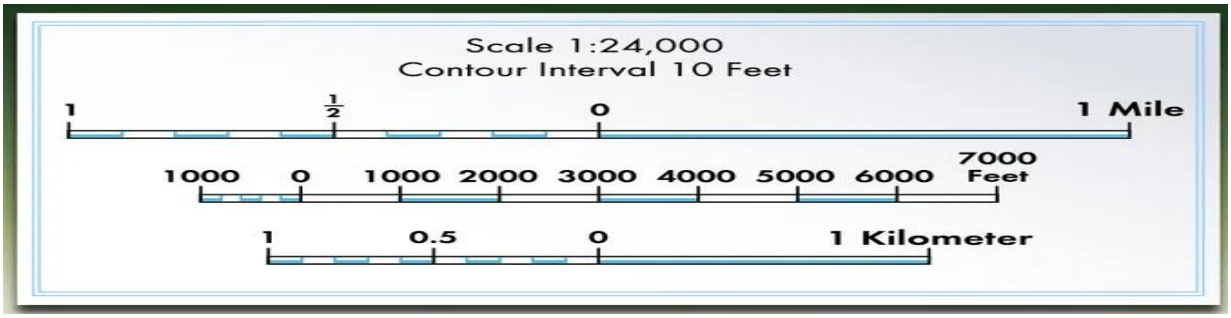


DID YOU KNOW?

- The Indian subcontinent divided into nine zones, and each zone will have its own set of grid references (e.g., 10A, 10B), making it easier to locate any specific point within that zone.

PART III: DEFINITION AND METHODS OF EXPRESSING A SCALE & SCALE LINE

8. **Definition of Scale**. Scale on the map is the ratio between the distance on a map and the corresponding distance on the ground. It's usually expressed as a fraction or ratio, such as 1:10,000 or as a bar scale. Scale is the proportion between the distance of the two points on the map to the distance between same two points on the ground. This means that the scales given on the map can help a cadet in calculating the actual distance between two points from the map itself. However, it must be kept in mind that scale is inversely proportional to details covered in the map that is to say bigger the scale lesser the details.



Types of Sclaes

9. **Methods of Expressing a Scale**. There are various methods of expressing a scale.

(a) **In Words**. This is expressed as 1" = 1 mile, it means that 1 inch on the map is equal to 1 mile on the ground.

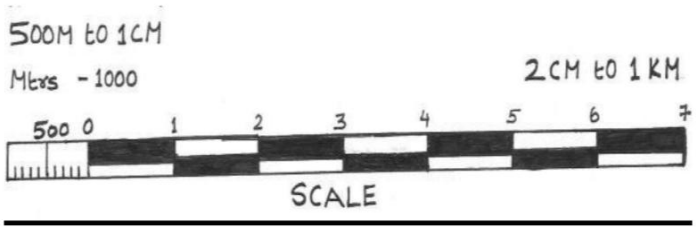
(b) **As a Representative Fraction (RF)**. In this method, scale is expressed in the form of a fraction, say 1/50,000, which means that one unit of the map represents 50,000 unit on the ground. It could mean that one centimetre on the map represents 50,000 cm (or 500 metres) on the ground. RF = Distance on map/Actual distance on ground.

(c)**Bar**. A map's scale can also be expressed graphically as a bar scale, which shows equal divisions marked with readings in miles or kilometres.

DID YOU KNOW?

- Datum followed by Survey of India is Everest Datum.
- Scale Lines on Topographical Maps are between 4 to 6 inches (10 to 15 cm) long.
- Topographical maps usually have scales ranging from 1:50,000 to 1:2,50,000.
- Area covered by 1:50,000 map sheet is 15'x 15' (15 minutes X 15 minutes), which on ground translates to 27.6 Km by 27.6 Km.
- Contour Interval on a 1: 50,000 map and 1: 25,000 map is 20 meters.

10. **Scale Line**. The large divisions of these scale lines are called primaries & small divisions on the left are secondaries. An example of the scale line for a scale "2 cm to 1 km" is as shown in below.



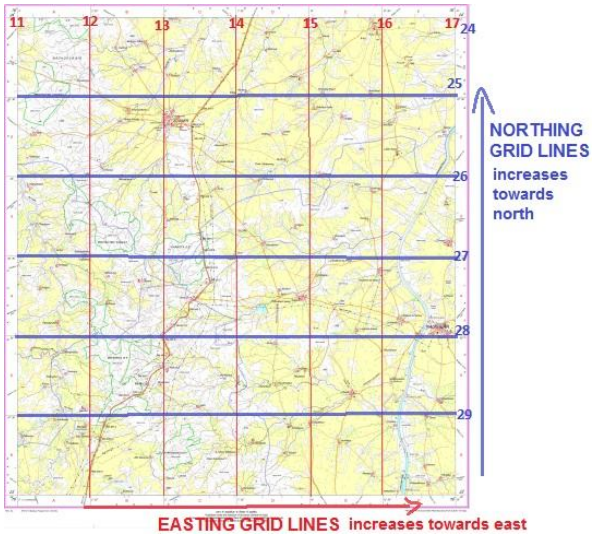
- Scale line lies at the centre of the bottom of the map.

PART IV: DEFINITION OF GRID AND AND METHOD OF FINDING GRID REFERENCES (GR)

11. **Grid System and Grid Lines**. A grid system is a regular pattern of horizontal and vertical purple or red colour lines intersecting each other at right angles forming squares or rectangles that would help to locate the place accurately. There are two types of grid lines- Easting grid lines and Northing grid lines.

(a) **Easting Grid lines**. These are vertical lines which increase towards the East. They are generally referred to as 'Eastings'.

(b) **Northing Grid lines**. These are horizontal lines which increase towards the North. They are generally referred to as 'Northings'.



DID YOU KNOW?

- For the ease of understanding Easting grid lines are called (*Khadi Rekha)* or vertical lines and Northing grid lines are called (*Padi Rekha)* or horizontal lines.
- Numbers of Easting and Northing grid lines will not always start with 0 or 1 in each map because starting grid lines numbers may be falling in adjacent map sheets.

12. **Method of Reading a Grid Reference**. On a 1:50,000 scale map, 2 cm are equal to 1km. Accordingly, a 4-figure grid reference (GR) on a 1:50,000 map covers an area of 1 km x 1 km, while a 6-figure GR depicts an area of 100 m x 100 m. Technically, an 8-figure GR would represent an area of 10m x 10m while a 10-figure GR would pinpoint an area of 1m x 1m, however both these GRs are difficult to be plotted on a 1;50,00 Map, hence Map Reading is restricted to using six figure GR only.

13. **The steps to read a GR of any point are as follows**.

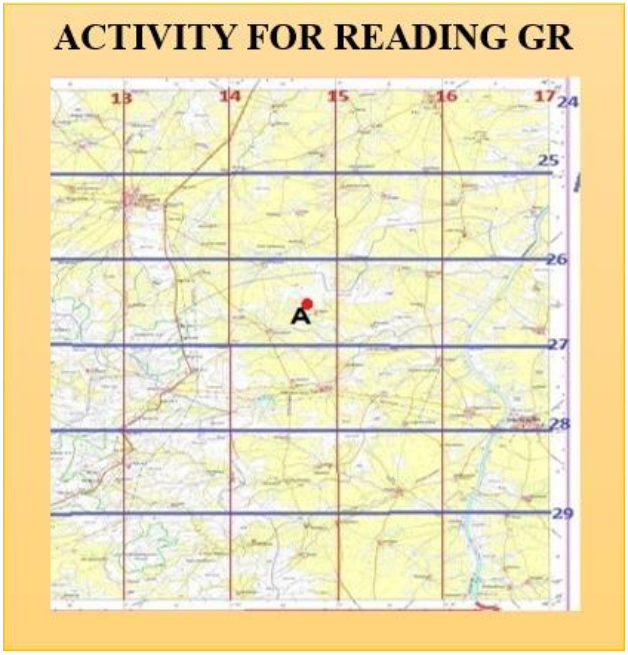
(a) A reference is always in even number of figures and start reading from south west corner.

(b) Grid References are of different types viz. 4- Figure, 6-Figure, 8- figure and 10- Figure.

(c) Mostly 6- Figure Grid Reference is used.

(d) For 4-figure GR of any point A in given activity, first read EASTING lines number from the WEST to EAST and then read NORTHING lines number from SOUTH to NORTH. Thus, 4-fig GR is1427.

(e) For 6-fig GR of point 'A', take nearest 10th part of sub-division of easting line number as 3rd digit and nearest 10th part of sub-division of northing line number as 6th digit. Thus, 6-fig GR is 147275.



(f) For six figure Grid Reference the third and the Sixth figure represent the divisions of 1000 meters square to the nearest 10th part, so they have to be estimated and for these figures a slight latitude is allowed.

(g) If a general Grid Reference is to be given or there is only one such object in one square e.g. bridge, temple, road junction then its identity and four figure grid reference would suffice.

HOT

Correlation Between Lat-Long Maps and Metric Maps. For the purpose of Map projection, Indian and adjacent countries encompassing areas from 4º N to 44º N Latitude and 44º E to 124º E Longitude have been covered as under:-

- **Grids of 4º by 4º**. Total area was divided into grids of 4º by 4º, which came to 180 such grids. Since 44 of these were in waterborne areas, balance 136 were projected on Map sheets with scale of 1:10,00,000(called Million Maps) or 1" : 16 Miles, each comprising an area of 441.6 Km by 441.6 km.

- **Grids of 1º by 1º**. Each grid of 4º by 4º was further sub-divided into 16 grids of 1º by 1º with scale of 1: 2,50,000(called Degree or Quarter Inch Maps) or 1": 4 Miles, each comprising an area of 110.4 Km by 110.4 Km. These were numbered from A to P. A total of 2176 (136x16)sheets were required to be made.

- **Grids of ½ º by ½º**. Further four sub-divisions of ½ º by ½ º or 30' by 30 'were made of each 1º by 1º sheet on a scale of 1:1,00,000(called Half Inch Maps) or 1"= 2 Miles, titled NE,NW,SE or SW, with each such grid depicting an area of 55.2 km by 55.2 Km. A total of 8704(136x16x4) such sheets were made.

- **Grids of 15' by 15'**. Further four sub subdivisions of each ½ º by ½ º Sheets were done. Since this also meant 16 sub divisions of the 1º by 1º grid, these were numbered as 1 to 16. The scale of these Maps was 1:50,000(called Inch Maps) or 1" = 1 Mile, each comprising an area of 27.6 Km by 27.6 Km. A total of 34816(136x16x4x4) such sheets of 15' by 15' were made. This is the sheet which is typically used for MR in NCC.

DID YOU KNOW?

- While reading a GR in topographical maps, Eastings are read first followed by the Northings, but for reading co-ordinates from Latitude- Longitude Maps, the latitudes (horizontal lines) are read first and then the longitude. For example, the location of India Gate in Delhi is given as 28.6129° N, 77.2295° E.

CONCLUSION

14. The proper knowledge of the map and its conventional signs is very essential to know about the unknown area and to relate the objects on ground with the objects on the map by their conventional signs. Conventional symbols help in detailed study of a map. It is important for everyone to relate conventional signs accurately for reading a map efficiently. The grid system and reading grid reference help in locating the different objects of ground on map within a span of 10 to 100 meters. Moreover, it enables us to find our own position and plays a fundamental role in navigation.

SUMMARY

- Map is a piece of paper on which objects on the ground are shown with the help of some conventional signs on a particular scale.
- Scale is a ratio of distance between any two objects on the map with respect of the actual distance on ground.
- Scale can be expressed In Words and as a Representative Fraction (RF).
- A grid system is a regular pattern of purple or red colour lines intersecting each other at right angles forming squares or rectangles that would help to locate the place accurately.
- Easting Grid lines are the vertical lines whose number increases towards east.
- Northing grid lines are the horizontal lines whose number increases towards north.
- There are different types of Grid reference- 4 figure, 6- figure, 8- fig and 10 fig depends upon requirement of accuracy in finding location in a range of area 10 m to 1000m.
- For reading GR of any location, first Eastings are read followed by Northings.

ASSESSMENT EXERCISE

Multiple choice questions

Q1. Map that represents boundaries of state & countries is?

(a) Political maps
(b) Statistical maps
(c) Geographical maps
(d) Relief maps

Q2. The map which shows the rock formation below the topsoil?

(a) Statistical maps (b) Geographical maps
(c) Topographical maps (d) Meteorological maps

Q3. Which of the following are small scale maps showing whole countries continents oceans or even world on one sheet?

(a) Atlas maps (c) Relief maps
(b) Rail/road maps (d) Geographical maps

Q4. The maps show the information about population, industries, minerals ores etc are?

(a) Meteorological maps
(b) Relief maps
(c) Topographic maps
(d) Statistical maps

Q5. Which colour is used to show river on a map?

(a) Yellow (b) Blue (c) Black (d) Red

Q6. The science of making maps is known as?

(a) Photography (c) Choreography
(b) Cartography (d) None of the above

Q7. Which maps are an actual model of the ground?

(a) Relief maps
(b) Rail/Road maps
(c) Statistical maps
(d) None of the above

Q8. The maps that are used in connection of roads

(a) Statistical maps
(b) Rail/Road maps
(c) Relief maps
(d) None of the above

Q9. What is the conventional sign of Broken ground, camping ground, vine on trellis?

(a) 
(b) 
(c) 
(d) 

Q10. The conventional sign of Riverbeds-dry, with stream, with island and rocks

(a) 
(b) 
(c) 
(d) 

Q11. Which map we use to study in NCC?

(a) Relief Map (c)Statistical map
(b) Topographical map (d) Atlas map

Q12. There are _____ types of grid lines.

(a) 3 (b) 5 (c) 2 (d) 4

Q13. What are Easting Grid lines?

(a) Black vertical line (c) Red vertical line
(b) Black horizontal line (d) Red horizontal line

Q14. Which colour is used to show sand features on map?

(a) Red (b) Blue (c) brown (d) Yellow

Q15. What is scale of 1:50000 means?

(a) 1 cm= 1km (b) 2cm= 1km (c) 1cm= 2km (d) 1cm=50km

Short Answer Type Questions

Q1. What do you understand by map? Mention Its types?

Q2. What are Grid Lines?

Q3. What is Scale? What is Scale Line?

Q4. What is RF?

Q5. What are limitations of Map?

Long Answer Type Questions

Q1. Define Conventional signs? Draw conventional signs for battlefield, Fort, Graveyard and Railway line?

Q2. Explain method or rules for calculating Grid Reference?

Q3. Explain Grid System. What are types of Grid lines?

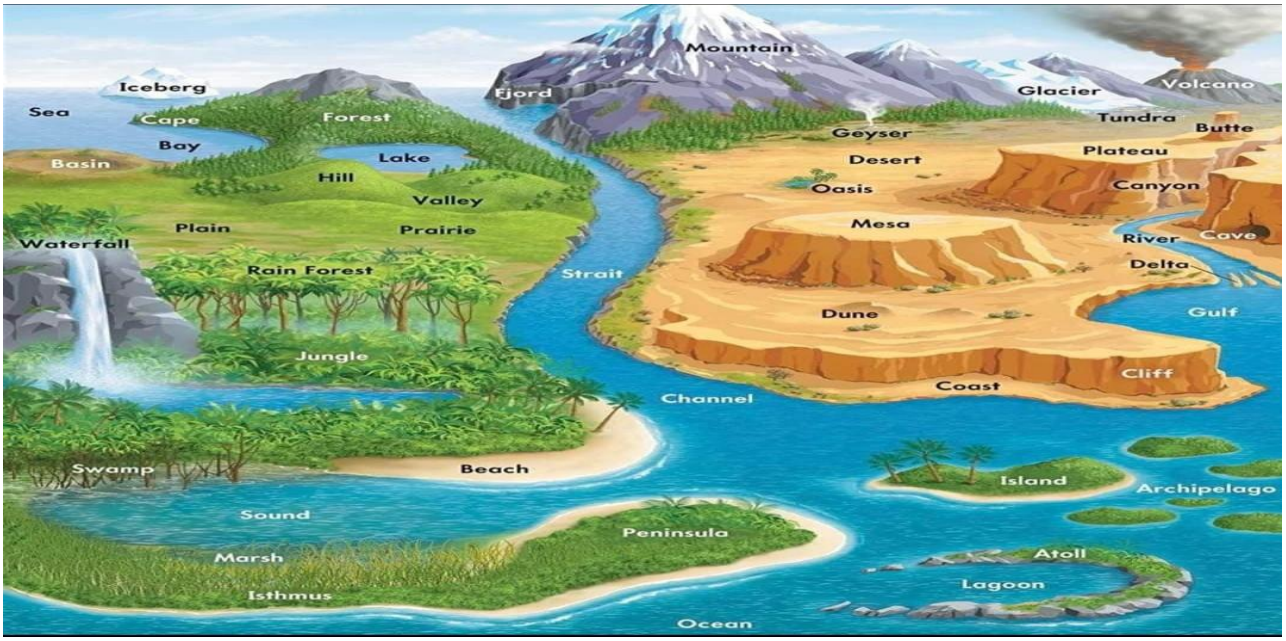
Q4. Calculate RF of 1: 250000.

Q5. What are colour code of conventional signs for different features?

MAP READING

CHAPTER MR-II: TOPOGRAPHICAL FORMS AND TECHNICAL TERMS

A map says to you, read me carefully, follow me closely, Doubt me not... I am the earth in the palm of your hand. Beryl Markham



TEACHING INSTRUCTIONS

Period : 03 (120 Mins)
Type : Lecture and Practice.
Year : 1st (01) and 2nd Year (02) SD/SW
Conducting Officer : Permanent Instructor.

Training Aids: Map, Compass and Pencil.

Time Plan

- **Introduction : 05 Mins**
- **Part I : 35 Mins**
- **Part II : 35 Mins**
- **Part III : 40 Mins**
- **Conclusion : 05 Mins**

INTRODUCTION

1. Topography is the study of natural and man-made features on ground. The topography of an area is description of the land forms and features on map of that area. The different types of structures, terrains and groups of shapes on surface of earth are termed as Topographical forms. In India, the topographical maps are prepared by Survey of India. Topographical maps are of different scales. As these maps are prepared by the same agency in India, for the ease of identification and assimilation, these maps have same conventional signs, symbols and colours for same physical features.

PREVIEW

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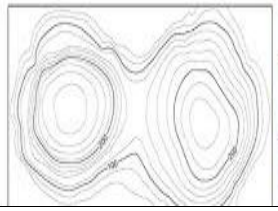
- **Part I: Topographical forms and their technical terms.**
- **Part II: Relief, contours, gradient, horizontal equivalent and vertical interval.**
- **Part III: Practice of reading gradient and contours.**

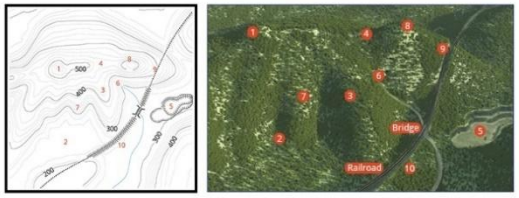
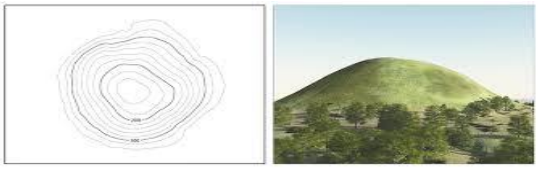
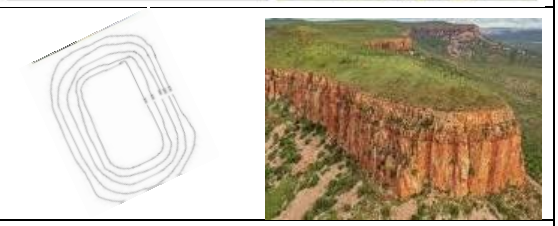
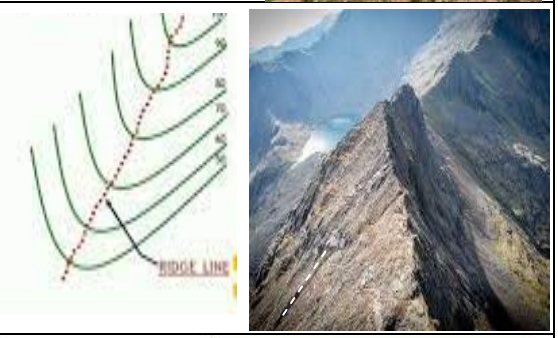
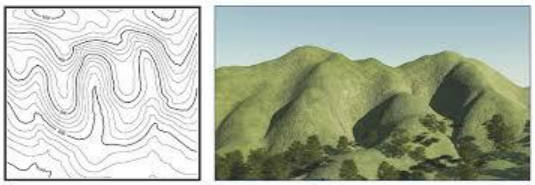
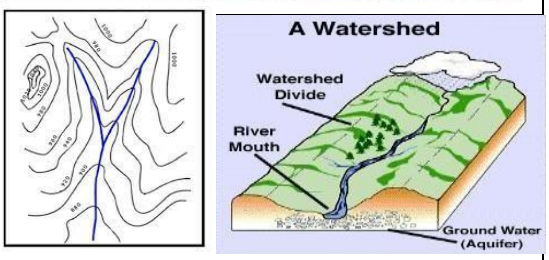
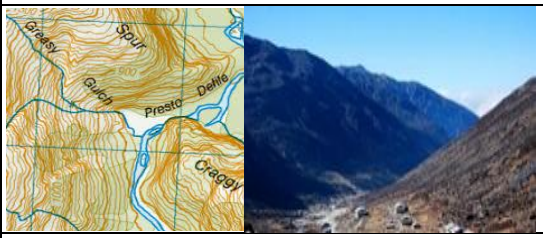
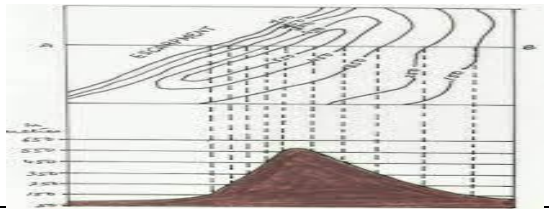
LEARNING OBJECTIVES

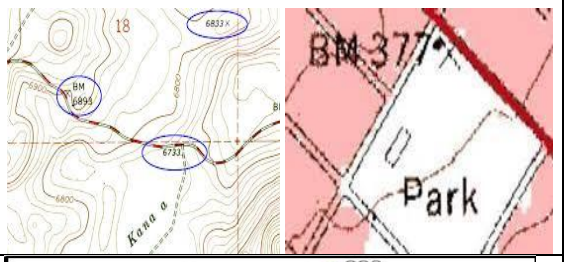
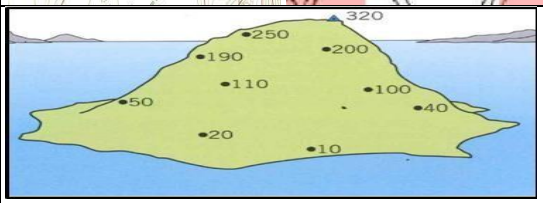
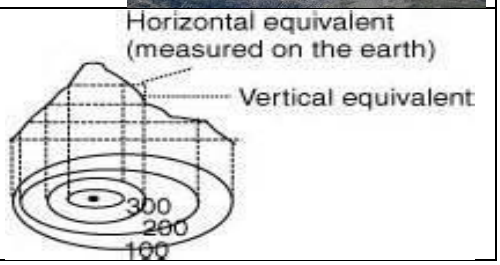
- **Understanding various topographical forms and their technical terms**.
- **To comprehend the Relief and its methods of representation**.
- **To acquaint with contour, gradient, horizontal, equivalent and vertical interval**.

PART I: TOPOGRAPHICAL FORMS AND THEIR TECHNICAL TERMS

2. The topographical forms are the physical features present on the earth. For instance, mountains, plateaus, rivers, lakes and plains are some of the major topographical forms of the earth. Some important topographical forms with their description and their respective ground and map view are given in the table below:-

Ser No	Topographical Forms	Map View	Ground View
(a)	**Basin**. The area which is fairly levelled either by the depression on hills or drained by a river and its distributaries is called Basin.		
(b)	**Col or saddle**. Two higher hills when joins form a narrow ridge of high land called Col or Saddle.		

(c)	**Crest**. The highest part of a mountain range or hill is joined by an imaginary line. The ground slopes down in opposite direction from this line.	
(d)	**Knoll**. A small isolated hill is called Knoll.	
(e)	**Plateau**. A plateau is an elevated tabled land and almost flat surface formed on top of a hill.	
(f)	**Ridge**. It refers to a long, narrow elevated area or crest of land, typically formed by the natural erosion of the landscape. Ridges are often the top parts of mountain ranges or hills and can be visualized as the "spines" of a range of mountains.	
(g)	**Spur**. The piece of high ground descending from range of hills into lower ground is called Spur.	
(h)	**Watershed**. The high part of a mountain range which separates the water flow of two different rivers is called a watershed. It is not necessary to be the highest part of a mountain or mountain range for the watershed.	
(i)	**Defile**. The narrow feature for which the troop has to reduce its front/ formation to cross is called Defile. Bridge is an artificial defile.	
(j)	**Escarpment**. The steep slope formed by a sudden drop in hill side separating comparatively two level or more gently sloping surface resulting from erosion is called Escarpment. It is usually from a plateau.	

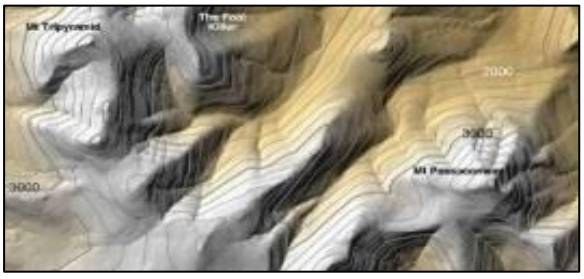
(k)	**Bench Mark**. Is a permanent mark usually cut into a wall recording exact height for future reference. On Ordnance Survey Maps it is marked as BM with height on it.	
(l)	**Spot Height**. The height of some points is determined by Survey methods and mentioned with black dot and a number giving exact height above sea level in meters.	
(m)	**Trig Height**. A point marked by a small triangle with the height on Ordinance Survey Maps. It is fixed during the triangulation at the beginning of a survey.	
(n)	**Horizontal Equivalent**. The distance measured between adjacent contour lines which varies according to the nature of the relief.	

DID YOU KNOW?

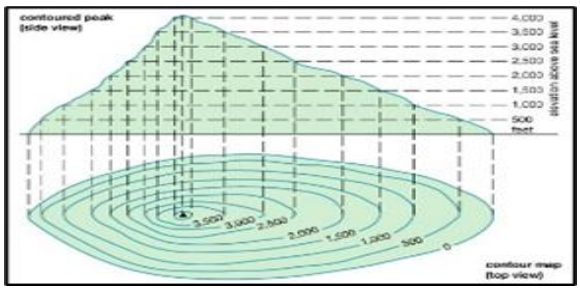
- Spot height and Trig height of a place are generally not the same.

PART II: RELIEF, CONTOUR, GRADIENT, HORIZONTAL EQUIVALENT &VERTICAL INTERVAL

3. **Relief**. It is undulating feature of the land that shows heights and shape of the ground above or below sea level. The relative heights of highlands and low lands is shown by **Relief** which is depicted by shading, form lines, hachure, contours, spot heights, layer tints, bench marks, trig heights and relative heights on a map.

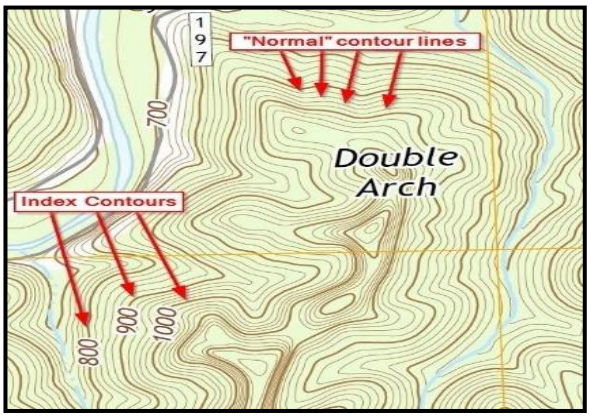


4. **Determining Relief of a map**. It is very simple to determine the relief of a map. The difference in elevation between the highest and lowest points in an area is calculated as Relief. For example, if a mountain has an elevation of 5,000 feet and an adjacent valley has an elevation of 1,000 feet. The relief between these two is calculated by subtracting the lowest



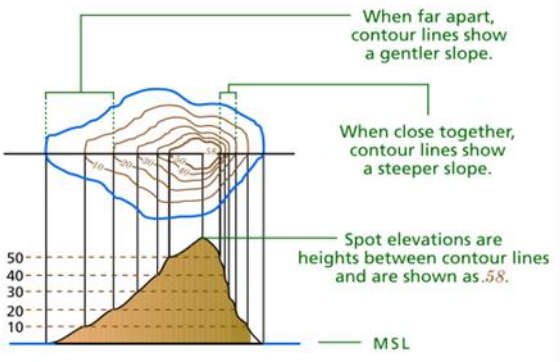
elevation from the highest and we find that the relief between the two is 4,000 feet.

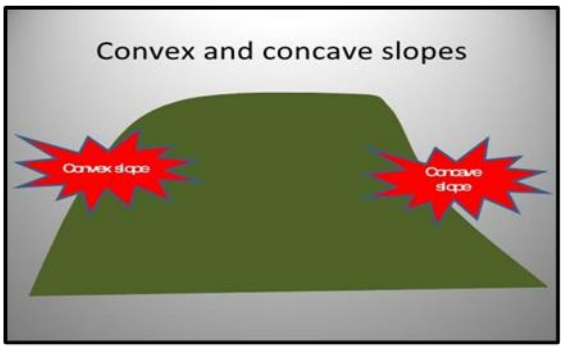
5. **Contours**. These are lines that joins all the points of the same height above sea level on a map. A person doesn't gains or loses elevation while walking along a contour line. The characteristics of Contours are as following:-



(a) They precisely depict height, shape, and slope of a place.
(b)In general, contours are represented in brown colour.
(c)Every fifth contour is known as the Index Contour as height is marked on it.
(d) The shapes made by contour lines differs.
(e) Contour lines never meet or cross each other.
(f) Distance between two contour lines is 20 meters.

6. **Slope**. It means a rise or fall of the Earth's surface. Slope generally follows the grain of the country starting from a higher point and moving towards a lower point. These slopes can be gradual as well as steep. When distance is less and contour lines are closer than the slope of the hill is steep. If contour lines are far apart, then slope down is gradual. The slopes are of two types:-

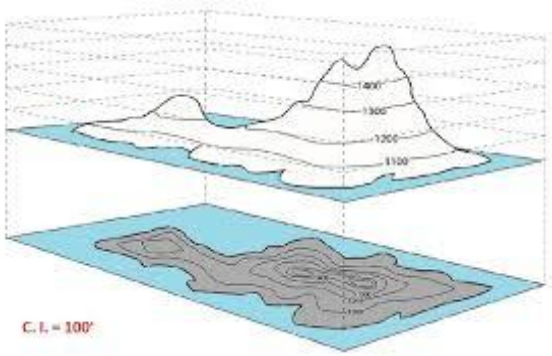




(a) **Regular slope**. The slopes with equal rise and fall are called regular slopes. They are also refer to as gentle slopes.

(b) **Irregular slope**. The slopes with unequal rise and fall of earth are called Irregular slopes. There are two types of irregular slopes: convex and concave slopes. A convex slope is one where from the top, its bottom cannot be seen as the slope bulges outwards and obstructs the view. The concave slope is one in which the slope curves inward and the view to the bottom is clear.

7. **Gradient**. It is the slope of a hill or height, which is expressed as a fraction. The ratio between vertical interval and horizontal equivalent are expressed as a fraction VI/HE. The vertical interval is taken as numerator and the horizontal distance is taken as the denominator. In this the vertical interval is equal to the vertical distance between two successive contours whereas the horizontal equivalent is the horizontal distance between the same adjacent contours.



The formula for Gradient = (Vertical interval)/(Horizontal distance).

Gradient = slope in degree/60

Calculating Gradient

If the lowest contour is 200 and highest contour is 500. The Vertical Interval is 300m. the Length from point A to B is 6 cm. Map Scale is 1cm to 2km Calculate the gradient.

Solution

- Change 6cm into ground (map scale 1cm to 2km)
- 6cm = 6 x 2 km = 12km
- 12km =12000m (1km =1000m)
- Gradient = (Vertical interval (V.I))/(Horizontal distance (H.D))
- =300m/12000m
- Gradient = 1/40

8. **Expressing Gradient**. The gradient can be expressed as ratio and percentage as given below:-

(a) **As a Ratio**. Example: If the rise is 100 meters and the run is 500 meters, the gradient is 1:5. This means for every 5 units of horizontal distance, there is a 1-unit rise.

(b) **As a Percentage**. Convert the ratio into a percentage by multiplying by 100: Gradient Percentage= (Vertical Distance/Horizontal Distance) × 100

Example: (100/500) ×100=20%

PART III: PRACTICE OF READING GRADIENTS AND CONTOURS

9. **Basic Gradient Calculation**. Learn to calculate the gradient between two points on a map.

Steps

(a) Two points were chosen on the topographic map with clear contour lines.

(b) Measure the vertical elevation change between the points using the contour interval (e.g. Point A = 200m, Point B = 300m; elevation change = 100m).

(c)Measure the horizontal distance between the points using the map scale.

(d) Calculate the gradient using the formula: Gradient = Vertical Distance / Horizontal Distance.

(e) Example: If the vertical distance is 100m and the horizontal distance is 500m, the gradient is 100/500=1/5=1:5

10. **Comparing Gradients**. Identify and compare gradients of different slopes.

Steps

(a) Select three different slopes on the map (e.g. hill, valley and ridge).

(b) Calculate the gradient for each slope using the method above.

(c)Compare which slope is steeper by analysing the gradient ratio (smaller ratios indicate steeper slopes).

11. **Identifying Steep and Gentle Slopes**.Visually interpret gradient steepness without calculation.

Steps

(a) Examine contour spacing on the map:

(i) Closely spaced contours indicate steep slopes.

(ii)Widely spaced contours indicate gentle slopes.

(b) Mark areas of steep and gentle slopes on the map.

(c) Validate your assessment by calculating gradients in selected areas.

12. **Gradient Application in Planning**. Use gradient calculations for route selection.

Steps

(a) Plan a hiking or cycling route using a topographic map.

(b) Identify areas with steep gradients to avoid or challenge.

(c)Calculate gradients along different paths to decide the most suitable route based on difficulty.

13. **Practical Benefits**. These exercises not only improve your understanding of gradients but also enhance skills in terrain analysis, route planning and decision-making in activities like hiking, surveying and environmental assessment.

CONCLUSION

14. It is essential to understand various aspects of topographical forms and their technical terms in detail for a successful map reading during military operations. By knowing relief, contour and gradient etc, as discussed in this chapter we will be able to understand the terrain layout, difficulty in traversing the terrain, calculate approximate time to navigate through that particular terrain and special stores that are required to carry to cross difficult objects during the navigation. Relief map over a topographic map will be of great help to understand the terrain as it shows the terrain in a 3D view. To achieve expertise in map reading such as contour lines reading and understand the layout of the terrain more practical knowledge is imperative.

SUMMARY

- Topographical forms are fundamental for understanding and interacting with the physical environment.
- They bridge the gap between theoretical map reading and real-world geography, enabling users to analyse, interpret and adapt to diverse terrains effectively.
- Mastery of topographical knowledge is essential for professionals and enthusiasts in various fields, fostering informed decision-making and a deeper appreciation of the Earth's landscape.
- Relief is undulating feature of the land that shows heights and shape of the ground above or below sea level.
- **Relief** is shown with means of, shading, form lines, hachure, contours, spot heights, layer tints, bench marks, trig heights and relative heights.
- Contour is an imaginary line that joins all the points of the same height above sea level.
- **Slope** means a rise or fall of the Earth surface.
- The slopes with equal rise and fall of earth are called **Regular slopes**.
- The slopes with unequal rise and fall of earth are called **Irregular slopes**. There are two types of irregular slopes: convex and concave slopes.
- Gradient is the slope of a hill, which is expressed as a fraction. The ratio between vertical interval and horizontal equivalent are expressed as a fraction VI/HE.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. _____ is the study of the forms and features of land surfaces.

(a) Topography (b) Geography
(c) Polygraphy (c) Radiography

Q2. A small isolated hill is _____.

(a) Mountain (b) High Ground
(c) Knoll (d) Saddle

Q3. Gradient is the slope of a hill and it is expressed as a _____.

(a) Number (b) Fraction
(c) Bearing (d) Degree

Q4. The angle formed by a line joining two points and the North and South line is called _____.

(a) Degree (b) North line
(c) Bearing (d) Contour

Q5. Bearings are always measured in which direction?

(a) Clockwise (b) Anti Clockwise
(c)Top to Bottom (d) North to South

Q6. _____are the lines drawn on topographical maps to join all places or areas with the equal height or altitude from the mean sea level.

(a) Northings (b) Eastings
(c) Relief (d) Contours

Q7. _____is the difference in height between the two places.

(a) Trigonometric Height (b) Vertical Interval (V.I)
(c) Horizontal Distance (d) Spot

Q8. True North is the direction of the _____Pole from the point.

(a) North (b) South
(c) Eash (d) West

Q9. A ‘V’ shape contour lines indicate a much _____ ridge.

(a) Gentle slope (b) Plain
(c) Steeper (d) River

Q10. The difference between True North & Magnetic North is called _____.

(a) Vertical Interval (b) Horizontal distance
(c) Magnetic Variation (d) Grid Variation

Q11. What is a Trigonometric Point on a map?

(a) A point that indicates the elevation of a place
(b) A point marked by a small triangle used during triangulation
(c) A point representing the highest point on a map
(d) A point indicating the location of a mountain range

Q12. What is the purpose of contours on a map?

(a) To join points of equal temperature
(b) To join points of equal height above sea level
(c) To indicate the distance between two points
(d) To mark boundaries of different countries

Q13. What does the Horizontal Equivalent represent in relation to contour lines?

(a) The distance measured between adjacent contour lines
(b) The total area covered by a map
(c) The average height between contour lines
(d) The total elevation of a region

Q14. Relief on a map refers to:

(a) The representation of vegetation types
(b) The depiction of elevation differences and shapes of landforms
(c) The boundaries of urban areas
(d) The distance between various landmarks

Q15. How is relief calculated on a map?

(a) By measuring the height of mountains
(b) By subtracting the lowest elevation from the highest elevation in an area
(c) By calculating the slope of a hill
(d) By adding all elevations in a region

Short Answer Questions

Q1. What are contours?

Q2. What is topography?

Q3. What is gradient?

Q4. What is the characteristic of contour lines on a map?

Q5. What does a regular slope indicate?

Long Answer Questions

Q1. Write any five topographical forms.

Q2. Write any three Technical Terms of Topography.

Q3. What are Trigonometric height and Spot? Explain with diagrams.

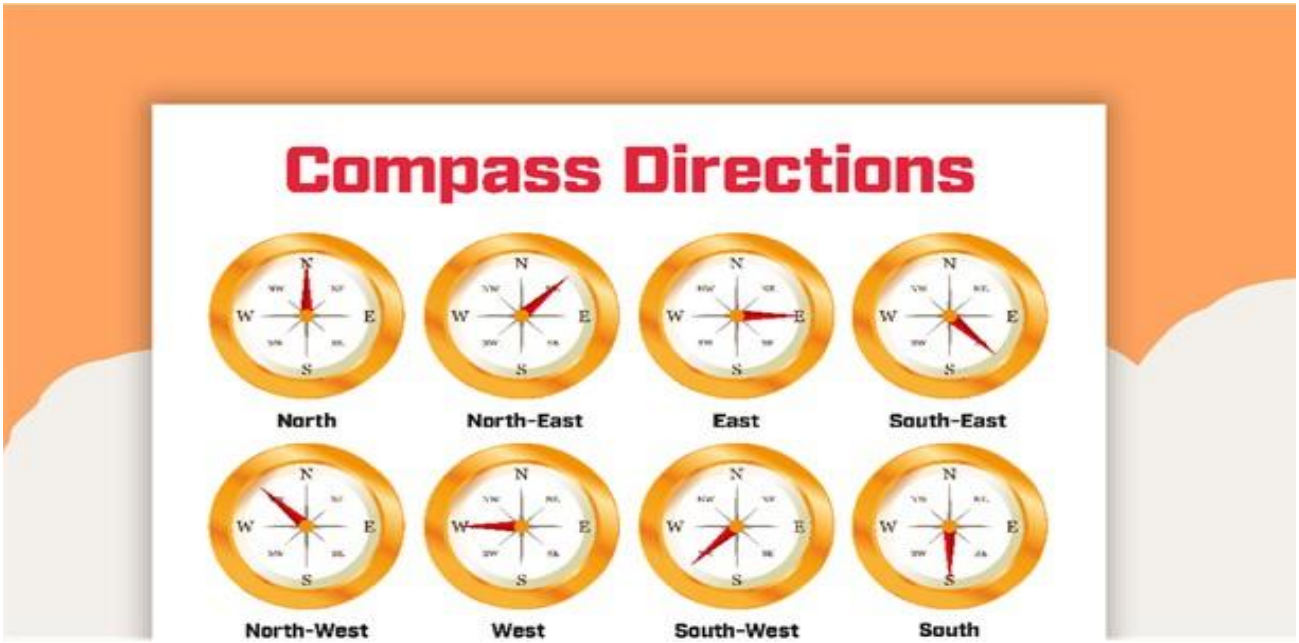
Q4. How is relief calculated on a map?

Q5. What are characteristic of a convex and concave slope?

MAP READING

CHAPTER MR-III: CARDINAL POINTS AND TYPES OF NORTH

"Mastering navigation starts with knowing cardinal directions and north types."



TEACHING INSTRUCTIONS

Period : 01 (40 Mins)
Type : Lecture/Practice
Year : 1st Year SD/SW
Conducting Officer : PI Staff

Training Aids: Class Room, Prismatic compass, Map, Screen, Pointer Staff, Presentation, Precis Flagged or Lesson Plan in File, Board, Markers, Graph Pencils, Display Charts.

Time Plan

- **Introduction : 02 Mins**
- **Part I : 08 Mins**
- **Part II : 05 Mins**
- **Part III : 10 Mins**
- **Part IV : 10 Mins**
- **Conclusion : 05 Mins**

INTRODUCTION

1. Knowing the directions is essential for every person to reach at respective destination. However, for Military purpose or for NCC cadets, it is the first step towards Map Reading. From the point of view of military action, the north direction is considered to be the most important of the four major directions. It is very important to find the directions on the ground and map. It is difficult to make and understand a map or a sketch without knowing the north direction. In the same way, on the ground, we also need to know the directions to march under different conditions. There are many good methods to find the north direction, in which the methods of finding the north direction are different during the night and day. Thus, when the north direction is known, we can find the remaining direction.

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I: Cardinal directions.**
- **Part II: Types of North.**
- **Part III: Variation Between North Lines.**
- **Part IV: Methods used for find out North.**

LEARNING OBJECTIVES

- **Understanding Cardinal Directions.**
- **Understanding types of north to learn the various types of north.**
- **To comprehend Magnetic variation, Grid Convergence.**
- **To Understand methods to find north with or without compass.**

PART I: CARDINAL DIRECTIONS

2. **Kinds Of Directions**. For learning, we can divide the directions into three parts:-

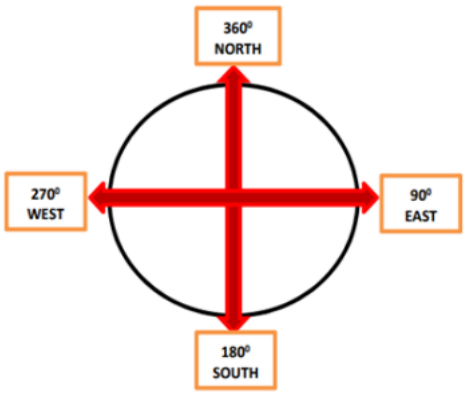
(a) Major directions or Cardinal directions:-

(b) Intermediate directions

(c)Minor directions (sub-intermediate direction)

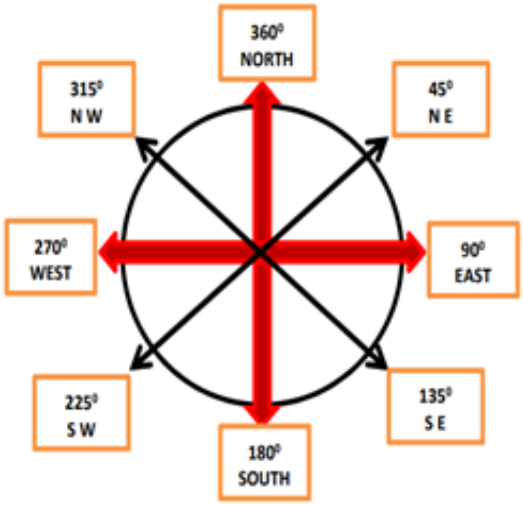
3. For the ease of understanding and assimilation, the explanation of cardinal, intermediate and minor directions with diagrams ia as given below:-

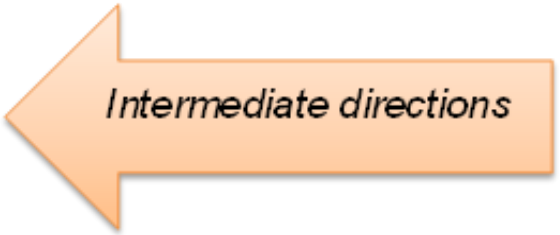
(a) **Cardinal Directions**. There are four main Cardinal directions which are North, South, East and West. The north direction is considered to be 360° because the bearings start from the north direction to measure the curve in the degrees. It takes 360° in turning from north to north. In this rotation, the east direction is at 90°, the south direction is at 180° and the west direction is at 270°.



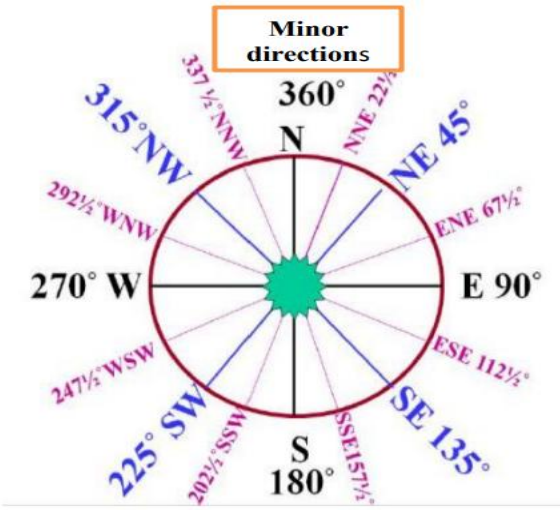
Cardinal Directions

(b) **Intermediate Directions**. The intermediate directions are also four (04). These directions fall between the cardinal direction, i.e. each cardinal direction is divided into two equal parts. These are named as North East (NE), South East (SE), South West (SW) and North West (NW).





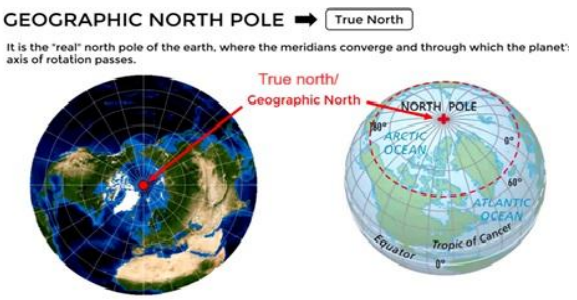
(c)**Sub-Intermediate Directions (Minor directions)**. There are eight (08) sub-intermediate directions between cardinal and intermediate directions, which are represented by three letters. Their names are given in such a way that the cardinal direction is named first and then the intermediate direction. Thus, a sub-intermediate direction is half of the angular distance between a cardinal direction and an intermediate direction, i.e., 22 and 1/2 degrees.



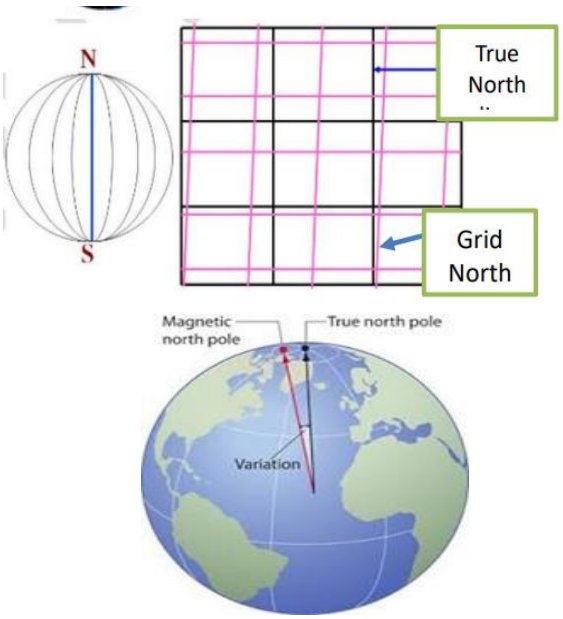
PART II: TYPES OF NORTH AND BEARINGS

4. **Types of North**. On ground, there is only one North, that is the True North. However, since there are no tools to depict it directly, same is done with the help of Grid North and Magnetic North. These three norths are explained below:-

(a) **True North**. It is the direction of the geographic North Pole of the Earth at 90-degree North Latitude. This is a fixed point on the Earth's surface and it doesn't change. On a map, True North is depicted by a solid black North-South line.



(b) **Grid North**. Grid North is the North depicted by the North-South lines of a map (grid). Since semi-circular lines of longitudes are depicted as straight lines on maps, an angular difference is created which makes them deviate from True North. The direction of Grid North is hence marked separately and depicted by red or purple lines on a map, called grid lines.



(c)**Magnetic North**. It refers to the direction to the Magnetic North Pole, as indicated by the north-seeking needle of a magnetic compass. The position of Magnetic North is not static as the magnetic field of the Earth keeps varying, so it does not align with True North. For this reason, it is also not depicted on a map.

5. **Bearing**. The angle formed between an object and North, with respect to a particular location or observer, is known as Bearing. It is always measured clockwise and is related to one of the three types of North. It is classified on two basis:-

(a) On the basis of North.

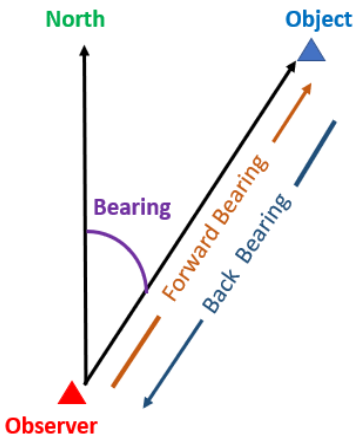
(b) On the basis of orientation (forward or back).

6. **Types of Bearing on Basis of North**. These are:-

(a) **True Bearing**. The clockwise angle formed between an imaginary line joining two points and the direction line of True North is called True Bearing.

(b) **Magnetic Bearing**. The clockwise angle formed between an imaginary line joining two points and the direction line of Magnetic North is called Magnetic Bearing.

(c)**Grid Bearing**. The clockwise angle formed between an imaginary line joining two objects and the direction line of Grid North is called Grid Bearing.



7. **Types of Bearing on Basis of Orientation**.

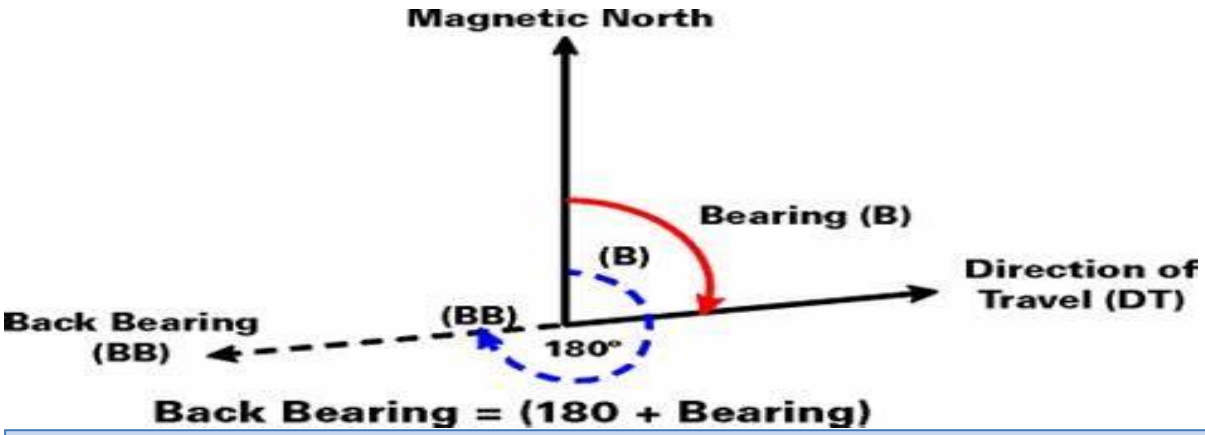
(a) **Forward Bearing**. The bearing of an object's position from the observer.

(b) **Backward Bearing**. It is the reverse of forward bearing and means the bearing of the observer from the object. It is obtained by adding or subtracting 180 degrees to/from the forward bearing.

8. **Relationship Between Forward Bearing And Back Bearing**. The Back Bearing of a line can be determined if its Forward Bearing is given and vice versa. If the Forward Bearing of a line is given as the whole circle bearing, then following is the way to calculate back bearing:-

- Back Bearing = Forward bearing + 180, if F.B. <180° and,
- Back Bearing = Forward bearing – 180, if F.B. > 180°.
- B.B. = F.B ± 180°

Let us understand the same with the help of below diagram.



Examples: Forward bearing of a tree and temple are measured 60° and 270° respectively. Find out their back bearing.

- Forward bearing of tree = 60°
- Back bearing of tree = 60° + 180° = 240°
- Forward bearing of temple = 270°
- Back bearing of temple = 270° - 180° = 90°

9. **Reasons for Non-Alignment among Types of North**. The earth spins on its axis, which passes through the North and South poles. **The North Pole with its Latitude as 90 Degrees North is the geographical or True North and is located in the middle of Arctic Ocean**. Since all Longitudes converge at this point, it does not have its own Longitude. The True North direction is fixed in space and time, i.e. it neither varies from place to place nor from time to time. Depiction of True North on a Map is done by black coloured 'Meridians', which are lines drawn from North Pole to South Pole and are true North-South lines.

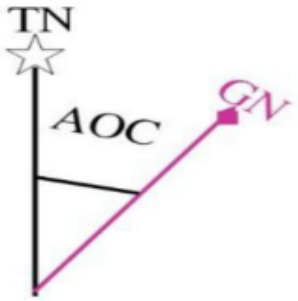
10. Grid lines, marked on the map for referencing, do not lie in the true North and South direction except along one standard meridian (00) and elsewhere they make an angle with the true North-South line. Since all grid lines marked on the map are parallel, they are conveniently used for drawing or measuring bearings. **The northern alignment of the North-South grid lines is hence recognized as Grid North**. On the map, they are shown in purple colour. Grid North and Grid Lines are also fixed in space as well as time i.e. their relative position with reference to True North and True North-South lines remains the same at a given place.

11. Magnetic North is the direction towards the Magnetic North Pole, which is a

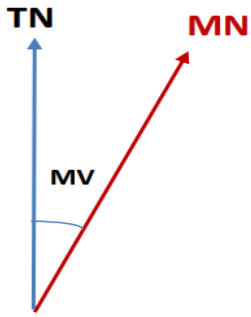
wandering point where the Earth's magnetic field goes vertically down into the planet. It is estimated that it is moving at a speed of about 40 kilometres per year and over the last century the Magnetic Pole has moved a remarkable 1100 kilometres. **The present location of Magnetic North Pole (in 2025) is 85.7 N and 138.6 E, about 1200 km south of the True North or the geographic North Pole and falls in Canadian Territorial waters, west of Greenland**. The needle of a Magnetic Compass points to this location and not to the True North. There are only a few places at which Magnetic North coincides with the True North. At other places it is at an angle to True North which varies yearly. Therefore, Magnetic North lines are not shown on topographical map

PART III: VARIATION BETWEEN TRUE, GRID AND MAGNETIC NORTH

12. **Variation Between True, Grid and Magnetic North**. These are as follows:-



(a) **Angle of Convergence (Grid Convergence)**. The angular difference between True North and the Grid North line at a given place is called the angle of convergence. Grid convergence remains constant and does not change for a given location. In topographical maps, it is generally specified on top of the map.



(b) **Magnetic Variation**. The angular difference between True North and Magnetic North is called Magnetic Variation. Magnetic variation varies not only between different countries but also within different regions of a single country. It changes gradually over time each year. Where the difference is negligible, it is treated as constant. The magnetic variation for the map's publication year is shown at the top and to update it, one can adjust the value annually based on the number of years since the map was published.

Example:

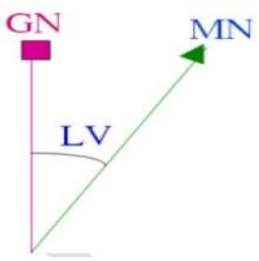
For a given location, Magnetic Variation was 2° East in the year 1968, decreasing every year by 2'. What will be the magnetic variation in the year 1998?

Ans. Total years 1998 – 1968 = 30 years

Decrease in 30 years 30 x 2 = 60 minutes or 1°

Therefore, MV during 1998 will be 2°-1° = 1° East.

(c)**Local Variation /Grid Magnetic Angle (GMA)**. Due to the difference between the Grid North and the Magnetic North line at a given place, the angular distance between them is called Local Variation. It is related to the indicated Magnetic North, which can be in either east or west direction, but the angle of convergence will always remain the same.

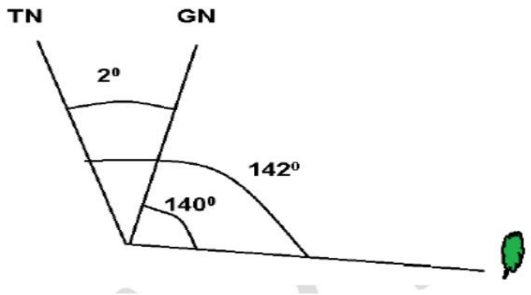


Local Variation

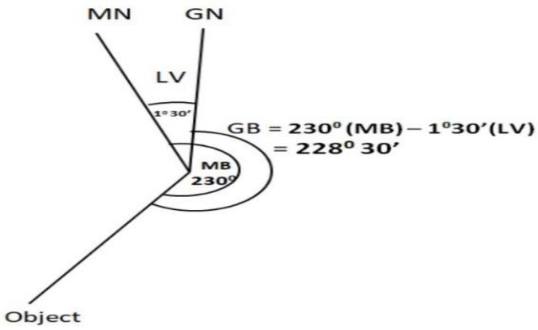
13. **Conversion of bearing**. Conversion of bearing is required for finding own position and to remove compass error. Examples of Conversion are given below:-

Examples of Conversion

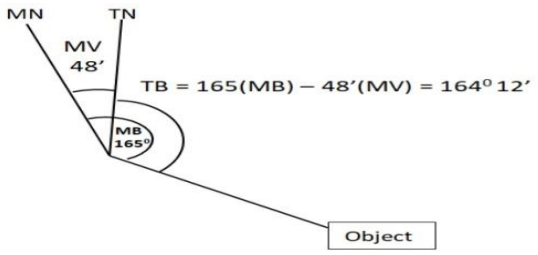
(a) **Grid bearing into true bearing**. A tree has a grid bearing of 140° from a point 'A'. If angle of convergence (GV) is 2° east, then what is true bearing.
Solution: The angle of convergence is 2° east i.e. true bearing is 2° more than grid bearing. True bearing of tree from point 'A'= 140°+ 2° = 142°.



(b) **Magnetic bearing into grid bearing**. The magnetic bearing of an object is 230°. is Grid bearing of that object, if magnetic north is in west of grid north and local variance is 1°30' west?
Solution: the local variance i.e. MN is in 1°30' west of GN. Grid bearing of the object is 1°30' less than magnetic bearing. GB= 230°-1°30' = 228°30'.



(c)**Magnetic bearing into true bearing**. The magnetic bearing of an object is 165°. What is true bearing of the object, if the magnetic variance is 48' west?
Solution: Magnetic north is in 48' west of True north i.e. True bearing is less than 48' than MB. True bearing of the object = 165°-48' = 164°12'.

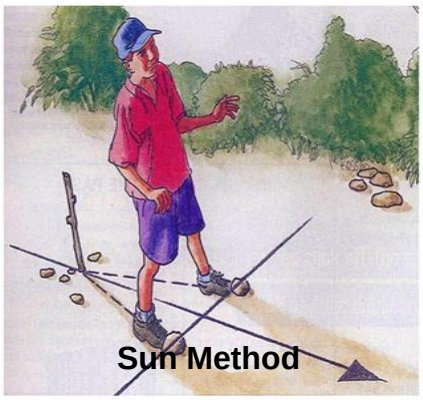


HOTS

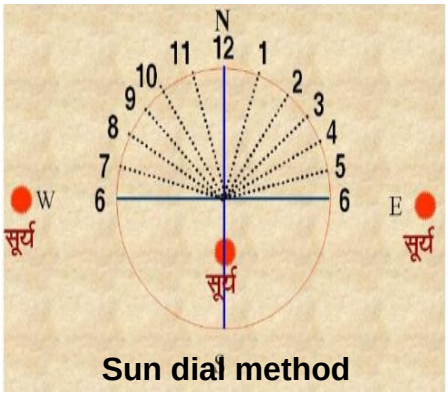
- Considering different methods to find north, which method would you recommend for a person in a dense forest without any modern tools? Justify your choice.
- During the day, if you are stranded without a compass and a map, explain how you can use the "Sun Dial" method to determine the north direction. What potential errors might arise from using this technique?

PART IV: METHODS USED TO FIND OUT NORTH

14. **Methods of finding North**. There are many methods to find the general direction of North during the day and night, which are as follows:-**By Compass**. The best method to know the north direction is the compass. The needle of compass always points towards the North Pole. The compass needle glows at night, so north can be find at any time of the day or night.


Sun Method

(a) **By Sun**. The north direction can be determined by the sun, as it rises in the east and sets in the west. When facing the sunrise, east is in front, west behind, north to the left and south to the right.



Sun dial method

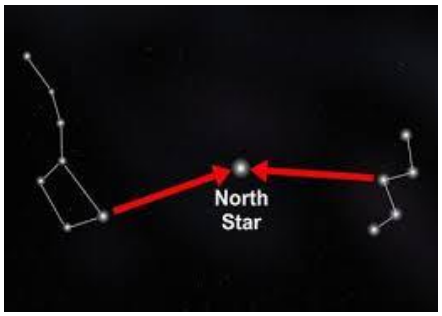
(b) **By Sun Dial**. Place a stick vertically in the ground in an open area. Mark the tip of the shadow (this will be the westernmost point). Wait for about 15–30 minutes, then mark the tip of the shadow again (it will move eastward). The line between the two marks runs roughly from west to east. Stand with the first mark (west) on your left and you will face north. At 12 noon, the stick's shadow will always align straight, with the north direction in front of you and south behind, revealing the true north-south line.

(c)**By Grave Yard**. In many cultures, graveyards follow a particular orientation where graves are placed with the head of the deceased facing **east** or **west**.

(d) **By Mosque**. In India, by standing with your back to the mosque's dome, you'll face west, with north on your right and south on your left, revealing the direction of true north. The two minaret walls and vaulted dome help indicate this orientation.

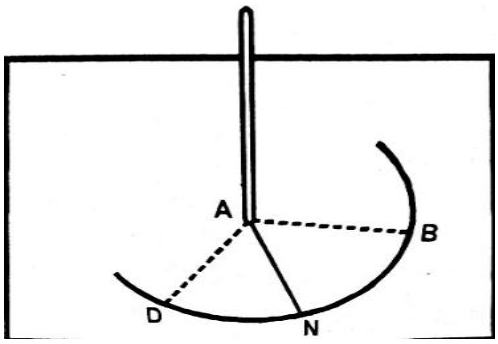
(e) **By Shivalay**. This method can help determine the approximate north direction using a Shiva Linga in Indian temples, which is believed to be connected to Mount Kailash. When water is poured onto the Shivling, it flows toward the north, revealing the direction, though this method is not highly reliable.

(f) **By Pole Star**. In the Northern Hemisphere, the North Star can help determine the north direction, as it always points to the north and remains stable within 2.25 degrees. This bright star is considered the indicator of True North in the Northern Hemisphere, marking the north position accurately within about 2 degrees.

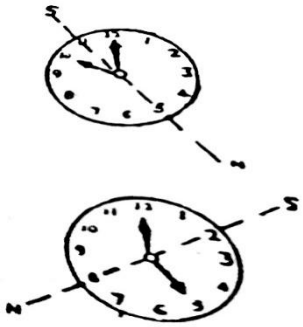


(g) **Equal Altitude Method**. This is a correct way to find north, but it takes a long time and cannot be done without sunlight. In this method, place a pencil in a vertical position and mark point 'A' at the base of the pencil. Mark point 'B' at the end of the

pencil's shadow. Measure AB and draw a half-circle. At noon, wait until the shadow of the pencil touches the boundary of the half-circle. Mark this as point 'D'. Then draw a line AN through the midpoint of the angle formed by the shadow lines. This AN line indicates True North.



(h) **Watch Method**. In the Northern Hemisphere, point the hour hand towards the sun and halfway between it and 12 o'clock will show south, with the opposite direction indicating north. In the Southern Hemisphere, point the 12 o'clock mark towards the sun and halfway between 12 and the hour hand will show north.



CONCLUSION

15. Understanding directions is fundamental for navigation and especially vital for military purposes, such as map reading for NCC cadets. The north direction is of paramount importance, as it serves as a reference point for both ground navigation and map interpretation. The various methods to find the north, including by compass, the sun, the pole star and even using specific cultural or environmental indicators, provide multiple options for determining direction in different conditions. Understanding the variations between all the three norths, is crucial for accurate navigation. Mastery of these methods ensures that individuals can effectively navigate and orient themselves, whether on land or whilst using maps.

SUMMARY

- Knowing directions is essential for reaching the right destination and is a fundamental skill for military purposes and NCC cadets, especially in map reading.
- There are different methods to find the north direction, which vary between day and night.
- **Cardinal Directions**. North, South, East and West (360°, 90°, 180°, 270° respectively).
- **Intermediate Directions**. are sub-directions between cardinal directions.
- **Minor Directions**. Eight directions between cardinal and intermediate, divided by 22.5°.
- Types of north are True North, Grid North, Magnetic North.
- **Angle of Convergence**. Difference between true north and grid north.
- **Magnetic Variation**. Difference between true north and magnetic north.
- **Local Variation**. Difference between grid north and magnetic north at a given place.

- There are many Methods to Find North during day and night.
- The sun rises in the east and sets in the west. Standing facing the sun, the directions are East (front), West (back), North (left), South (right).
- **By Pole Star**. In the Northern Hemisphere, the Pole Star points to the north. It's a stable reference for north.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. What are the three types of directions?

(a) Cardinal, Sub-cardinal and Intermediate
(b) Major, Minor and Intermediate
(c) Major, Intermediate and Minor
(d) North, South and West

Q2. Which of the following is a major cardinal direction?

(a) North-East (c)North-East-West
(b) South (d) West-South

Q3. How many cardinal directions are there?

(a) 2 (b) 8(c)4 (d) 16

Q4. Which of the following is NOT an intermediate direction?

(a) North-East (b) South-East (c) West (d) North-West

Q5. What is the angular distance between cardinal directions and intermediate directions?

(a) 15° (b) 22.5° (c) 45° (d) 90°

Q6. What is the total number of directions when combining cardinal, intermediate and sub-intermediate directions?

(a) 4 (b) 8 (c) 16 (d) 12

Q7. What is True North?

(a) The north direction pointed by a compass
(b) The north indicated by grid lines on a map
(c) The north determined by the pole star
(d) The north direction during the day

Q8. Which type of north is represented by the red or purple line on map?

(a) Magnetic North (c) True North
(b) Grid North (d) Local North

Q9. What does the term “magnetic north” refer to?

(a) The north indicated by the pole star
(b) The north indicated by grid lines on a map
(c) The direction the compass needle points to
(d) The north shown at the equator

Q10. What is the angle of convergence?

(a) The difference between magnetic north and true north
(b) The angular difference between grid north and magnetic north
(c) The angular difference between true north and grid north
(d) The difference between cardinal and intermediate directions

Q11. How are intermediate directions named?

(a) Based on the division of cardinal directions into four equal parts
(b) By combining adjacent cardinal directions
(c) By dividing the Earth into equal geographical regions
(d) Based on magnetic properties

Q12. What is the main feature of sub-intermediate directions?

(a) They divide the Earth into two equal halves
(b) They are named after the cardinal direction
(c) They are located at 45° intervals between cardinal and intermediate directions
(d) They are used for space navigation

Q13. How is the back bearing calculated if the forward bearing is 60°?

(a) Back Bearing = 60° + 180° = 240°
(b) Back Bearing = 60° - 180° = -120°
(c) Back Bearing = 60° + 90° = 150°
(d) Back Bearing = 60° + 45° = 105°

Q14. What is the "angle of convergence" in navigation?

(a) The difference between true north and magnetic north
(b) The angular difference between true north and grid north
(c) The angle between the observer and the object
(d) The angle between the two geographic poles

Q15. Which method can be used to find the north direction using the Sun?

(a) By watching the Sun's reflection in water
(b) By noting where the Sun rises and sets
(c) By using the shadow of a vertical stick at noon
(d) By observing the position of the moon

Short Answer Questions

Q1. What are the four main cardinal directions?

Q2. How many intermediate directions are there and how are they named?

Q3. What is the angle difference between cardinal and intermediate directions?

Q4. What is the angle of convergence (Grid Convergence)?

Q5. What are different types of bearing?

Long Answer Questions

Q1. Explain the concept of cardinal directions, intermediate directions and sub-intermediate directions. How are they categorized and what is the total number of directions?

Q2. Describe the three types of north (True North, Grid North and Magnetic North) and explain their differences?

Q3. What is the "Variation Between North Lines"? Discuss the three types of variations: Angle of Convergence, Magnetic Variation and Local Variation?

Q4. Explain different types of bearings.

Q5. What are methods of finding north? Give any 03 methods.

MAP READING

CHAPTER MR-IV: TYPES OF COMPASS & ACQUIRING BEARING



TEACHING INSTRUCTIONS

Period : 02 (80 Mins)
Type : Lecture/Practice
Year : 1st Year SD/SW
Conducting Officer : PI Staff

Training Aids: Classroom, Prismatic compass, Map, Screen, Pointer Staff, Presentation, Precis Flagged or Lesson Plan in File, Board, Markers, Display Charts.

Time Plan

- **Introduction : 03 Mins**
- **Part I : 07 Mins**
- **Part II : 10 Mins**
- **Part III : 10 Mins**
- **Part-IV : 05 Mins**
- **Conclusion : 05 Mins**
- **Practice : 40 Mins**

INTRODUCTION

1. The Prismatic compass is a navigating tool used to find magnetic bearings. It has been used extensively in the army, ships and aircraft for navigation and to find and maintain direction. There are two types of prismatic compass with which one magnetic bearing can be measured on the ground. Compass error is there due to material used in its manufacturing which causes deviation of magnetic needle from actual north.



Liquid Prismatic compass

Dry Lensatic compass

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I- Magnetic Compass and its types.**
- **Part II- Parts of Compass.**
- **Part III - How to take bearing.**
- **Part-IV - Compass error.**

LEARNING OBJECTIVES

- **To familarise with different types of Magnetic Compasses & understand their difference**.
- **To learn the parts of Parts of Prismatic Compass**.
- **To understand the mechanics of taking Bearing**.
- **To learn the reasons for Compass error and to factor the same whist taking bearing**.

PART I: MAGNETIC COMPASS AND ITS TYPES

2. **Magnetic Compass**. A magnetic compass is a navigational instrument used to determine direction relative to the Earth's magnetic poles. It works based on the Earth's magnetic field, which causes a magnetic needle inside the compass to align with the Earth's magnetic north.

DID YOU KNOW?

- During ancient history, magnetic compass called "*Matsya yantra"* was used for navigation by our ancestors in the 4th and 5th centuray AD. A prismatic compass can be used to measure all bearings from 0° to 360°.

3. **Types of Compasses**. Compass is mainly classified into two types: Liquid Prismatic compass and Dry Lensatic compass.

(a) **Liquid Prismatic Compass**. The Prismatic compass is a navigating tool with a transparent lid and magnetic needle, used for find directions and to measure bearings between two points by sighting through a prism. Without moving the eye, it is possible to sight the object and read bearing on the graduated dial at the same time.



(b) **Dry Compass (Lensatic Compass)**. The Dry Compass is a round box shaped mechanism which is set on a flat base plate helps in finding direction or bearing on ground. It is also used for orienting the map with the ground and helps in navigation. It is very sensitive and is used as a standard compass. To read bearing on Dry compass, open the compass so that the cover is at 90 degrees to the base and the lens bracket is at about 30 degrees with the vertical.



4. A dry military compass is better than other compasses because of its few special features, which are as follows:-

(a) **Induction Damped Needle**. These have been imposed for slowing down the rotation of the magnetic needle.

(b) **Aluminium Body Frame**. Aluminium metal frame makes this compass light weight (5.3 ounces with Line yard) but strong.

(c) **Waterproof**. It can be used even during rains as it is waterproof.

(d) **Climate Capacity**. Climate capacity of Military Compass is -500°F+1500°F. Due to this feature, this compass can be used in any weather conditions.

PART II - PARTS OF PRISMATIC COMPASS

5. **Parts of Prismatic Compass**. The parts of Prismatic Compass with their characteristics are as follows:-

(a) **Lid**. It is used as cover of the compass which comprises of tongue, tongue notch glass window.

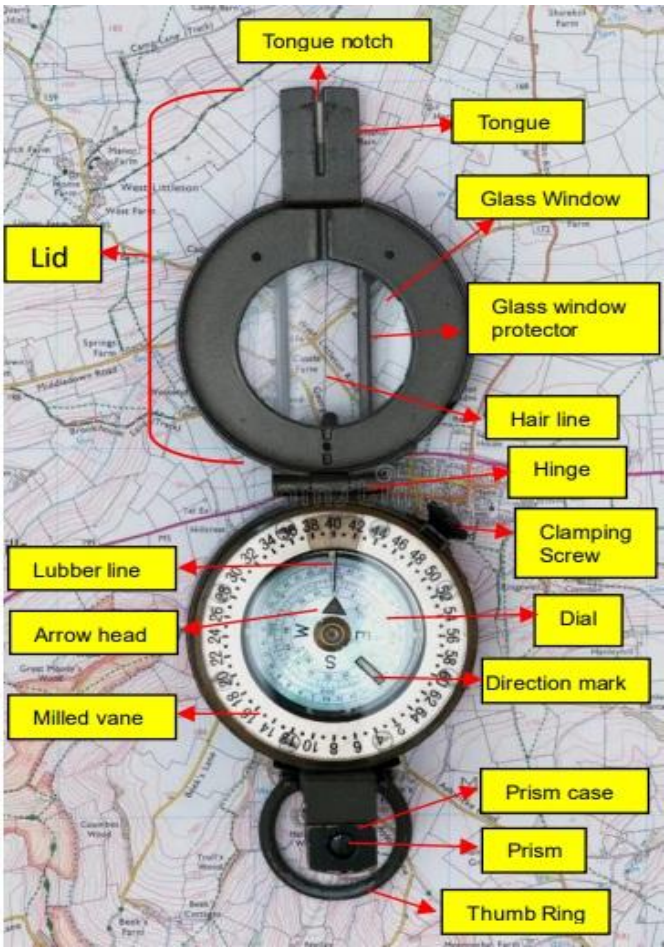
(b) **Tongue**. It is fitted to the Lid, which is used to lift the lid from outside and to keep the dial clear when the compass is not in use. It is also used during night march.

(c) **Tongue Notch**. It is a cut at the centre of tongue used to align the compass with target and used in setting of map.

(d) **Glass window**. It is the transparent part of lid used to see the target.

(e) **Glass window protractor**. It is the metallic part of the lid used to protect glass window.

(f) **Hair line**. It is used to align the compass with the target to read the bearing precisely.



(g) **Hinge**. It is used to join lid of compass with compass box.

(h) **Card or Dial**. To read the glass directly, the four cardinal directions are marked on dial with degrees and reverse bearing are marked to read the prism.

(j) **Glass plate or Milled Vane**. It is movable and marked with degrees, which may be set for marching on a bearing.

(k) **Arrowhead**. It indicates magnetic north and always points in the direction of north.

(l) **Prism**. It is positioned near the graduated circle for accurate reading of bearings. Magnifies the readings and ensures clear visibility of measurements.

(m) **Prism case**. It covers the prism to protect it from dust and moisture and raised and lowered to focus and read the bearing on dial.

(n) **Clamping Screw**. It is used to clamp the glass plate and then milled vane is easy to move and it is easy to set it with lubber line during night march.

(o) **Lubber line**. The lubber's line is a line on the compass used to indicate the direction in which the individual is heading during night march.

(p) **Direction Mark**. This is made up of radium which glows in dark therefore is used during night march.

(q) **Thumb ring**. It is used to hold the compass while taking bearing of an object.

INTERESTING FACTS

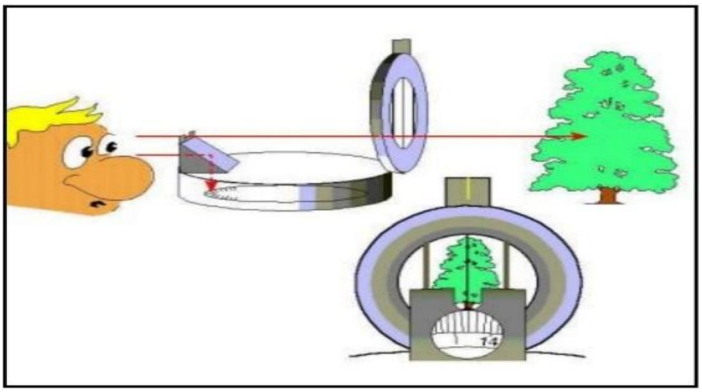
- The Liquid compass is used as a steering compass as oscillations of magnetic needle are damped in the liquid.
- The dial of compass is painted with radium to read bearing during night.

PART III: HOW TO TAKE BEARING WITH MAGNETIC COMPASS

6. **Reading bearing with a compass**. The magnetic bearing can be measured by magnetic compass in following steps:-

(a) Stand facing the direction of the object to read bearing with the compass.

(b) Open the window at 90 degrees and keep the prism over the milled vane and held the compass box with the thumb in thumb ring and the forefinger near hinge supporting lid and align the hairline with the target.



(c) At the same time look at the dial scale through the prism by raising or lowering its case and read the required bearing when arrowhead comes to rest.

(d) When see through the prism, readings increase from right to left as seen through the prism.

(e) This bearing will be a forward bearing.

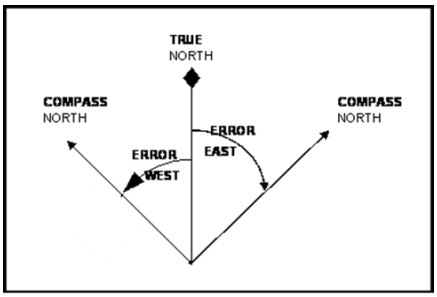






PART IV – COMPASS ERROR

7. **Compass Error**. Compass error means that the compass needle is pointing slightly right or left instead of exactly at $360^{o}/0^{o}$ degrees. Based on where the compass needle is pointing left or right will define the type of compass error. If the needle points left(west), the error will be subtracted and if the needle points right (east), compass error will be added to obtain the correct bearing.



8. **Local Magnetic Attraction (Deviation)**. Local Magnetic deviation is a magnetic interference in the compass reading caused due to local magnetic attraction or magnetic fields generated by things like electrical systems or ferrous materials. This local magnetic attraction can interfere with the compass needle, cause deviation from the true magnetic north and lead to erroneous reading.



9. If there is an error on the right or left of the compass, then the compass should be read as follows:-

(a) **East Compass Error**. If there is a mistake of 2 degrees east in the compass, then the compass bearings will be 2 degrees less at the time of the bearing, because the needle will start the bearings from 2 degrees ahead. Adding 2 degrees to the compass bearing with a fault of 2 degrees east will reveal the correct magnetic bearing.

(b) **West Compass Error**. If there is a mistake of 2 degrees west in the compass, then the compass bearings will be 2 degrees more at the time of the bearing, because the needle will start the bearings from 2 degrees ahead. Subtracting 2 degrees to the compass bearing with a fault of 2 degrees west will reveal the correct magnetic bearing.

Example 1: If the magnetic bearings of an object is 118° with a compass having east compass error of 2° east. What will be the correct bearing of that object?
Solution: With east compass error of 2° magnetic bearing of object = 118°
Correct bearing = 118° + 2° = 120°

Example 2: If the magnetic bearings of an object is 86° with a compass having west compass error of 2° west. What will be the correct bearing of that object?
Solution: With west compass error of 2° magnetic bearing of object = 86°
Correct bearing = 86° - 2° = 84°

Example 3: If the magnetic bearings of an object is 48° taken with a compass with 1° east error and a magnetic attraction of 2° to the west. What is correct magnetic bearing?
Solution: With east compass error of 1° magnetic bearing of object = 118°
Correct bearing = 48° + 1° = 49°

Correct bearing with mag attraction of 2° to the west = 49°- 2° = 47°

DID YOU KNOW

- Error in the reading of compass occurs due to local magnetic attraction or compass error. Compass error occurs due to two reasons:
  - Manufacturing defects
  - Mishandling

HOT

- Evaluate the importance of checking the compass for correctness and errors before use. What could be the consequences of neglecting this step?
- What could be the potential consequences of using a Prismatic Compass with a malfunctioning needle due to interference from a nearby magnetic field?

CONCLUSION

10. Prismatic compass is most important for soldiers, cadets and any individual to have command on Map Reading. The user should know the types of compasses, different parts of compass and its use i.e. how to navigate on compass bearing. While using the compass ensure that there is no iron object nearby as it may cause deviation. Before using a compass, errors and correctness should be checked.

SUMMARY

- The Prismatic compass is a navigating tool used for determine directions and to measure bearings or angles between two points by sighting through a prism used in surveying.
- Two types of compasses are liquid Prismatic compass and Dry compass.
- The Liquid compass is used as a steering compass.

- The dry card compass is very sensitive and light weight.
- The main parts of compass are Lid, tongue, Hing, Clamping screw, Glass window, Hair line, Glass plate, Dial, Arrowhead, Lubber line, Direction, Prism, Prism case, Thumb ring.
- To read bearing on compass, open the window at 90 degrees and hold the compass box with the thumb ring and align the hairline with the target.
- Compass error means that the compass needle is slightly right or left instead of exactly at 360 degrees.
- The Compass Error occurs mainly due to local magnetic attraction or magnetic fields generated by electrical systems or ferrous materials.

ASSEMENT EXERCISE

Multiple Choice Questions

Q1. Prismatic Compass is a _____ tool.

(a) Aluminium
(b) Survey
(c) Navigation
(d) None of these

Q2. Compass is used to find _____ bearing.

(a) Backward
(b) Forward
(c) True
(d) Grid

Q3. _____ is used as cover of compass.

(a) Glass van
(b) Lid
(c) Glass window protractor
(d) Glass window

Q4. Bearing is set on _____ for marching.

(a) Milled vane
(b) Lid
(c) Dial
(d) Lubber line

Q5. What is the weight of Dry Military Compass (including lanyard)?

(a) 5.2 ounces
(b) 4.3 ounces
(c) 5.3 ounces
(d) 4.2 ounces

Q6. The Arrow head indicates–

(a) Direction of marching
(b) Magnetic variation
(c) Deviation
(d) Magnetic north

Q7. The movable part of the compass which is marked with degrees is called–

(a) Hinge
(b) Prism
(c) Dial
(d) Glass Plate

Q8. Which line is used to indicate the direction, on the compass, in which the individual is heading during the night march?

(a) Hair Line
(b) Grid Line
(c) Lubber Line
(d) Direction Mark

Q9. What is the primary function of a prismatic compass?

(a) To measure distances between two points
(b) To calculate the elevation of a point
(c) To determine directions and measure bearings or angles
(d) To calculate the area of a land parcel

Q10. Which of the following is NOT a part of the prismatic compass?

(a) Hairline
(b) Steering wheel
(c) Prism case
(d) Lubber line

Q11. When reading the bearing on a prismatic compass, which of the following should be aligned with the target?

(a) Thumb ring
(b) Hair line
(c) Direction mark
(d) Lubber line

Q12. What is meant by 'Compass Error'?

(a) The angular difference between true north and magnetic north
(b) The difference between magnetic and true south
(c) The alignment of the compass dial
(d) The error in the bearing due to a malfunction in the compass

Q13. The Dry compass is also known as–

(a) Prismatic compass
(b) Survey compass
(c) Lensatic compass
(d) None of these

Q14. Compass Error occurs due to _____.

(a) Magnetic north and true north
(b) Earth’s magnetic field and magnetic interference
(c) Electrical systems and ferrous materials
(d) Both (b) and (c)

Q15. What does "magnetic variation" refer to?

(a) The interference caused by magnetic fields from electrical systems
(b) The difference between true north and magnetic north
(c) The shift in the compass needle due to Earth's gravitational pull
(d) The misalignment between compass north and geographic north

Short Answer Questions

Q1. Write five parts of a Prismatic Compass.

Q2. What is Dial?

Q3. What is magnetic deviation?

Q4. What is the use of lubber line?

Q5. What causes Compass Error?

Long Answer Questions

Q1. What are the types of Prismatic Compass?

Q2. What are the characteristics of Dry Compass?

Q3. What is Compass Error and its types?

Q4. How to take bearing with Prismatic Compass?

Q5. What is the use of Clamping Screw and Direction mark?

MAP READING

CHAPTER MR-V: SERVICE PROTRACTOR & ITS USES



TEACHING INSTRUCTIONS

Period : 01 (40 Mins)
Type : Lecture/Practice
Year : 1st Year SD/SW
Conducting Officer : PI Staff

Training Aids: Class Room, Prismatic compass, Map, Screen, Pointer Staff, Presentation, Precis Flagged or Lesson Plan in File, Board, Markers, Graph Pencils, Display Charts.

Time Plan

- **Introduction : 05 Mins**
- **Part I : 10 Mins**
- **Part II : 20 Mins**
- **Conclusion : 05 Mins**

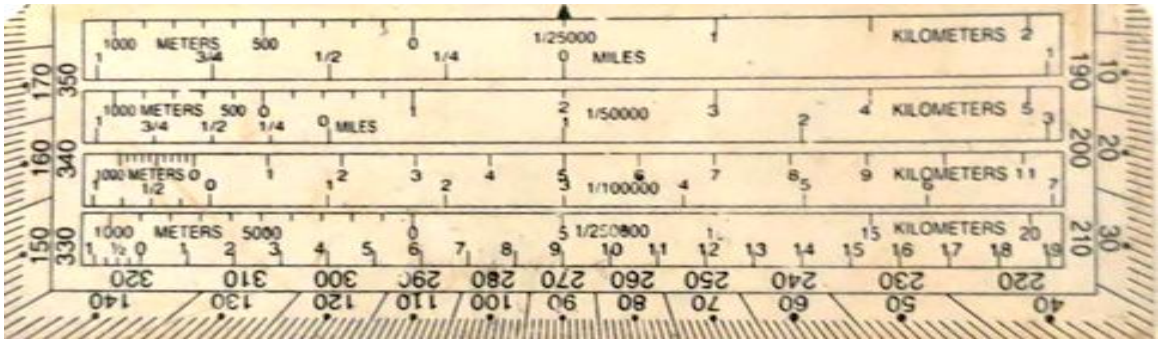
INTRODUCTION

1. The Service Protractor is a essential device which is used in map reading. It can be used to read grid bearing of an object and convert it into a magnetic bearing to locate the object on ground. Conversely, after taking the magnetic bearing of an object on ground it helps in converting the same into grid bearing and subsequent plotting on the map. The protractor provides not only angle measurements, but also had a variety of scales including reference scale and distance scale which helps in finding distance between two objects on the map. The scale lines on the service protractor are drawn exactly the same manner as given at the bottom of the map.

PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I : Service Protractor and its uses.** ➢ **Part II: Reading and Plotting of Grid bearing with Service Protractor.**	➢ **Understanding Service Protector and its uses.** ➢ **To learn Reading and Ploting Grid Bearing with Service Protractor.**

PART I: FAMILIARISATION WITH SERVICE PROTRACTOR

2. The service protractor is an instrument used for plotting and measuring grid bearing as well as distance between any two objects on the map. It is the essential link between the compass and the map, as magnetic bearings measured with compass can be plotted as grid bearing on the map. Likewise, magnetic bearing of an object obtained by a compass can be converted into grid bearing and plotted on the map. Service Protractor also has a scale similar to the one similar to the one made at the bottom of the map. With this scale the distance of two points on the map can be measured and thereafter, converted to actual distance of two objects on the ground. Service protector is made of either card board, metal or ivory. It is a simple device which is 6 inches long and 2 inches wide. The front face of the protractor is marked with 360° degrees around its three edges. In case of necessity an experienced cadet can make this protector with careful measurement at home also.



Front view of Service Protractor

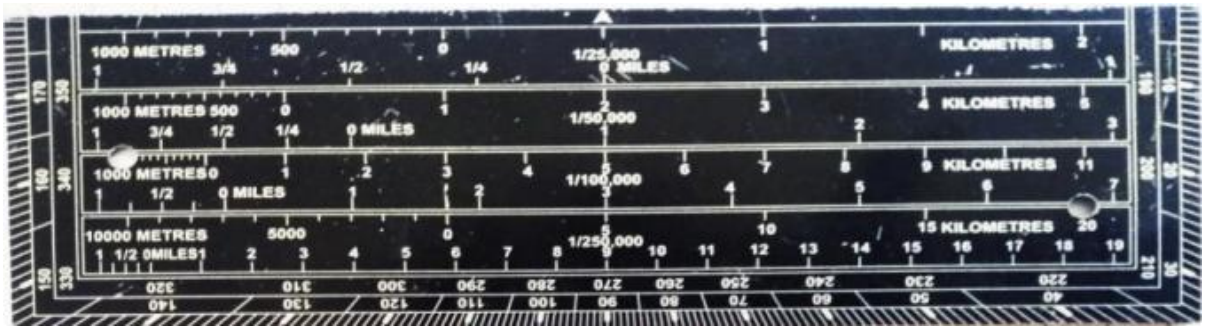
INTERESTING FACTS

- The degrees are marked in a clockwise direction, on outer side starting from 0° to 180° and from 181° to 360° on the inner side of a service protector.
- We can plot and measure grid bearing of any object by keeping small arrow, which is at the centre of the fourth side of the protector, on the object.

3. **Description of Service Protractor**. Service protector is made of either card board, metal or ivory. It is a simple device which is 6 inches long and 2 inches wide. It has two faces, namely front side and back side having different markings. These are explained as under:-

(a) **Front Face**. The front face of the protractor is marked with 360° degrees around its three edges. On the underside where the degrees are marked three (3) double scale lines and two (2) single scale lines were formed. There are primary and secondary parts. The distribution of divisions is given in the table below:-

SCALE	PRIMARY DIVISION	SECONDARY DIVISION
1/50000 .789 Mile = 1”	1 KM 1Mile	100 Metre ¼ Mile
1/100000 1578 Mile = 1”	1KM 1Mile	100 Metre ¼ Mile
1/200000 3156 Mile = 1”	5 KM 1Mile	1 KM ¼ Mile
1/25000 1/250000	500 Metre 1Mile	100 Metre ¼ Mile



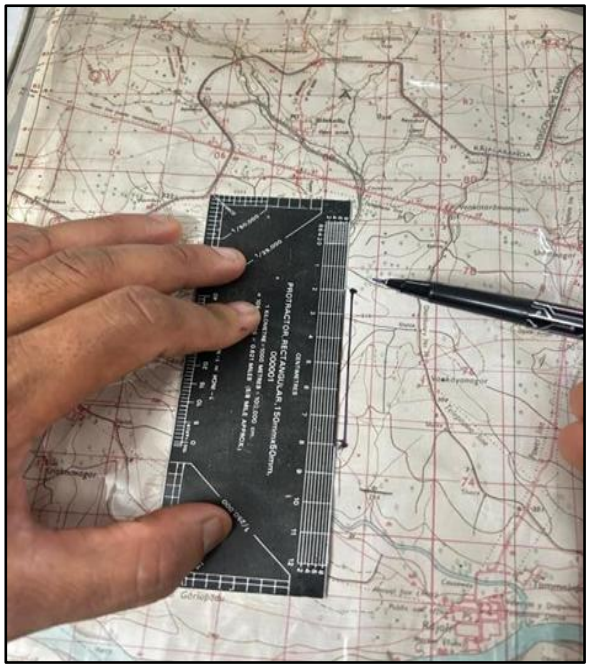
(b) **Back Face**. On the other side of it, there are total six scales are marked. One centi-meter scale, one inch scale and four diagonal scales of 1/50,000, 1/25,000, 1/1,00,000 and 1/2,50,000 with sub divisions marked on both edges of service protractor for finding correct grid reference of respective scales. Its name and mark are written in the empty space in the middle.



PART II: READING AND PLOTTING OF GRID BEARINGS WITH SERVICE PROTRACTOR

4. **Method of Using Service Protractor**. Before using a service protractor, it is important to know that the object is at which side of map, from which bearings are to be read. See whether it is in the east on the right or in the west on the left. If the object is in the east, the arrow in the centre of service protractor will be placed at the object and place the edge marked with degrees in the east. If the object is in the west, the arrow will be placed at the object and place the edge marked with degrees in the west.

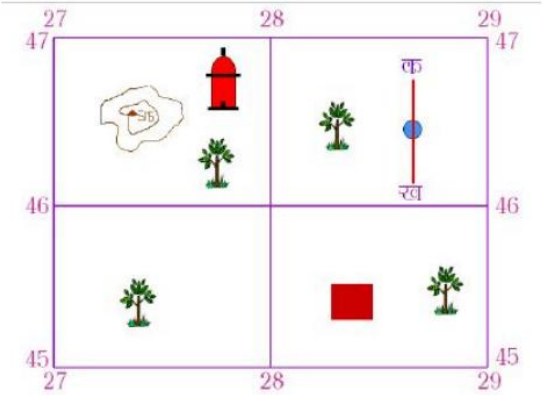
5. **Reading bearing with Service Protractor**. If you want to read the bearings of one place from another place (source), then draw a line with sketch or pencil joining both places. Keep service protractor parallel to easting grid lines.



(a) If the object whose bearing is to be detected is in the east of another object, then place the arrow (zero edge) on source object and read outer bearing (ranging from 0° to 180°).

(b) If the object whose bearing is to be detected is in the west of another object, then place the arrow (zero edge) on source object and read inner bearing (ranging from 181° to 360°).

For measuring grid bearing of temple from contour, we place arrow (zero edge) of service protector on centre of contour



USES OF SERVICE PROTRACTOR

- For reading and plotting grid bearing of the map.
- Can read forward bearings and back bearings together.
- Can measure distance between two objects on the map.
- Diagonal reference scale can used for finding 6 figure grid reference accurately.
- Can read and plot 0 to 360 degrees on map.
- Can find degree of slope.
- Helps draw parallel lines on maps.
- Helps measure hundredth part of 1 cm or an Inch.

6. **Plotting Bearings on Map**. For plotting bearing on map, first convert magnetic bearing into grid bearing than plot grid bearing on map. For example:-

(a) In given figure if magnetic bearing of church at GR 167287 from a fort at GR 152305 is 145°

(b) The difference between grid north and magnetic north is 5°.

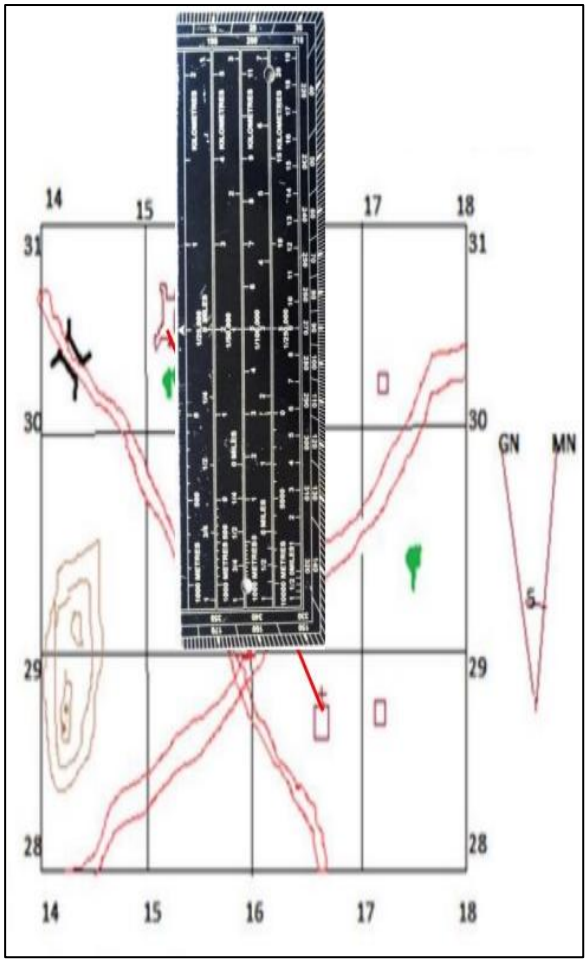
(c) By adding 5°, convert magnetic bearing into grid bearing.

(d) Add 5° to 145°.

(e) Grid bearing of church from fort is 150°.

(f) Put arrow (zero edge) of service protractor on fort, parallel to easting grid lines.

(g) Mark a point on 150°and draw a line from fort to that point which will meet at church.



HOT

- Imagine you are using a Service Protractor to plot the bearing of a distant object. What steps would you follow to ensure that the bearing is correctly plotted on the map.
- Evaluate the importance of understanding the positioning of an object (east or west) when using the Service Protractor. How could incorrect placement of the arrow impact the results?
- Compare the use of the Service Protractor on maps with different scales (e.g., 1/50000 vs. 1/100000). How does the scale affect the precision of plotting bearings?

CONCLUSION

7. Service Protractor is a crucial tool for accurately converting magnetic bearings to grid bearings, facilitating the process of finding and plotting the bearings of objects on a map. The protractor's various scales, including those for different map ratios, further enhance its utility by aiding in grid reference calculations. Through its method of use, the Service

Protractor ensures that bearings are correctly read and plotted, making it an indispensable instrument for navigational tasks that require precision and reliability.

SUMMARY

- Service Protractor a tool that converts magnetic bearings to grid bearings, linking a compass to a map.
- Used to find and plot the bearing of an object by converting it into a magnetic bearing, which can be set on a compass to locate the object.
- Includes a protractor that offers angle measurements and a variety of map scales.
- The underside features markings for degrees and multiple scales (primary and secondary divisions) for measuring distances on maps.
- Different scales (e.g., 1/50000, 1/100000, 1/200000, 1/25000) on Service Protractor are used for distance calculations on maps.
- These scales measure distances in miles, kilometres and meters, with sub-divisions for precision.
- The protractor also features scales for centimetre, inch and diagonal measurements (e.g., 1/50,000, 1/25,000).
- For using the Service Protractor determine the location of the object (east or west) on the map. Place the zero edge (arrow) at the object, aligning the degree-marked edge accordingly (east for objects in the east and west for objects in the west).
- Read the bearing on the protractor by matching the object's position relative to the grid lines.
- For objects east of the source: Read the outer bearing (0° to 180°).
- For objects west of the source: Read the inner bearing (181° to 360°).
- Ensure the zero edge of the protractor is aligned parallel to the grid lines for accuracy and now.

ASSESSMENT EXCERCISE

Multiple Choice Questions

Q1. What is the use of the Service Protractor?

(a) To measure the distance between two points
(b) conversion of magnetic bearings into grid bearings
(c) To calculate the area of a map
(d) To locate north on a compass

Q2. What materials are used to make a Service Protractor?

(a) Wood and plastic
(b) Cardboard, metal and ivory
(c) Glass and rubber
(d) Steel and plastic

Q3. What does the Service Protractor help in measuring between two points on a map?

(a) Temperature (b) Elevation (c) Bearing (d) Area

Q4. Which of the following is NOT a scale marked on the Service Protractor?

(a) 1/50,000 (b) 1/25,000 (c) 1/5,000 (d) 1/100,000

Q5. How many primary and secondary scale lines are on the underside of the Service Protractor?

(a) Four primary and three secondary
(b) Three primary and two secondary
(c) Five primary and four secondary
d) Two primary and one secondary

Q6. Which scale is used for measuring 1 KM on a 1/50000 scale map?

(a) 1 Mile (b) ¼ Mile (c) 100 Metres (d) 1 meter

Q7. What is the purpose of the diagonal scales on the Service Protractor?

(a) To measure distance in kilometres
(b) To plot bearing points
(c) To find the correct grid reference of respective scales
(d) To measure elevation

Q8. How should the Service Protractor be placed when measuring bearings from an object in the east?

(a) Place the zero edge in the west
(b) Place the arrow on the object and mark the degree scale in the east
(c) Place the zero edge in the west
(d) Place the Service Protractor at the top of the map

Q9. If an object is in the west, where should the Service Protractor's arrow be placed?

(a) On the top of the map
(b) At the west of source object
(c) On the bottom of the map
(d) In the middle of the map

Q10. What is the range of the outer bearing on the Service Protractor when the object is in the east?

(a) 0° to 90° (b) 0° to 180° (c) 180° to 360° (d) 90° to 180°

Q11. What is the range of the inner bearing on the Service Protractor when the object is in the west?

(a) 0° to 90° (b) 90° to 180° (c) 181° to 360° (d) 360° to 540°

Q12. What must be done first when plotting a bearing on a map using the Service Protractor?

(a) Draw a line between two points
(b) Convert magnetic bearing into grid bearing
(c) Mark the map with elevation data
(d) Mark the distance of two objects

Q13. What is the grid bearing of an object if the magnetic bearing is 135° and the difference between magnetic north and grid north is 5° and grid north is in west of magnetic north?

(a) 150° (b) 140° (c) 145° (d) 135°

Q14. What is the purpose of the protractor on the Service Protractor?

(a) To measure distances on the map
(b) To convert grid bearings into magnetic bearings
(c) To provide angle measurements
(d) To determine the area of the map

Q15. How many scales are there on back side of Service Protractor?

(a) 4 (b) 6 (c) 3 (d) 7

Short Answer Questions

Q1. What is the primary function of a Service Protractor?

Q2. What materials are used to make a Service Protractor?

Q3. What scales are marked on the Service Protractor?

Q4. How do you use the Service Protractor to measure bearings?

Q5. What is the first step when plotting a bearing on a map with the Service Protractor?

Long Answer Question

Q1. What is Service Protractor?

Q2. What are uses of Service Protractor?

Q3. How to plot bearing on map?

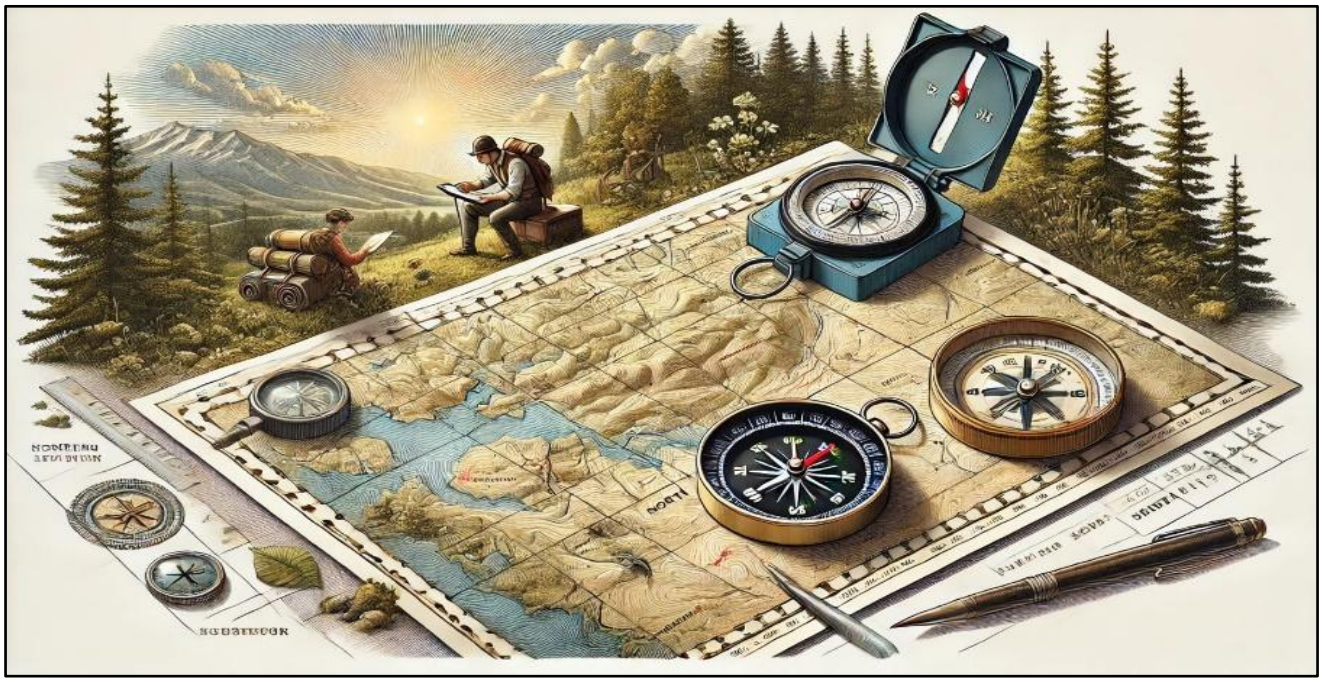
Q4. Give description of Service Protractor?

Q5. If magnetic bearing is 266° and difference between grid north and magnetic north is 5°. Grid north is in east of magnetic north. Calculate grid bearing.

MAP READING

CHAPTER MR-VI: SETTING OF A MAP, METHODS OF FINDING OWN POSITION

“Map setting, north orientation and position are crucial for navigation.”



TEACHING INSTRUCTIONS

Period	**:**	**06 (240 Mins)**
Type	**:**	**Lecture/Practice**
Year	**:**	**1st (02), 2nd (02) &3rd Year (02) SD/SW**
Conducting Officer	**:**	**Permanent Instructor**

Training Aids: Map sheets, Compass, Service Protractor, Pointer, Charts, Black board & Chalk.

Time Plan

- **Introduction : 05 Mins**
- **Part I : 35 Mins**
- **Part II : 35 Mins**
- **Conclusion : 05 Mins**
- **Practice : 160 Mins (04 periods)**

INTRODUCTION

1. This lecture aims to equip NCC Cadets with the fundamental skills required for effective map reading and navigation. These abilities are crucial for interpreting terrain, planning routes and safely navigating during field operations or outdoor activities. The lecture will focus on practical techniques for setting a map, identifying directions and determining one's location, providing cadets with confidence and precision in the field. Cadets will learn two key methods for setting a map: using a compass to align magnetic north with the map and using visible landmarks to match terrain features with the map. The lecture will also cover methods like resection, which involves bearings to known landmarks and inspection, which relies on comparing terrain features, to determine their position on the map.

PREVIEW

The lecture will be conducted in the following parts:-

- **Part I: Methods of Setting a Map.**
- **Part II: Finding Own Position on the Map.**

LEARNING OBJECTIVES

- **Understanding and applying methods to set a map**.
- **Learn techniques to determine own position**.
- **Understanding Resection and Inspection methods**.
- **Developing practical navigation skills**.

PART I: METHODS OF SETTING A MAP

2. **Setting of a Map**. To "set" a map means to orient it so that it matches the real-world direction on the ground. More specifically, it means ensuring that **North** on the map aligns with **North** on the ground. When a map is properly set, the physical features on the map will match up correctly with the actual features on the ground, making it much easier to read and use.



DID YOU KNOW?

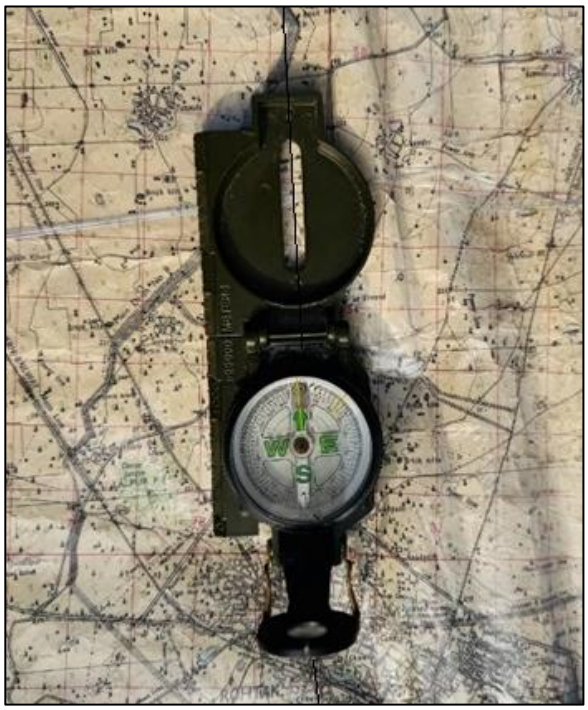
- Setting a map ensures that directions and bearings are correctly interpreted. It makes it easier to plan routes and understand your surroundings.

3. **Methods of Setting a Map**. There are two primary methods for setting a map:-

(a) By Compass.

(b) By Objects on the Ground.

4. **Setting a Map Using a Compass**. The compass method is one of the most common and reliable ways of setting a map. Here's how you can do it:-



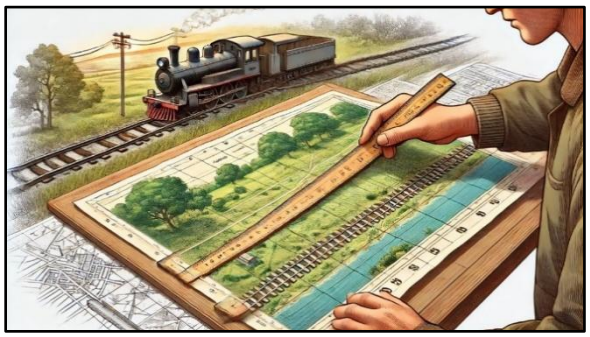
(a) Spread the map on a levelled ground. On Topographical maps besides grid lines, 3-4 Black lines are drawn vertical and horizontal.

(b) These lines are longitudes and latitudes. The vertical lines indicate true north.

(c) Now open the compass and put it on one of this longitude (vertical black line) in such a way that its tongue notch, hairline and thumb ring notch cover the black line, while putting compass on a line, it is ensured that compass lid should point north of the map (grid north).

(d) Now without moving compass, move map in such a way so that arrow mark on the compass come in line with hair line at 360 degrees. Now map is set.

5. **Setting a Map Without a Compass**. If you don't have a compass, there are still ways to orient your map based on visible landmarks. These methods rely on natural features around you to set the map.



(a) **Using a Straightedge (eg, Railway Line)**. Recognize a well-defined, straight feature on the map, such as a railway line, road,or river. Find this feature on the ground and align your position along it. Hold the map in such a way that the object you see on the ground aligns with the corresponding feature on the map. This will allow you to orient the map to the ground.

(b) **Recognizing and Joining Landmarks**. Identify one object on the map and the corresponding object on the ground. It could be a mountain, tower, or other prominent feature. Line up your position on the map with the landmark on the ground. Holding the map steady, ensure that when you look in the direction of the landmark, it lies in a straight line between you and the map. This sets the map in the correct orientation.

(c) **Parallel Method**. This method involves recognizing two landmarks visible on both the map and on the ground. These could be roads, rivers, or other distinct features. Even if these landmarks are not visible continuously, you can choose two landmarks that are located on the map and on the ground. Hold the map so that the two landmarks on the map are parallel to the two corresponding landmarks visible on the ground. This will give you a rough orientation of the map.



(d) **On/Near Line Joining Two Points**. Identify two landmarks that are nearby, either on the map or on the ground. Stand on an imaginary line joining these two points. Once you do this, you can position the map so that the two landmarks on the map are aligned with the real-world counterparts.

DID YOU KNOW?

- As magnetic variation in India is very small so setting of map is almost correct without considering magnetic variation.
- Compass Error and Local Magnetic Attraction should be taken care of while using a Compass.

PART II: FINDING OWN POSITION ON THE MAP

6. **Finding Your Own Position on the Map**.Knowing your position on a map is vital for navigation. Knowing own position is the primary step of learning map reading as well as developing basic navigation skills with the help of map and compass. In simple words the term 'Finding own position' refers to locating one's own position on the map exactly as per actual position on ground.



7. **Methods of Finding Own Position**. Finding own position is a relatively easier task if the location of the observer is close to or is at a clearly identifiable features/objects which can be seen both on ground as well as on map. In such cases, inspection method can be use to find the own location. However, a cadet may be in a situation where he finds himself in a unknown area, not having any clearly identifiable and known features/objects which are available both on map as well as ground. In such cases resection method must be used to find own position. There are two methods to find own position described in succeeding paragraph.

8. **By Inspection Method**. Align the map with visible features on the ground. Compare the map's details (e.g., rivers, buildings) with the surroundings to estimate your position. This is the easiest as well as most reliable method of finding own position for which it is essential to know how to identify manmade and natural features of ground on map, also one must have good practice of identifying the feature on ground through the help of information provide by map. For which following things are necessary:-

(a) Have a good knowledge of map reading.

(b) One must have proper estimation of the area in which the own position is required to find.

(c) There should be a land mark around the place to be found in the map.

(d) The cadet should carry out continuous map reading whilst arriving at the new location to use inspection method effectively.



Inspection method consists of:

- Setting of maps
- Recognition of general area of own position on the map.
- A close study of the ground details.
- Continuous map reading

9. The inspection method in map reading is a simple technique used to approximate your location by visually matching the terrain features around you with corresponding features on a map. Unlike more precise methods like resection, the inspection method relies on observation and recognition rather than measurements with a compass.

10. **Advantages of Inspection Method**.

(a) **Quick and Easy**. No special tools (like a compass or protractor) are needed.
(b) **Practical**. Useful for casual navigation when precision isn't critical.

11. **Disadvantages of Inspection Method**.

(a) **Less Accurate**. The method is approximate and can be prone to errors.
(b) **Dependent on Visibility**. It requires clear visibility of recognizable landmarks.
(c) **Subjective**. Estimation of position varies depending on the user's observation skills.

12. **When to Use the Inspection Method**.

(a) In good weather with clear visibility.
(b) When you need a quick approximation of your location.
(c) In areas with distinct and easily identifiable terrain features.

13. This method is commonly used by hikers, scouts and in casual navigation scenarios when precision isn't necessary. However, for more accurate navigation, combining it with tools like a compass or GPS is recommended.

14. **Resection Method (With Compass)**. Identify two or three visible landmarks on the map as well as on the ground. Use a compass to find their bearings, then plot back bearings on the map. The intersection of the lines is your location. Resection Method (With Compass) explained in detail in next chapter.

15. **Advantages of Navigation by Compass**.

(a) **Simplicity**. A compass is easy to use and does not require batteries or advanced technology. It works reliably in almost any condition.

(b) **Global Use**. It can be used anywhere on Earth, including remote areas, forests, mountains or at sea, making it a versatile tool for travellers, hikers and sailors.

(c) **Durability**. A compass is generally a rugged, durable tool that can withstand harsh outdoor environments, such as rain, wind and cold temperatures.

(d) **No Need for Satellites**. Unlike GPS, a compass doesn’t rely on satellites or signals, making it a reliable tool in areas with poor satellite coverage or when technology fails.

16. **Disadvantages of Navigation by Compass**.

(a) **Magnetic Interference**. The compass can be affected by magnetic fields from nearby metal objects, electrical equipment, or even large geological deposits, leading to inaccurate readings.

(b) **Requires Skill**. Effective use of a compass requires knowledge of how to read it properly, understand maps and orient the compass with the environment, which can be challenging for beginners.

(c) **Limited Information**. Unlike GPS, a compass only shows direction and does not

provide information on specific locations, distances, or real-time updates, requiring additional tools like maps.

(d) **Not Suitable for Urban Areas**. In densely built environments, such as cities with large buildings and metal structures, the compass can be less reliable due to interference from surrounding materials.

DID YOU KNOW?

- Sun's Position: Ancient travellers used the sun and shadows to determine direction, a technique still taught in survival training.
- North Star: Polaris, a fixed point in the northern sky, has guided navigation for centuries, aiding explorers and sailors.

HOT

- How would you determine your position on a map if you were lost in a dense forest with no compass or visible landmarks?
- Compare the accuracy of the Equal Altitude Method and the Resection Method. Which one would you prefer in varying terrains, and why?
- How can traditional navigation methods (e.g., watch method, star navigation) complement GPS technology in remote or challenging environments?

INTERESTING FACTS

- **Star Navigation: Ancient sailors and explorers, including Egyptians and Polynesians, used the North Star (Polaris) and the night sky for navigation long before modern tools became available**.

CONCLUSION

17. The lecture provides a comprehensive understanding of the essential skills for map reading and navigation. By learning how to set a map using a compass or landmarks, determine true north using natural indicators and find their own position on a map, cadets will be equipped with the practical skills needed for effective navigation in the field. These foundational skills are crucial not only for military operations but also for personal safety and confidence when traveling in unfamiliar or remote areas. Mastering these techniques ensures cadets can navigate independently, regardless of the availability of modern technology or equipment.

SUMMARY

- **Setting a Map:** Align the map with the real-world environment to ensure its features correspond to the terrain accurately.
- **Using a Compass:** Learn to draw a magnetic north line, adjust for magnetic variation and align the map with magnetic north.
- **Using Landmarks:** Orient the map by aligning visible landmarks (roads, rivers, buildings) with corresponding map features, useful without a compass.
- **Pole Star:** Identify true north using Polaris in the Northern Hemisphere.
- **Finding Position with Compass:** Use bearings from three visible landmarks to determine the exact location on the map.
- **Finding Position without Compass:** Use tracing paper to project lines from landmarks and identify their intersection as the location.
- **Position by Inspection:** Match terrain features like hills, rivers, or roads with the map to estimate your position.
- **Practical Applications:** Techniques are crucial for navigating in challenging environments with or without modern tools.
- **Outcome:** Cadets will gain confidence and precision in setting maps, finding directions and identifying their positions.
- Navigation by compass is simple, low cost, versatile covers boarder areas.
- Navigation by a compass is affected by magnetic and electric fields which cause compass error and magnetic deviation.
- Unlike GPS, a compass doesn't rely on satellites or signals, making it a reliable tool in areas with poor satellite coverage or when technology fails.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What does it mean to "set" or "orient" a map?

(a) To draw features on the map
(b) To align the map with real-world objects
(c) To measure distances on the map
(d) To colour the map according to terrain types

Q2. Which of the following is NOT a method of setting a map?

(a) By compass
(b) By using objects on the ground
(c) By drawing landmarks
(d) By using the sun for orientation

Q3. How do you orient a map using a compass?

(a) By aligning the map with true north
(b) By aligning the map's magnetic north with the compass needle
(c) By aligning the map with visible stars
(d) By drawing a magnetic line on the map

Q4. What method can you use to find true north without a compass in the Northern Hemisphere?

(a) Watch method
(b) Equal Altitude method
(c) Using a smartphone GPS
(d) Using a road map

Q5. What is the main advantage of the Equal Altitude Method?

(a) It is quick and easy
(b) It works well on cloudy days
(c) It accurately determines true north
(d) It requires only basic knowledge of the stars

Q6. Which star is used to determine true north in the Northern Hemisphere?

(a) Sirius (b) Orion
(c) Polaris (North Star) (d) Betelgeuse

Q7. Which of the following is a method of finding your position on a map?

(a) Map inspection
(b) Resection method with a compass
(c) Checking local weather patterns
(d) Using satellite imagery

Q8. In the resection method, what is used to determine the position on a map?

(a) The position of a compass needle
(b) Bearings from visible landmarks
(c) A GPS device
(d) A topographic map

Q9. What is the purpose of the "By Inspection" method for finding your position?

(a) To calculate precise coordinates
(b) To study and match ground features with those on the map
(c) To draw a new map based on surroundings
(d) To use a compass to measure distances

Q10. Which of the following methods would you use if a compass is unavailable?

(a) Inspection method (b) Parallel method
(c) Tracing paper method (d) Magnetic line method

Q11. What is first step for finding own position?

(a)Select four objects on ground
(b)Take grid bearing of identified objects
(c) Setting of maps
(d)Take magnetic bearing of identified objects

Q12. Which of the following can be used to align a map with the ground when no compass is available?

(a) A compass rose
(b) A straightedge or visible landmarks
(c) A protractor
(d) A ruler

Q13. Which feature on a topographical map helps identify true north?

(a) Latitudes (b) Longitudes (c) The compass rose (d) Grid lines

Q14. When setting a map using a compass, where should the compass lid point?

(a) Towards the south of the map
(b) Towards the east of the map
(c) Towards the north of the map
(d) Towards the west of the map

Q15. What is the second method for setting a map without using a compass?
(a) Using a GPS device
(b) Using visible landmarks
(c) Using a straightedge
(d) Using a magnifying glass

Short Answer Questions

Q1. What does it mean to "set" a map?

Q2. What is magnetic declination?

Q3. What is the main purpose of the equal altitude method for finding north?

Q4. In the Resection Method with a compass, what is done after identifying landmarks?

Q5. What feature is primarily compared to locate one's position using the Inspection Method?

Long Answer Questions

Q1. Explain the process of setting a map using a compass?

Q2. Describe how you would determine your position on a map using the resection method with a compass?

Q3. How does the watch method help determine the direction of true north?

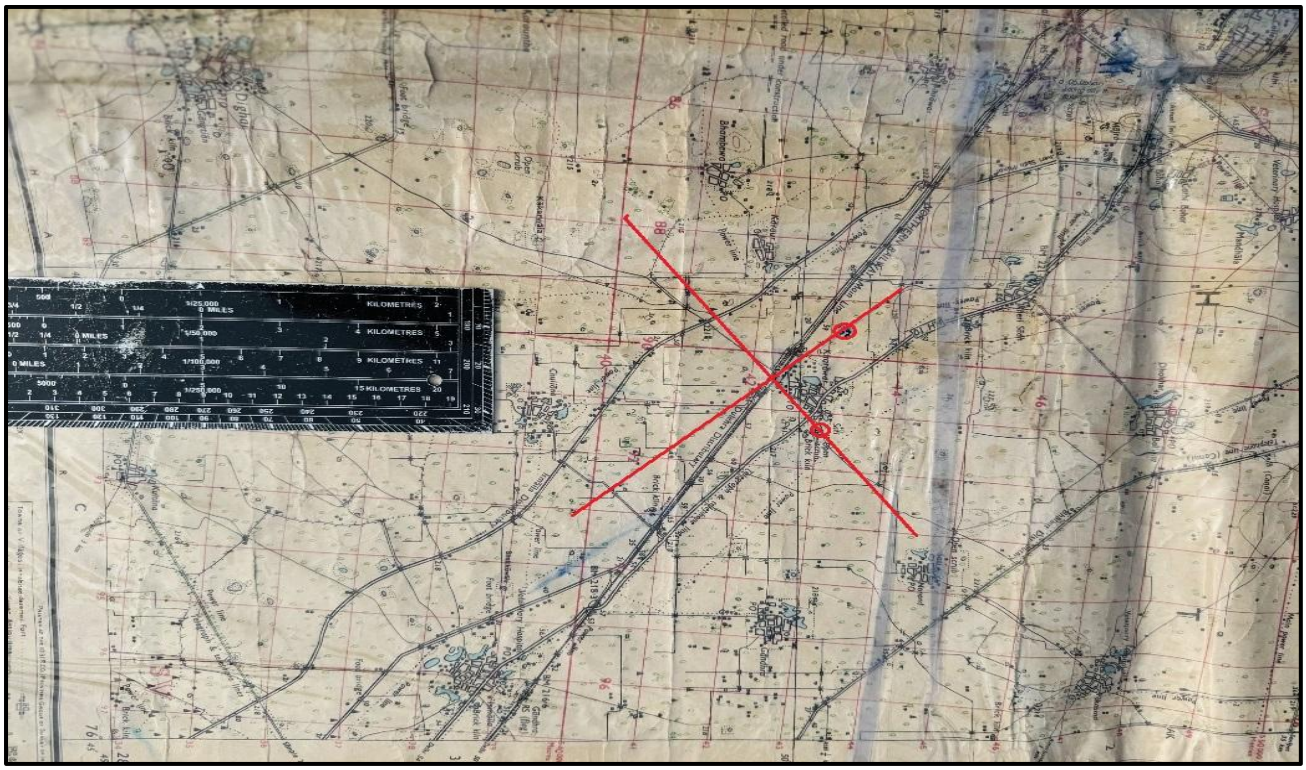
Q4. Explain equal altitude method for finding north.

Q5. Describe inspection method for finding own position.

MAP READING

CHAPTER MR-VII: RESECTION METHOD WITH COMPASS

"Sometimes, in order to move forward, we must take a back bearing, retracing our steps to understand how far we've come."



TEACHING INSTRUCTIONS

Period : **04 (160 Mins)**
Type : **Lecture and Practice.**
Year : **2nd(02) and 3rd Year(02) SD/SW**
Conducting Officer : **Permanent Instructor.**

Training Aids : **Map, Compass, service protector, GPS, Activities, Work Sheets, Assignments.**

Time Plan

- **Introduction : 05 Mins**
- **Part I : 30 Mins**
- **Conclusion : 05 Mins**
- **Practice exercises(01 period for 1st year) (02 period for 2nd year) : 120 Mins**

INTRODUCTION

1. In this Chapter we will cover concepts such as resection method with compass and inspection method. Back bearing is a concept commonly used in navigation, both in land and marine settings. It refers to the reverse direction of a bearing or course that one has previously taken. The resection method is a fundamental technique in navigation and orienteering, primarily used to determine one's position on a map or in the field. This method involves using known landmarks or reference points in the environment and measuring their bearings (angles) from the observer's current location. By drawing lines from these landmarks on a map based on their respective bearings, the intersection of these lines identifies the observer's position.

PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I: Resection method with compass.** ➢ **Part II: Practice exercise.**	➢ **Resection with Compass Method to find own position**. ➢ **Inspection Method when no navigational tools are available**. ➢ **Practice exercises**.

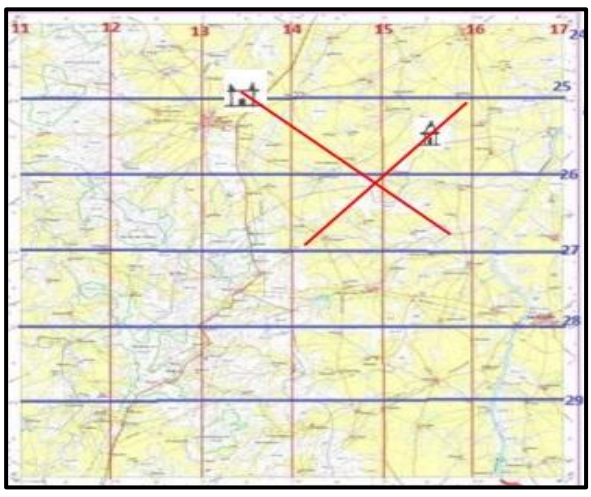
PART I: RESECTION METHOD WITH COMPASS

2. The resection method in map reading is a technique used to determine your own position on a map by identifying and using known landmarks visible in the terrain. This method is particularly useful when you do not know your exact position but can see recognizable features around you. In practice we can calculate bearing of the observer without moving to observer. In other words in map reading and navigation, back bearing (or back azimuth) refers to the direction opposite to a given bearing. It is useful when retracing a path or checking your position by sighting an object in reverse.

3. **Steps in the Resection Method**. Steps for resection method are as follows:-

(a) Identify at least two (preferably three) prominent, identifiable landmarks or objects on the ground (e.g., a hill, a tower, or a building) and find their corresponding locations on the map.

(b) The angle between these three prominent objects must not be more than 100 or less than 30 apart. These objects should be clearly visible from a distance.



(c) Use a compass to measure the magnetic bearing (forward bearing) of each landmark(object) from your current position.

(d) Convert the forward magnetic bearing into Grid bearing with the help of magnetic and grid variation given on the map.

(e) Convert Grid bearing into grid back bearing by adding or subtracting 180°.

(f) The lines of the back bearing of these objects must be plotted on the map with the help of service protector.

(g) The point where these lines intersect indicates your approximate location. If you use three landmarks, the intersection will form a triangle and your position is the centre of triangle.

(h) This method is especially useful when you don't have a direct way of measuring distances to the reference points, but you can see them from your location.

TIPS FOR ACCURACY

- Use distant, distinct landmarks to reduce errors caused by small angle deviations.
- Ensure the compass is held steady and level when taking bearings.
- Double-check the landmarks to avoid misidentifying features.

INTERESTING FACTS

- In marine navigation, back bearings are crucial when returning to a port or retracing a route in open water, helping sailors avoid mistakes in direction and drift.
- The Resection method has been used for centuries in both land and maritime navigation. Early explorers and surveyors relied on this technique to pinpoint their exact location using visible landmarks.

DO YOU KNOW?

Practical Uses of the resection method is commonly employed in:-

- Navigation during hiking, mountaineering, orienteering, and military operations.
- Situations where GPS devices are unavailable or unreliable

PRACTICE EXERCISES

4. **Basic Back Bearing Calculation**. Learn how to calculate a back bearing from a given compass bearing.

Instructions:-

(a) Choose a starting bearing, for example, 75°.

(b) Add 180° to this bearing: 75° + 180° = 255°.

(c) Check if the resulting bearing exceeds 360°. If it does, subtract 360° (e.g., 400° would be reduced to 40°).

(d) The back bearing for 75° is 255°.

Challenge. Try different initial bearings like 120°, 225°, or 310° and calculate the corresponding back bearings.

5. **Using a Compass in the Field**. Practice using a compass to find a back bearing in the field.

Instructions:-

(a) Go outdoors with a map and a compass.

(b) Pick a landmark that you can identify clearly (e.g., a mountain peak or a distant building).

(c) Take a bearing to the landmark using your compass.

(d) Calculate the back bearing by adding or subtracting 180°.

(e) Use the back bearing to walk in the opposite direction and see if you can return to your starting point.

Challenge. Try this with multiple landmarks and ensure you can accurately find your way back.

6. **Landmark Identification Using Back Bearings**. Identify your location based on back bearings from known landmarks.

Instructions:-

(a) Stand at an unknown location and choose two landmarks that you can see.

(b) Take a bearing to each landmark.

(c)Reverse the bearings (find their back bearings).

(d) Using a map, plot the back bearings and identify your approximate location where the lines intersect.

Challenge. Increase difficulty by adding more landmarks or practicing in areas with more complex terrain.

7. **Map and Compass Challenge**. Use a map and compass to practice back bearings and navigate.

Instructions:-

(a) On a map, mark a starting point and a destination.

(b) Take the bearing from your starting point to your destination and then calculate the back bearing.

(c) Walk along the route to the destination.

(d) At the destination, use the back bearing to return to your starting point.

Challenge. Repeat the exercise with different locations and bearings, ensuring accuracy in both directions.

8. **Basic Resection method with known landmarks**. Learn to determine your location (Own Position) by taking bearings from two known landmarks.

Instructions:-

(a) Stand at an unknown location and identify two or more visible landmarks whose positions are known (e.g., mountain peaks, towers, or buildings).

(b) Use a compass to take a bearing of the first landmark.

(c) Record the bearing and repeat the process for the all chosen landmarks.

(d) Calculate the Back Bearings of all the chosen landmarks

(e) On a map, plot the back bearings from each landmark.

(f) The point where these lines intersect is your Own Position.

Challenge. Try this exercise with different combinations of landmarks and vary your position to see how the accuracy changes.

9. **Identifying Terrain Features for Inspection Mehthod**. Practice recognizing terrain features using a topographic map.

Steps.

(a) Select a topographic map of an area.

(b) Locate features like hills, valleys, ridges, saddles and depressions.

(c) Match these features with their contour patterns (e.g., concentric circles for hills, U- or V-shapes for valleys).

10. **Estimating Elevation**. Use contour lines to determine the elevation of specific points.

Steps:

(a) Identify a starting point and a target point on the map.

(b) Count the contour intervals between them.

(c) Add or subtract based on the direction of travel to estimate the elevation.

11. **Finding the Direction**. Determine bearings and directions using the map.

Steps:

(a) Orient the map using a compass.

(b) Select two points on the map.

(c) Draw a line connecting the points and measure the angle using a protractor or compass rose.

CONCLUSION

12. Back Bearing refers to the direction opposite to the forward bearing of a line or a course. It is used to determine the direction back to the starting point or reference line. Mastery of back bearings is essential for accurate and reliable navigation in both fieldwork and map reading. Resection is a map-reading technique used to determine one's exact position on a map by identifying known landmarks and measuring their bearings. The compass method is the most common to perform resection. Resection method is used in orienteering, military navigation, search and rescue operations and outdoor expeditions.

SUMMARY

- Back Bearing helps in retracing the path or determining the reverse direction to locate the starting point.
- It is calculated by adding or subtracting 180° from the forward bearing.
- Resection method with Compass allows the surveyor or hiker to pinpoint their location by triangulation.
- Resection method determines a person's position on a map by measuring bearings to known landmarks.
- The technique relies on visible landmarks and is especially useful when the exact position is unknown.
- Inspection Method is used to ensure the map and navigational tools are accurate and functioning properly during the journey.
- It prevents errors in navigation by allowing the user to confirm their route and position.
- Together, these methods aid in accurate navigation and position determination, especially in areas with little to no direct reference points.
- Each method has its advantages and specific uses depending on the navigation context and practical exercises are provided to help improve understanding and skills in using these techniques.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the back bearing of a direction if the forward bearing is 120°?

(a) 240° (b) 300° (c) 180° (d) 60°

Q2. A bearing of 270° corresponds to which back bearing?

(a) 90° (b) 180° (c) 360° (d) 270

Q3. A back bearing of 155° is given. What is the corresponding forward bearing?

(a) 35° (b) 175° (c) 335° (d) 315°

Q4. If a bearing of 180° is measured, what is the back bearing?

(a) 0° (b) 90° (c) 360° (d) 180°

Q5. What is the primary purpose of the resection method in map reading?

(a) To determine the direction of a map
(b) To locate the position of a point on a map
(c) To calculate distances between landmarks
(d) To find the contour of the terrain

Q6. In the resection method, which of the following tools is commonly used for determining the angle of the position?

(a) Compass (b) Protractor (c) Clinometer (d) Ruler

Q7. When performing resection, how many known points on the map are typically required for accurate positioning?

(a) One (b) Two (c) Three (d) Four

Q8. What is minimum angle between two reference points in resection method?

(a) 50° (b) 20° (c) 30° (d) 25°

Q9. The resection method is most effective with how many reference points?

(a) 02 (b) 01 (c) 03 (d) none of above

Q10. To use the Resection method effectively, you should:

(a) Align the map with the compass needle.
(b) Use a ruler to measure distances between landmarks.
(c) Identifying two objects as reference points on map and on ground.
(d) Measure angles to landmarks using a protractor.

Q11. What is the primary use of back bearings in navigation?

(a) To find the forward bearing
(b) To determine the opposite direction of a given bearing
(c) To identify landmarks
(d) To calculate distances between two points

Q12. Which of the following is a true statement about the relationship between forward bearing and back bearing?

(a) They are always the same direction
(b) The back bearing is 180° opposite to the forward bearing
(c) The forward bearing is always greater than the back bearing
(d) Back bearing is always calculated by adding 360°

Q13. How is the back bearing calculated when the forward bearing is less than 180°?

(a) Back bearing = Forward bearing - 180°
(b) Back bearing = Forward bearing + 180°
(c) Back bearing = Forward bearing x 2
(d) Back bearing = Forward bearing ÷ 2

Q14. In the resection method, which of the following steps is NOT involved?

(a) Taking bearings of landmarks from the current location
(b) Drawing lines of position based on the bearings
(c) Using a GPS device to track your position
(d) Identifying landmarks on a map

Q15. What is a major disadvantage of the inspection method in navigation?

(a) Requires a compass for precise directions
(b) It is time-consuming
(c) It is less accurate and dependent on visibility
(d) It requires multiple tools like a map, compass and protractor

Short Answer Questions

Q1. How do you calculate a back bearing If the forward bearing (FB) is less than 180°?

Q2. If your forward bearing is 135°, what is your back bearing?

Q3. If forward bearing to a mountain peak is 270°. What is the back bearing?

Q4. What are types of bearing?

Q5. What are advantages of inspection method?

Long Answer Questions

Q1. What is back bearing in map reading and explain with three examples?

Q2. What is the purpose of resection with compass method and explain the method to know own position by using this method.

Q3. What is inspection method and explain how it is carried out along with its purpose.

Q4. Describe Resection method of finding own position?

Q5. When to use inspection method?

MAP READING

CHAPTER MR-VIII - MAP TO GROUND AND GROUND TO MAP

"A good navigator can see the map in their mind, but the best navigators see the ground with their eyes and heart."



TEACHING INSTRUCTIONS

Period : 04 (160 Mins)
Type : Lecture/Practice
Year : 2nd (02) & 3rd (02) year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Map Sheets, Compass, Service protractor, Pointer, Charts, Black board & Chalk.

Time Plan

- **Introduction : 10 Mins**
- **Part I : 30 Mins**
- **Part II : 30 Mins**
- **Conclusion : 10 Mins**
- **Practice : 80 Mins**

INTRODUCTION

1. Map reading is a fundamental skill for navigation, used by hikers, travellers, soldiers, surveyors and outdoor enthusiasts. It involves interpreting the symbols, scale and features of a map to understand the terrain and identifying your current location or a feature in the physical environment and finding its corresponding representation on the map. This chapter focuses on the two key processes in map reading: **Map to Ground and Ground to Map**. These techniques form the backbone of effective navigation and situational awareness. Together, these skills empower individuals to interpret landscapes, orient themselves confidently and make informed decisions in both familiar and unfamiliar environments. Mastering these concepts is crucial for ensuring safety, accuracy and efficiency in navigation.

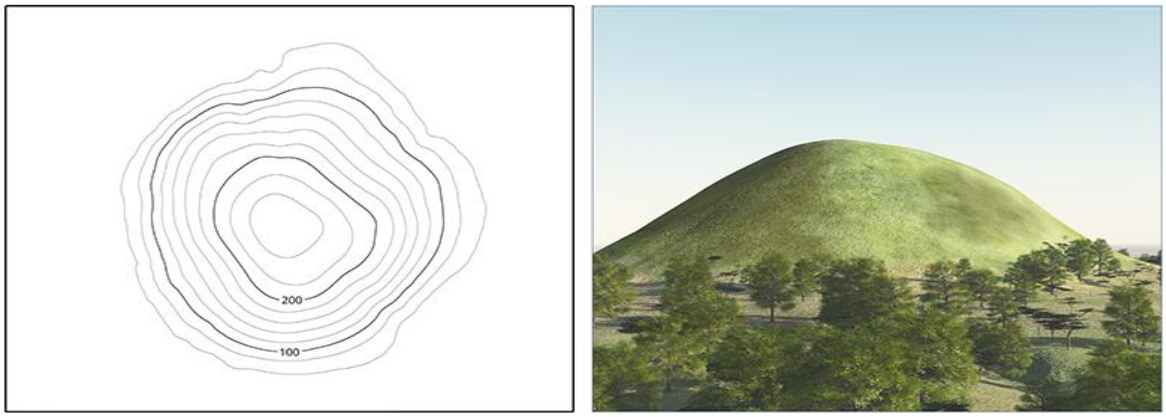
PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I: Map to ground** ➢ **Part II: Ground to map**	➢ **Learn to Identify features from ground to map** ➢ **Learn to identify objects from map to ground**

PART I: MAP TO GROUND

2. Map-to-ground is a skill to find out the details of features, landmarks, or terrain of the map on the ground in the real world. This process is essential for navigation, in outdoor activities such as hiking, orienteering and military operations. It helps in locating the exact position of objects on ground and on map. It also requires to judge distance. The essence of map reading training is to identify the objects and features on ground from a map and to identify the objects and features on the ground from the map.

3. Following methods are used to identify objects from map to ground:-

(a) **Bearing and Distance Method**. It is a fundamental technique in navigation used to determine or confirm a position on a map or in the field. This method involves taking a compass bearing from a known point to a visible landmark or feature and measuring the distance between the two points. The bearing provides the direction relative to the cardinal points, while the distance, often calculated using pacing, a measuring device, or map scales, allows for precise plotting on the map. By combining these two pieces of information, navigators can pinpoint their location, plan routes, or confirm their progress along a path. This method is particularly effective in areas with clear landmarks and is often supplemented with other navigation techniques for increased accuracy.



Map Contour seen on Ground

(b) **Direction and Distance Method**. In this method after setting of maps, draw a line on the map from own position to object to be identified on ground. Calculate distance and using any of the following methods find the direction of the object:-

(i) With the help of a sight rule find the ground direction of the object.

(ii) With the help of two points on the map estimate the ground direction.

(iii) Place a foot ruler /pencil at own position and align it with line of the map.

(iv) Place a pin each at the object and at the own position on the map. Align both pins and find general direction.

(c) **By Estimation Method**. The object is identified with the help of details around the object without measuring bearing, distance and direction by this method. The **estimation method in map-to-ground navigation** involves interpreting map information to predict or identify features in the terrain without needing precise tools. It's a useful technique for navigating efficiently when you need to relate features on a map to the real-world environment. Here are ways to apply estimation for **map-to-ground navigation**.



(i) **Map Orientation**. Use a compass to align the map with magnetic north, or match visible landmarks (e.g., a hill or lake) with their positions on the map to orient it. Once the map is oriented, estimate the direction of travel by visualizing a straight line between your position and the destination.

(ii) Using Linear Features. Identify roads, rivers, ridgelines, or trails on the map and locate them in the terrain. Use these as guides to stay oriented.



HOTS

- Plan and mark safe and easier route for hiking to the top of the hill from the base.

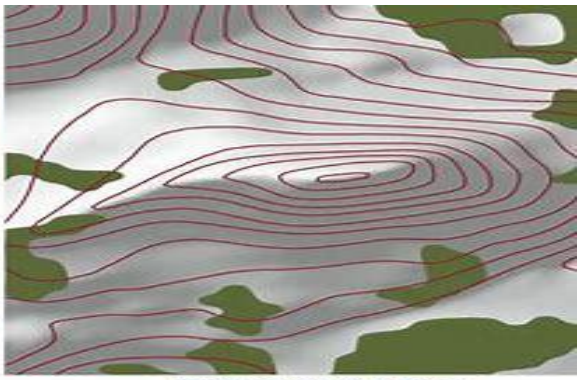
INTERESTING FACTS

- Linear features like rivers, trails, or roads can act as handrails to guide you while you navigate.
- Even without a compass, you can orient a map using the sun, stars, or simply by aligning with visible landmarks.
- When two paths or rivers cross, they are often marked distinctly on maps, making them reliable waypoints to confirm your position.

PART II: GROUND TO MAP

4. Ground-to-map navigation involves identifying your current location or features in the environment and then finding their corresponding positions on a map. This skill is essential for determining your position, planning your route, or verifying your progress in real-time. Ground-to-map complements map-to-ground navigation and ensures accurate situational awareness. To identify a ground object on the map is called ground to map. Methods to identify ground objects on map are:-


Hilltop on the Ground


Hilltop on the Map

(a) **Simple Method Using Bearing**. After Setting the map, find own position. Find out the magnetic bearing and the distance of the object. Convert magnetic bearing to grid bearing and plot grid bearing from own position by drawing a line. Mark the distance measured with service protractor on that line. The object on the map will be in proximity of the mark.



(b) **Intersection Method**. The intersection method is used to locate items that are farther away or in hilly terrain. In intersection method any object or place is identified by drawing line from two prominent feature of ground on map. Select minimum two prominent landmarks which can be easily identified on the ground. Read the bearing of object, we need to identify from two different places/ objects by compass. Convert the magnetic bearing in Grid bearing. Plot these two objects of ground on Map with help of GB and draw straight lines touching these two points. Find the object in the proximity of the intersection of the two lines.

(c) **By Direction Method**. After setting the map, find own position. Find the grid bearing of object on the map. Plot grid bearing on map and draw a line from own position in that direction. Mark the estimated distance of the object on the line. The object will be in the proximity of the marked point:

(i) Align own position with direction of object by placing a foot ruler /pencil at own position.
(ii) Place a pin at own position and second pin is placed in the direction of the object.
(iii) Mark the object on the map and find direction with the help of details around the object.
(iv) With the help of sight rule find exact direction of the object.

(d) **By Estimation Method**. An object can be located on map by knowing the magnetic bearing and distance of the object on ground. Observe features or distances on ground and estimate where they correspond on the map. This method is useful when trying to locate your position or confirm your route while moving in unfamiliar terrain. Here's a breakdown.

DID YOU KNOW?

- By aligning a map with visible landmarks (like rivers or mountains), you can navigate without tools.
- The sun’s position or shadows can also help determine directions in the absence of a compass

CONCLUSION

5. Map-to-ground and ground-to-map navigation are complementary techniques essential for successful navigation in outdoor and unfamiliar environments. Together, they enable individuals to effectively interpret maps, relate terrain features to map symbols and determine their position or path with confidence. Map-to-Ground navigation helps with planning routes, estimating distances and predicting terrain challenges using a map. This skill is critical for efficient movement and decision-making before and during travel. Ground-to-Map navigation is vital for confirming one's position, reorienting if lost and adapting plans based on the terrain. This technique ensures navigators can connect observed features in the environment back to the map. By mastering these skills, individuals can explore and operate in diverse environments, whether for recreational, professional, or emergency purposes, with reduced risk and greater efficiency.

SUMMARY

- Map-to-ground is an essential skill to find out the details of features, landmarks, or terrain of the map on the ground in the real world.
- Map to Ground is the essence of map reading training to enable the cadets to identify the objects and features on ground from a map and vice versa.
- Bearing and Distance Method, Direction and Distance Method and Estimation Method are used to identify objects from map to ground.
- Bearing and Distance Method is a fundamental technique in navigation used to determine or confirm a position on a map or in the field.
- Draw a line joining own position and object to be identified on the map in the Direction and Distance Method. Than measure distance between them and find the direction of the object.
- The estimation method involves interpreting map information to predict or identify features in the terrain without needing precise tools.
- To locate an object on the map that is indicated on the ground is called ground to map.
- Bearing, Intersection Method, Direction Method and Estimation Method are used to locate objects from ground to map.
- Ground-to-Map navigation is vital for confirming one's position, reorienting if lost and adapting plans based on the terrain.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. What does the term "orientation" refer to on a map?

(a) The size of the map
(b) The alignment of the map with respect to the cardinal directions
(c) The symbols used on the map
(d) The scale of the map

Q2. Which of the following is not a primary cardinal direction?

(a) North (b) South (c) East (d) Northwest

Q3. Which of the following is used to measure distances on a map?

(a) Compass (b) Scale ruler(c) Protractor (d) Legend

Q4. What is the term used to describe the vertical distance between contour lines on a map?

(a) Elevation (b) Relief (c) Gradient (d) Slope

Q5. On a map, the scale is given as 1 cm = 5 km. If a river is 3 cm long on the map, what is the actual length of the river?

(a) 5 km (b) 10 km (c) 15 km (d) 20 km

Q6. What does the contour interval on a topographic map represent?

(a) The horizontal distance between two points
(b) The vertical distance between contour lines
(c) The depth of rivers
(d) The elevation of the highest peak

Q7. What does "map-to-ground" navigation involve?

(a) Locating your position on a map based on ground features
(b) Matching map symbols to real-world landmarks
(c) Identifying the scale of a map
(d) Using a compass to determine the cardinal directions

Q8. Which of the following is the most reliable feature for finding your position on a map?

(a) Contour lines
(b) Temporary landmarks (e.g., parked cars)
(c) Permanent landmarks (e.g., mountains, rivers, buildings)
(d) Grid lines

Q9. What does "Alignment of the map" mean?

(a) Turning the map so that north on the map aligns with true north on the ground
(b) Marking your current position on the map
(c) Calculating the distance to the next checkpoint
(d) Using the map to find a hidden trail

Q10. What is the purpose of using a resection method in map reading?

(a) To draw a new map.
(b) To identify your location on a map by triangulating from ground features.
(c) To plot a straight-line route on a map.
(d) To measure distances between two points.

Q11. What is the primary purpose of the Map-to-Ground navigation technique?

(a) To create new maps from the ground
(b) To identify features and landmarks on the ground based on a map
(c) To estimate distances between locations
(d) To convert magnetic bearings into grid bearings

Q12. Which method involves using a compass bearing and distance to locate a position on a map or in the field

(a) Direction and Distance Method
(b) Bearing and Distance Method
(c) Estimation Method
(d) Intersection Method

Q13. What does the "Direction and Distance Method" help to identify?

(a) The direction of a moving object
(b) The general direction and distance of an object
(c) The position of the map
(d) The exact location of an object based on latitude and longitude

Q14. Which of the following is NOT part of the "Direction and Distance Method"?

(a) Using a foot ruler to align with a line on the map
(b) Using a sight rule to find the ground direction of an object
(c) Calculating the exact elevation of an object
(d) Plotting grid bearings from known positions

Q15. What does the "Estimation Method" involve in Map-to-Ground navigation?

(a) Using precise tools to measure distances
(b) Estimating features in the terrain based on map information
(c) Drawing straight lines between two objects
(d) Measuring magnetic bearings to locate features

Short Answers Type Question

Q1. Explain Direction Method in finding object from Ground to Map

Q2. Define Estimation Method in Ground to Map?

Q3. What is the key feature of the "Intersection Method" in Ground-to-Map navigation?

Q4. In the "Simple Method Using Bearing," how is the object on the map located?

Q5. What is Intersection Method used to locate objects from ground to map?

Long Answers Type Questions

Q1. What are the methods of finding object from map to ground Explain in detail?

Q2. Explain the Direction and Distance Method in finding object from map to ground?

Q3. Explain Bearing and Distance method in finding object from map to ground.

Q4. What methods are used for Map-to-Ground?

Q5. What methods are used for Ground-to-Map?

MAP READING

CHAPTER MR-IX - NAVIGATION BY COMPASS

“Knowledge is the compass that guides us through the unknown.”



TEACHING INSTRUCTIONS

Period : 03 (120 Mins)
Type : Lecture/Practice
Year : 1st (01),2nd (01) &3rd (01) Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Compass, Computer slides, Pointer, Charts, Blackboard & Chalk.

Time Plan

- **Introduction : 05 mins**
- **Part I : 20 mins**
- **Part II : 10 mins**
- **Conclusion : 05 mins**
- **Practice : 80 mins**

INTRODUCTION

1. Compass navigation is a vital skill for traversing uncharted terrain, whether for exploration, military operations, or outdoor adventures. It combines the use of a magnetic compass with map reading and precise calculations to maintain direction, particularly in environments with limited visibility or scarce landmarks, such as dense forests, deserts, or during night time. Effective compass navigation involves understanding key concepts like bearings, magnetic variation and compass errors. Navigators must calculate grid bearings from maps, adjust for local variations to obtain magnetic bearings and account for compass errors to ensure precision. By following a structured procedure—setting the compass, aligning the magnetic needle and using landmarks or celestial references—individuals can navigate reliably. Team roles such as pacer, guide and recorder further enhance accuracy by measuring distances, maintaining course and documenting progress, making compass navigation a cornerstone of successful field navigation.

PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I: Navigation by Compass** ➢ **Part II: Composition of navigation party**.	➢ **Learn to Master Compass Navigation** ➢ **To understand team roles like pacer, guide and recorder for accurate navigation**. ➢ **Learn to apply navigation skills to diverse terrains and conditions**.

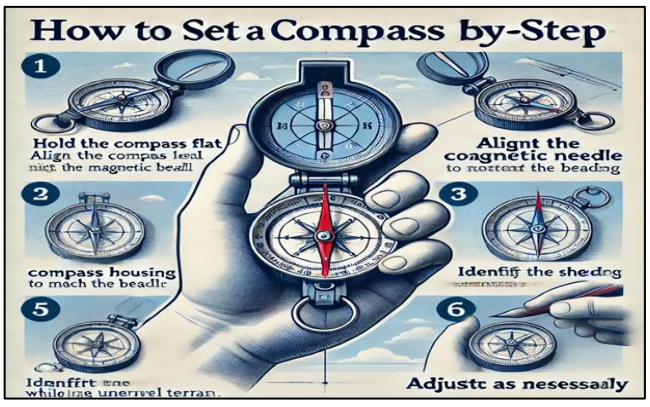
PART I: NAVIGATION BY COMPASS, CONCEPT OF PACER, GUIDE & RECORDER

2. **Navigation by Compass**. Navigation by compass is a critical skill in outdoor activities like trekking, military operations and orienteering. It involves determining and following a specified direction using a magnetic compass. This skill is particularly useful for maintaining a course over unfamiliar terrain during the day or night.

3. **Setting of Compass on Given Bearing**. Steps for setting bearing on a compass are as follows:-

(i) **Step 1**. Set the bearings of landmark on the compass to march.

(ii) **Step 2**. Place the compass on the palm of the left hand in such a way that the tongue notch is on the middle of the finger and the ring notch is straight to the wrist.



(iii) **Step 3**. Keep the left hand in the middle line of the belt in front of the body, then move it to the right or left accordingly so that the direction mark and the arrow mark come in a straight line.

(iv) **Step 4**. Now, by marching straight to the tongue notch, you will march on the bearings of the that landmark.

(v)**Step 5**. Regularly check the compass to stay on course, especially if terrain or obstacles require deviations.

Calculation of Grid Bearing

Grid bearing calculation is foundational for accurate navigation.

Example Calculation:

- Grid Bearing: 135°
- Local Variation: +5° (Magnetic north is east of grid north)
- Compass Error: -2°

Magnetic Bearing = 135° + 5° = 140°
Adjusted Magnetic Bearing = 140° - 2° = 138°

Set your compass to 138° and proceed.

4 **Procedure of Marching on a Compass Bearing**. Steps for marching on compass bearing during day and night are given below:-

(a) **During the day**. Steps for marching during day are as follows:-

(i) **Step 1**. **Determine the Destination** Identify the endpoint of your route on the map.

(ii)**Step 2**. **Calculate the Grid Bearing** Measure the angle between the grid north (vertical lines on the map) and the line pointing towards your destination using a protractor.

(iii) **Step 3**. **Convert Grid Bearing to Magnetic Bearing** Account for the local variation (difference between grid north and magnetic north, provided on the map legend).

(b) **Formula**. Magnetic Bearing=Grid Bearing ±Variation Add the variation if magnetic north is east of grid north. Subtract if it is west.

(i) **Step 4**. **Adjust for Compass Error** Compasses may have a built-in error or declination. Incorporate this into the magnetic bearing.

(ii) **Formula**. Adjusted Magnetic Bearing=Magnetic Bearing ±Compass Error

(iii) **Step 5**. **Set the Compass** Rotate the compass housing until the desired bearing aligns with the direction of travel arrow. Turn yourself until the magnetic needle aligns with the orienting arrow inside the compass housing.

(iv) **Step 6**. **Follow the Bearing** Identify a fixed point (e.g., a tree or rock) in your line of travel and walk towards it. Repeat as needed.



(v) **Step 7**. **Monitor Pacing** Use the concept of pacer (explained below) to measure distance accurately.

(c) **During the Night**. Follow the same steps as above but with modifications:-

(i) Use visible objects like stars, a flashlight beam, or reflective markers as reference points. Maintain frequent checks on your bearing to avoid deviation. Use a guide and recorder to support navigation (detailed below).

PART II: COMPOSITION OF NAVIGATION PARTY

5. **Composition of Navigation Party**. In any MR team there is a dedicated Pacer, Guide and Recorder. These roles are crucial for organized and accurate navigation, especially in a group.



(a) **Pacer**. Tracks the distance covered by counting steps. Knows the individual's average pace length (distance covered in one step). Calculates distance using the formula:

Distance Covered = Number of Steps × Pace Length

(b) **Guide**. Leads the group by following the compass bearing. Identifies intermediate landmarks and ensures the group remains on course.

(c) **Recorder**. Keeps a log of the bearings, distances and landmarks for reference. Acts as a backup for navigation data in case of errors.

6. **Tips for Effective Compass Navigation**. Some tips for effective compass navigation are given below:-

(a) **Practice Pacing**. Regularly measure and note your average pace length for accuracy in distance measurement.

(b) **Understand Local Variation**. Update your variation knowledge from recent maps or local sources.

(c) **Work as a Team**. Assign roles like pacer, guide and recorder to enhance efficiency and reduce errors.

(d) **Night Navigation**. Use luminous markers or reflective materials for better visibility and coordination.

HOT

- Analyse and evaluate the importance of accounting for local variation and compass error when navigating in unfamiliar terrain. How can errors in these calculations impact the overall accuracy of navigation?
- Given a scenario where a compass bearing is 90°, the local variation is -3° (magnetic north is west of grid north), and the compass error is 4° WEST, calculate the adjusted magnetic bearing. How would you ensure the team remains on course while navigating this bearing in a forested area?
- Design a strategy for a group trek at night, incorporating roles such as pacer, guide, and recorder, while considering challenges like reduced visibility and potential deviations from the planned route. How would teamwork enhance navigation efficiency in such conditions?

CONCLUSION

7. Navigation by compass is an essential skill that combines precision, planning and adaptability. By understanding the principles of bearings, accounting for variations and compass errors and mastering practical techniques for day and night navigation, individuals can confidently traverse diverse terrains. The roles of pacer, guide and recorder further enhance the accuracy and efficiency of navigation, fostering teamwork and reliability. Whether in challenging environments or routine expeditions, the ability to navigate effectively ensures not only safety but also a deeper connection with the art of exploration. With practice and knowledge, compass navigation remains a timeless and invaluable tool for adventurers and professionals alike.

SUMMARY

- Bearings and types of bearings, conversion of bearings
- Compass and Map Integration: Combine a magnetic compass with a map to determine and maintain a specific course, especially in challenging terrains.
- Grid to Magnetic Bearings: Calculate grid bearings from the map and convert them to magnetic bearings by accounting for local variation and compass errors.
- Precise Alignment: Set the compass to the calculated bearing and align the magnetic needle for accurate navigation.
- Night time Navigation: Use celestial objects, reflective markers, or artificial light to maintain direction in low-visibility conditions.
- Team Roles: Assign roles like pacer (measures distance), guide (follows bearing) and recorder (logs journey details) to enhance accuracy.
- Distance Measurement: Employ pacing to calculate distance by multiplying step counts with individual pace length.
- Adaptations for Conditions: Adjust navigation techniques for rugged terrain, low visibility, or dense vegetation.
- Landmark Guidance: Use visible landmarks to stay on course during daytime navigation.
- Practical and Continuous Practice: Master skills through consistent practice and real-world application to build confidence.
- Universal Utility: Apply compass navigation skills across diverse scenarios, benefiting adventurers, military personnel and outdoor enthusiasts.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary purpose of compass navigation?

(a) To find water sources in uncharted terrain
(b) To determine and follow a specific direction accurately
(c) To estimate the distance of travel in rough terrain
(d) To avoid the use of maps during navigation

Q2. What is the formula to calculate magnetic bearing?

(a) Magnetic Bearing = Grid Bearing × Variation
(b) Magnetic Bearing = Grid Bearing ± Variation
(c) Magnetic Bearing = Compass Error + Variation
(d) Magnetic Bearing = Grid Bearing ÷ Variation

Q3. What is the role of the pacer in compass navigation?

(a) Leading the group by following the compass bearing
(b) Measuring the distance travelled using step counts
(c) Recording bearings, distances and landmarks
(d) Identifying landmarks along the route

Q4. Which of the following steps is part of setting a compass?

(a) Using a protractor to calculate the bearing
(b) Adjusting for compass error on the map legend
(c) Aligning the magnetic needle with the orienting arrow
(d) Measuring the angular distance between cardinal directions

Q5. If the grid bearing is 120°, local variation is +3° and compass error is -2°, what is the adjusted magnetic bearing?

(a) 121° (b) 122° (c) 123° (d) 125°

Q6. Which of the following is NOT a step-in marching on a compass bearing?

(a) Determine the destination
(b) Adjust the compass to true north
(c) Identify and follow landmarks
(d) Monitor pacing regularly

Q7. What does a compass error of 2° EAST mean?

(a) The compass points 2° west of magnetic north
(b) The compass points 2° east of magnetic north
(c) The compass points 2° east of true north
(d) The compass is misaligned by 2° from grid north

Q8. What is the angular distance between cardinal and intermediate directions?

(a) 22.5° (b) 30° (c) 45° (d) 60°

Q9. Which role ensures documentation of bearings, distances and landmarks during navigation?

(a) Pacer (b) Guide (c) Recorder (d) Navigator

Q10. What is the primary adjustment made for night navigation?

(a) Avoid using reflective markers
(b) Use visible objects like stars or a flashlight as reference points
(c) Measure bearings using grid north instead of magnetic north
(d) Eliminate the need for compass error adjustment

Q11. Which of the following is a type of bearing based on North?

(a) Forward Bearing

(b) Backward Bearing

(c) True Bearing

(d) Object Bearing

Q12. What is a Grid Bearing?

(a) The angle between two objects measured from Magnetic North
(b) The angle between two objects measured from True North
(c) The angle between two objects measured from Grid North
(d) The angle between two objects measured from any random direction

Q13. Which of the following is true about Magnetic Bearings?

(a) Magnetic Bearings are based on True North
(b) Magnetic Bearings are based on Grid North
(c) Magnetic Bearings are based on Magnetic North
(d) Magnetic Bearings do not require any reference direction

Q14. What is the angle of convergence in bearing conversion?

(a) The difference between magnetic north and grid north
(b) The difference between true north and magnetic north
(c) The difference between true north and grid north
(d) The angular difference between two objects on the map

Q15. When converting a Grid Bearing to a True Bearing, if the angle of convergence is 2° east, what is the True Bearing?

(a) True Bearing is 2° less than Grid Bearing
(b) True Bearing is 2° more than Grid Bearing
(c) True Bearing remains the same as Grid Bearing
(d) True Bearing cannot be determined with this information

Short Answer Types Questions

Q1. What is the difference between grid north and magnetic north?

Q2. What is the role of a pacer in compass navigation?

Q3. Why is it important to adjust for compass error?

Q4. How is a Magnetic Bearing converted into a Grid Bearing when the magnetic north is west of grid north with a local variance of 1° west?

Q5. What should you do if obstacles cause deviations from the course when marching on a compass bearing?

Long Answer Types Questions

Q1. Explain the process of converting a grid bearing to a magnetic bearing, including how to account for local variation and compass error?

Q2. Describe the roles of pacer, guide and recorder in a navigation team and how they contribute to successful navigation?

Q3. How do navigation techniques differ during nighttime compared to daytime and what are the key challenges in night time navigation?

Q4. When converting a Grid Bearing to a True Bearing, if the angle of convergence is 2° east, what is the True Bearing?

Q5. Describe bearing and its types.

MAP READING

CHAPTER MR-X: PRACTICE OF MAP READING

"Maps are the mirrors of the world and map reading is the art of seeing through them to find your way"



TEACHING INSTRUCTIONS

Period : 04 (160 Mins)
Type : Practice.
Year : 1st (01), 2nd (01) & 3rd (02) Year SD/SW
Conducting Officer : Permanent Instructor.

Training Aids: Map with case, Compass, Writing material, Service Protractor and Pointer Staff.

Time Plan

- **General Guidelines of Doing Map Reading : 20 mins**
- **Practice on ground : 140 mins**

GENERAL GUIDELINES FOR DOING MAP READING

1. Map reading is a vital skill that allows individuals to navigate and interpret geographic locations, plan routes and understand the landscape. Whether you're hiking, traveling, or navigating urban environments, map reading provides a way to make sense of the terrain and find your way from one place to another.

2. **General Guidelines for Reading a Map**. Some general guide lines are for map reading are given below:-

(a) **Understand Map Features**.Familiarize with key map features like conventional symbols, scale (distance) and contour lines (elevation).

(b) **Setting of the Map**. Always orient the map so that the top corresponds to the true north or magnetic north. Use the compass for accurate orientation.

(c)**Identify Key Landmarks**. Look for prominent features such as rivers, roads, or peaks to identify your location or your route.

(d) **Use the Scale**. To measure distance, use the map scale and convert it to real-world distance (e.g., 1 cm on the map = 1 km in reality).

(e) **Use of Magnetic Compass**. Keep the compass level in hand to allow the magnetic needle to move freely. Point the compass in the direction object and then read the bearing from the compass housing. Bearings are measured in degrees from 0° to 360°, with north being 0°/360°, east 90°, south 180° and west 270°.

(f) **Follow the Bearing**. Keep the compass steady, aligning the magnetic needle with the orienting arrow as move towards the destination.

(g) **Reading Grid Bearing with Service Protractor**. for reading grid bearing between two objects, mark points with chinagraph pencil or by sketch on these objects. Draw a line joining these points and put the service protractor so the zero edge is over one of the point. Use the protractor to read the degree of the bearing.

(h) **Adjust for Declination (if necessary)**. If converting to magnetic bearings, adjust for the Grid convergence by adding or subtracting the declination value (found on the map).

(i) **For finding own position**. Find two or more landmarks that are visible and can be identified on the map.

(j) **Take Bearings to Each Landmark**. With your compass, take bearings from your location to each of the landmarks and calculate back bearing.

(k)**Plot Bearings**. Mark the back bearings on the map after converting to Grid bearing.

(l) **Determine Position**. The point where the lines of bearing intersect represents your location.

(m) **Map to ground**. Read the given Grid Reference (GR) on the map and determine its grid bearing relative to own position. Convert this grid bearing (GB) into a magnetic bearing (MB), then locate the corresponding object on the map in the direction of the magnetic bearing.

(n) **Ground to map**. Take MB of given object shown on the ground. Convert it into GB and plot on map from own position. Identify the object on line of this GB.

3. These steps will help you effectively navigate using both a magnetic compass and a service protractor for map reading and bearing measurements.

4. **Suggested charts for these exercises are as under**

(a) **Exercise chart for Own Position**

Ser No	Landmark name (Grid Reference)	Distance	Magnetic Bearing (MB)	Grid Bearing (GB)	Grid Back Bearing (GBB)	Own Position (GR)

(a) **Exercise chart for Map to Ground**

Ser No	Grid Reference (GR)	Grid Bearing (GB)	Magnetic Bearing (MB)	Distance	Name of landmark

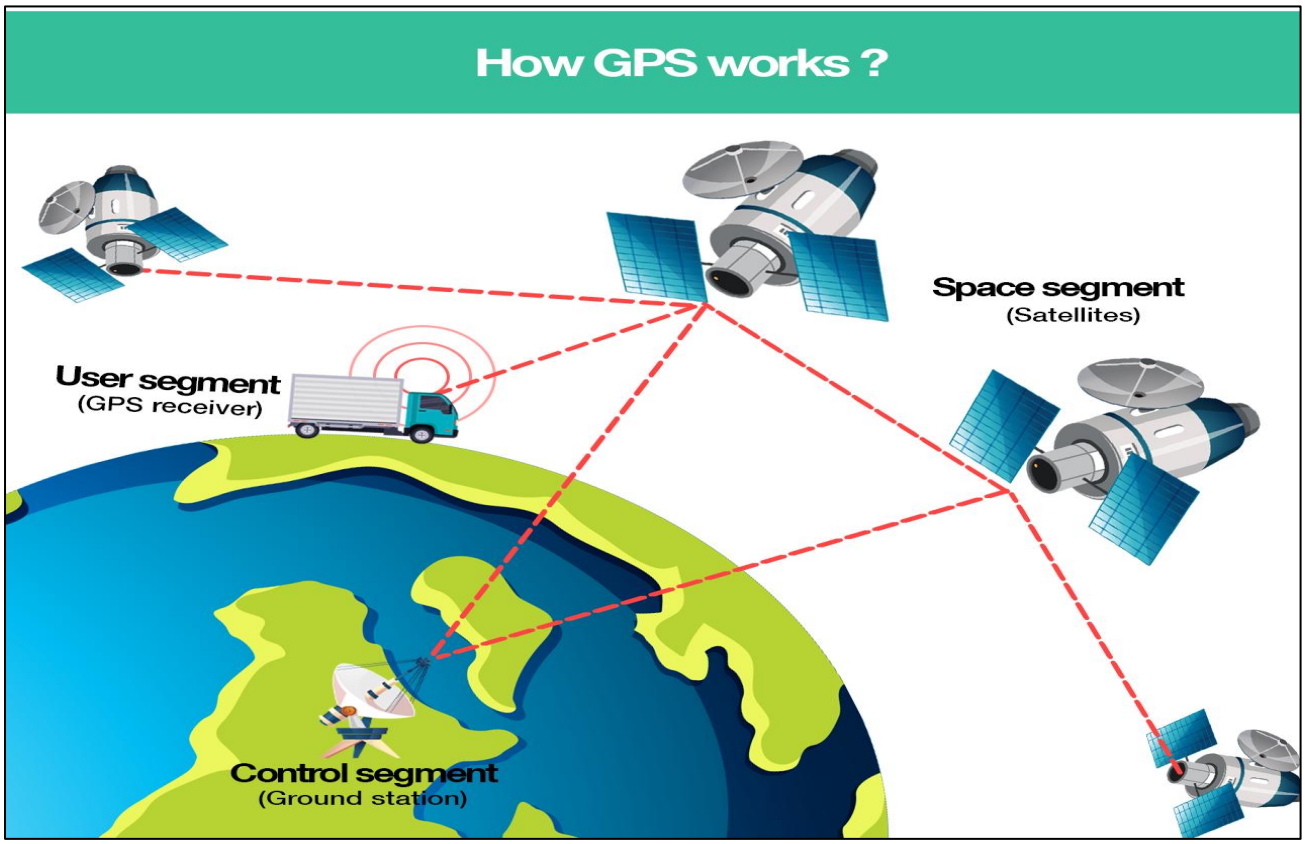
(c) **Exercise chart for Ground to Map**

Ser No	Landmark	Distance	Magnetic Bearing (MB)	Grid Bearing (GB)	GR

MAP READING

CHAPTER XI: GLOBAL POSITIONING SYSTEM

"Modern navigation isn't just finding a way—it's finding the best way."



TEACHING INSTRUCTIONS

Periods : 04 (160 Mins)
Types : Lecture & Practice
Year : 1st (01),2nd(02) & 3rd (01) Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: OHP, Board Chalks & Marker

Time Plan:

- **Introduction : 05 Mins**
- **Part I : 30 Mins**
- **Part II : 25 Mins**
- **Part III : 25 Mins**
- **Part IV : 30 Mins**
- **Conclusion : 05 Mins**
- **Practice : 40 Mins**

INTRODUCTION

1. Global Positioning System (GPS) refers to the satellite-based navigation system by which the precise location of people and devices on the earth is pinpointed. Therefore Navigation, now has become very facile with the advent of GPS in the Armed forces as well as civil life worldwide. Global Positioning System was originally developed by the U.S. Department of Defence (DoD) in the 1970s for military purposes Modern technology has made the GPS receivers small portable, very handy, accurate and are also equipped in many vehicles which helps fishermen, drivers and hikers for navigation.

PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I:Global Positioning System** ➢ **Part II: Important features of GPS** ➢ **Part III:Map Reading and Navigation by GPS** ➢ **Part IV: Salient Aspects of Navigation by Compass**	➢ **Understanding about GPS** ➢ **To comprehend its functions and various features** ➢ **To understand Map Reading and Navigation with GPS** ➢ **To Learn about latest trends in GPS**

PART I: GLOBAL POSITIONING SYSTEM



2. Global Positioning System was originally developed by the U.S. Department of Defence (DoD) in the 1970s for military purposes, GPS has become essential for civilian applications worldwide.

3. **Satellite System Architecture**. The GPS architecture is divided into three major segments:-

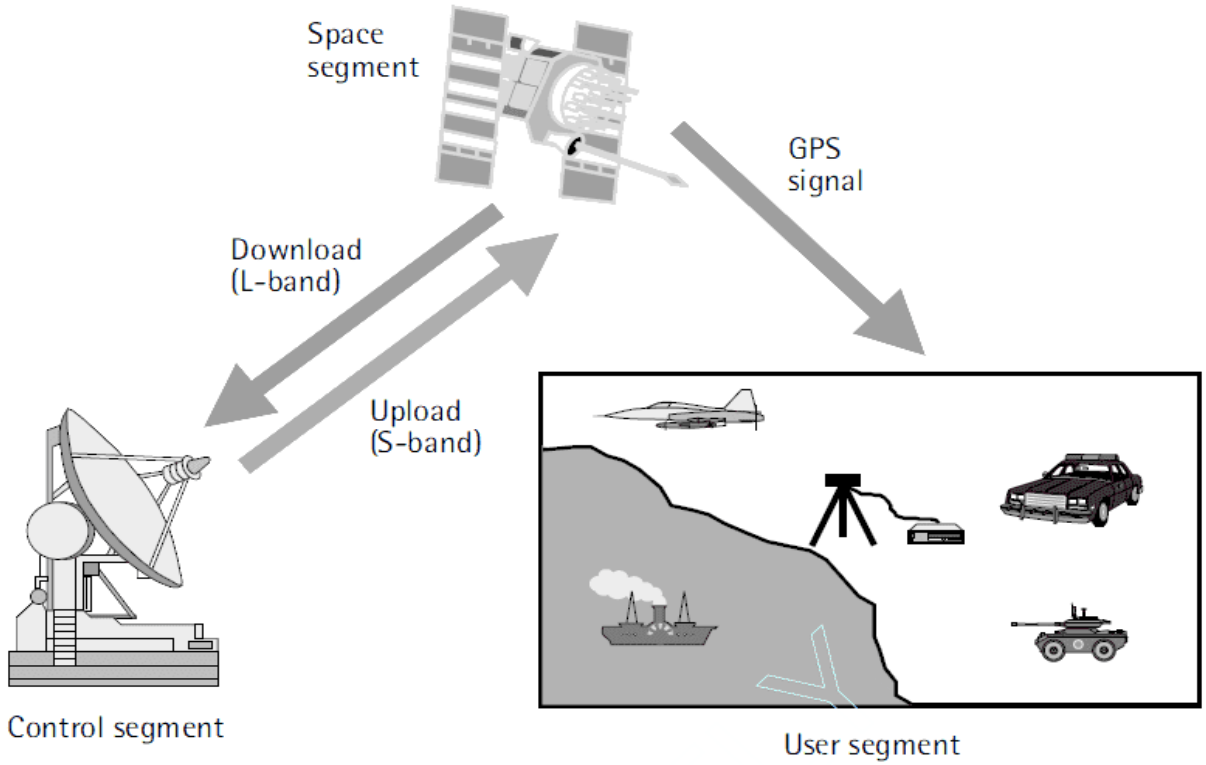
(a) **Space Segment**. Comprises a constellation of 24 to 32 satellites orbiting the Earth at an altitude of 20,200 km in six orbital planes. Each satellite completes an orbit in approximately 12 hours and is equipped with atomic clocks for accurate timing. Satellites continuously broadcast navigation signals containing Ephemeris data (precise satellite position) and Almanac data (general health and location of all satellites).

(b) **Control Segment**. A network of ground control stations responsible for monitoring, managing and updating satellites. It comprises of Master Control Station (MCS) that processes satellite data and ensures operational integrity, Monitor Stations to track satellite signals and relay data to the MCS and Ground Antennas to send updates (eg, corrections) to the satellites.

(c) **User Segment**. Includes any device with a GPS receiver, such as Smartphones, car navigation systems and drones (civilian use). Military vehicles, aircraft and personnel tracking systems. GPS receivers calculate position by triangulating signals from at least **four satellites** using the time delay of signals.

DID YOU KNOW?

- **Master Control Station**. Falcon Air Force base in Colorado, Colorado spring, America
- **Monitor Station**. Falcon Airforce base in Colorado, Cape Canaveral in Florida, Hawaii, Ascension island in Atlantic ocean - Diego Garcia island Indian ocean – Kwajalein island in Southern pacific ocean



Satellite System Architecture

4. **Usage of GPS**. GPS today has become indispensable in today's scenario and has diverse applications across industries, which include the following:-

(a) **Navigation**. GPS is primarily used for road, air and sea navigation for individuals and commercial transport. It is also used for guidance for autonomous vehicles.

(b) **Surveying and Mapping**. GPS is utilized for land surveying. Geographic Information Systems (GIS) for urban planning and resource management.

(c)**Agriculture**. GPS helps farmers with precision farming, allowing them to manage their land efficiently and optimize crop yields.

(d) **Military**. GPS helps in troop movement, target tracking, missile guidance and movement coordination.

(e) **Emergency and Disaster Management**. Locating survivors and coordinating relief efforts during natural disasters.

(f) **Geotagging**. GPS is used to add location data to photos, social media posts and videos, so people know where they were taken.

INTERESTING FACTS

- There are total 6 orbital planes with 4 space vehicles in each, that has equal space of 60° apart and inclined about 55° to the equatorial plane.
- Atomic clocks loose only one second in 3,00,000 years
- GPS is free worldwide however cost of maintaining the system is over $ 1 billion annually

PART II: IMPORTANT FEATURES OF GPS

5. **Key features of GPS**. Global Positioning System (GPS) Offers several critical features that make it indispensable for navigation, mapping and precise location determination. Below are the key features, including the specific ability to provide continuous own position and grid reference (GR) availability up to 10 figures:-



(a) **Global Coverage**. GPS works worldwide, providing location data regardless of where you are on Earth, whether in urban areas, remote regions, or at sea.

(b) **Real-Time Location**. GPS provides instant, real-time location information, helping users navigate and track their position continuously.

(c) **High Accuracy**. GPS offers high accuracy, typically within 5–10 meters, depending on conditions, allowing precise navigation and location tracking.

(d) **Grid Reference (GR) Availability (High-Precision Coordinates)**. GPS provides location data in various coordinate systems, including latitude/longitude and military-style Grid References (GR). A 10-figure GR provides accuracy up to 1 meter, ideal for military and precision tasks (e.g. 12345 67890). Civilian applications often rely on 6 to 8-figure GR, sufficient for general navigation and mapping.

(e) **Multi-Satellite Coverage**. At least four satellites are required to determine a 3D position (latitude, longitude and altitude) with precise time synchronization. Redundancy: With 24-32 satellites in orbit, the system ensures multiple satellites are visible at any location at any time, improving reliability.

(f) **Time Synchronization**. GPS provides highly accurate time data, synchronized to within nanoseconds, critical for financial systems, telecommunications and military operations.

(g) **Speed and Direction**. GPS can calculate speed which is useful for vehicles, aircraft and runners. It also indicates the direction of travel, crucial for navigation systems.

(h) **Dual-Use (Civilian and Military)**. Free access to basic GPS services for individuals and industries worldwide. Encrypted, higher-precision capabilities reserved for authorized military users, including anti-jamming and anti-spoofing protections.

HOT

- The local government is planning to develop a new community park in your city. Using Google Maps, students are tasked with identifying a suitable location, planning the layout, and evaluating the potential impact of the park on the community.

PART III: MAP READING AND NAVIGATION BY GPS

6. **Map Reading and Navigation by GPS**. The GPS method of map reading allows real-time tracking of your location on digital maps, showing nearby landmarks and routes. It eliminates the need for manual interpretation, offering turn-by-turn directions and automatically adjusting routes if you stray off course. This technology simplifies navigation by providing accurate, updated information instantly. Here's how working with GPS aids in navigation, including obstacle avoidance and route optimization:-

(a) **Turn on the Device**. Ensure the GPS is powered on and has a clear view of the sky to receive satellite signals.

(b) **Set Destination**. Input your destination or select it from pre-loaded maps.

(c) **Follow Directions**. The GPS provides step-by-step directions for reaching your destination, which may include turn-by-turn navigation.

(d) **Monitor Position**. GPS updates your position in real-time, showing your movement and recalculating routes if necessary.

DO YOU KNOW?

- GPS devices are used for a wide range of applications, including navigation, mapping, surveying, and tracking

7. **Working with a GPS**. Working with GPS is explained as under:-



(a) **Route Distance and Time Planning**. GPS offers pre-calculated routes to destinations based on various parameters, such as shortest distance, fastest time and preferred terrain (e.g. avoiding steep gradients or rough roads). Digital maps and navigation software uses topographic data to suggest routes with manageable slopes or gradients, making travel easier. Time Planning to arrive at the destination (ETA) is based on the speed, route and traffic data for vehicles, apps like Google Maps or Waze use live traffic information to refine time estimates.

(b) **Waypoints**. A **waypoint** is a specific location or coordinate that you mark or save for navigation. Waypoints are essential for multi-stage navigation, hiking, or exploring unfamiliar areas. A waypoint can represent landmarks, checkpoints, or points of interest. Waypoints can be marked using latitude/longitude, or Grid References.





(c)**Feeding/Entering Waypoints**. Feeding waypoints into a Garmin GPS device is a fundamental skill for navigation, especially in field exercises. Here's a step-by-step guide to inputting waypoints into most Garmin GPS devices:-

(i) Turn on Garmin GPS unit by pressing the **Power button** (usually located on the side or top).

(ii) On most Garmin devices, press the "Mark" or "Menu" button (depending on your model).

(iii)Select "Waypoints" or "Create New Waypoint" from the available options.

(iv) To name a waypoint usually a short code or reference, like "WP1" or "Check Point A." Some models allow for more detailed names, while others limit characters.

(v) After naming the waypoint, the next screen will allow to enter the latitude and longitude of the waypoint.

(vi) Garmin typically uses degrees, minutes and seconds (DMS) or decimal degrees (DD). Enter the coordinates accurately, following the format required by GPS device.

(vii) Coordinates can be entered manually or use the device's map to move a cursor to the marked location. Once the coordinates are entered and any other necessary information, press "Save" or "Enter" to save the waypoint to your GPS.

- To input multiple waypoints, repeat the process for each one. Many Garmin GPS devices allow to store hundreds of waypoints.
- To navigate to a waypoint, go to the **"Waypoints"** list, select one and then choose **"Navigate"**. This will provide directions on how to reach the waypoint.

(d) **Navigating Between Waypoints**. Point-to-point navigation follows a straight or optimal route between two waypoints. Waypoint Sequencing helps plan multi-stage routes by arranging waypoints in order. Visual and auditory cues guide users to each waypoint, displaying the distance, direction and estimated time to reach it.

(e) **Track Back Features**. The Track Back feature (also called Backtrack) helps users retrace their path to the starting point, essential in scenarios like getting lost or returning along the same route. Track Back feature works as GPS continuously records your trail as you move, storing it as a track log. While retracing steps the Track Back feature uses this recorded track log to guide you back along the same path displaying turn-by-turn directions, showing the exact route, you took.

8. **Uses of GPS in the Army**. The Global Positioning System plays a critical role in modern military operations, offering real-time positioning, navigation and timing data that enhance the precision, efficiency and effectiveness of various tasks. Here's how GPS is specifically used in the Army, with a focus on its application in artillery fire direction:-

(a) **Precise Targeting**. GPS allows artillery units to calculate exact target coordinates, improving the accuracy of their fire. Forward observers with GPS devices can relay precise positions of enemy targets, minimizing collateral damage.

(b) **Fire Control Systems**. Modern artillery systems use GPS to automate calculations of azimuth, elevation and range to target. This reduces errors from manual calculations and environmental factors, ensuring accurate targeting.

(c) **Movement and Positioning of Artillery**. GPS helps artillery units position themselves accurately, enabling quick setup and precise fire. Mobile artillery can reposition without losing targeting accuracy, offering enhanced operational flexibility.

(d) **Navigation and Movement**. GPS assists soldiers in navigating unfamiliar terrain, reducing reliance on traditional methods. It also helps military vehicles follow optimized routes, avoiding hazards and improving convoy coordination.

(e) **Reconnaissance and Surveillance**. GPS provides accurate geolocation data for reconnaissance, helping map enemy positions. It coordinates drones and reconnaissance teams, ensuring effective surveillance and targeting.

(f) **Logistics and Supply Chain Management**. GPS tracks military supplies, ammunition and personnel, reducing delays and preventing losses. It enables automated monitoring of supply convoys, ensuring efficient and timely delivery.

PART IV: MODERNS TRENDS IN GPS

9. **Modern Trends in GPS Technology**. GPS devices are now integrated with other GPS technology has expanded beyond navigation and military uses, integrating with other technologies for fitness, weather tracking, communication and more.



(a) **GPS and Radio Set Combos**. GPS-radio combos enable simultaneous navigation and communication, improving troop coordination and reducing friendly fire risks.

(b) **Smartphone Compatibility**. Modern GPS devices sync with smartphones, enhancing apps like Google Maps and Strava for real-time navigation, fitness tracking and route planning.

(c)**Integration with Active Weather Forecasting**. Many GPS devices now include real-time weather updates and barometric altimeters, providing essential weather data for outdoor activities and military operations.

(d) **Fitness and Health Monitoring**. GPS devices paired with fitness trackers monitor activity, track calories burned and provide heart rate data for more accurate health and workout insights.



(e) **Calorie-Burnt and Heart-Rate Monitors**.GPS technology in fitness trackers tracks movement, speed and elevation to monitor calorie burn and heart rate. Many devices also include optical sensors or chest strap compatibility for accurate heart rate data, providing insights into workout intensity and recovery.

10. **Advantages of GPS**.The advantages of GPS are given below:-

(a) **High Accuracy**. Provides precise location information, often within 5-10 meters.

(b) **Global Coverage**. Works worldwide, even in remote or difficult terrains.

(c) **Ease of Use**. Simple to operate, with automatic route calculation.

(d) **Real-Time Navigation**. Provides real-time directions and distance calculations.

(e) **Time Synchronization**. Offers accurate time for various industries.

(f) **Humanitarian Assistance and Disaster Relief (HADR)**. Enhances safety by aiding in rescue operations and monitoring.

11. **Disadvantages of GPS**. The disadvantages are given below:-

(a) **Signal Dependency**. Accuracy can be affected by weather, obstructions, or signal loss in certain areas.

(b) **Vulnerable to Jamming**. GPS signals can be jammed or hacked, disrupting functionality.

(c)**Privacy Concerns**. GPS can be used for tracking, raising potential privacy issues.

(d) **Over-Reliance on Technology**. Excessive use can lead to a decline in traditional navigation skills.

(e) **Cost of High-End Devices**. Specialized GPS units can be expensive.

CONCLUSION

12. The Global Positioning System (GPS) has evolved from a military tool to a crucial technology used in various industries like navigation, agriculture and emergency response. With its global coverage and high accuracy, GPS supports precise tasks such as surveying, military operations and logistics. Modern advancements, including integration with smartphones, wearables and weather forecasting, continue to expand its functionality, making it indispensable in daily life and specialized fields. These developments ensure GPS remains a vital tool in shaping the future of navigation and connectivity.

INTERESTING FACTS

- GPS was originally developed by the U.S. Department of Defence for military navigation. It became available for civilian use in the 1980s.
- GPS works anywhere in the world, 24/7, and in all weather conditions, thanks to a constellation of at least 24 satellites orbiting Earth.
- GPS relies on extremely accurate atomic clocks on each satellite, which provide precise timing down to a billionth of a second. This precision is crucial for accurate positioning.
- GPS technology helps monitor tectonic movements and can detect shifts in the Earth's crust caused by earthquakes, aiding in early warning systems.

SUMMARY

- Global Positioning System (GPS) is developed by the U.S. Department of Defence in the 1970s, initially for military use, GPS is now crucial for civilian applications globally.
- Satellite System Architecture consists of Space Segment**,** Control Segment, User Segment
- GPS is used in Navigation**,** Surveying & Mapping, Agriculture**,** Military, Emergency & Disaster Management, Time Synchronization**,** Scientific Research**,** Alternative Systems.
- Important Features of GPS are Continuous Position Tracking, Grid Reference (GR), Global Coverage, High Accuracy, Ulti-Satellite Coverage, Time Synchronization.
- **Real-Time Positioning**: GPS overlays the user's location on a digital map, updating in real time.
- **Waypoints**: Users can mark specific locations (coordinates or grid references) for navigation. This is useful for multi-stage travel or exploration.
- **Track Back**: Allows users to retrace their steps by following a recorded trail, providing safety if lost.
- **Artillery Fire Direction**: GPS enables precise targeting and coordination of artillery fire, minimizing collateral damage.
- GPS technology in fitness trackers tracks movement, speed and elevation to monitor calorie burn and heart rate.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. Which GPS feature allows users to save their current position for later use?

(a) Waypoint marking (c) Satellite mapping
(b) Geo fencing (d) Rerouting

Q2. What type of coordinates does GPS provide to determine a location?

(a) Elevation and temperature (b) Latitude and longitude
(c) Distance and speed (d) Altitude and velocity

Q3. Why might your GPS device show an inaccurate position?

(a) Low battery on the device
(b) Poor satellite visibility due to tall buildings or trees
(c) Using it while stationary
(d) Overheating of the receiver

Q4. How does a GPS receiver determine its position?

(a) By using signals from at least three satellites
(b) By analysing weather conditions
(c) By connecting to Wi-Fi hotspots
(d) By measuring the speed of travel

Q5. Which feature of GPS helps to measure the distance between two points?

(a) Route optimization (c) Geo fencing
(b) Ranging (d) Real-time tracking

Q6. Which of the following is not a typical application of GPS?

(a) Navigation for vehicles (b) Tracking the location of smartphones
(c) Sending and receiving text messages (d) Mapping and surveying

Q7. How does the army use GPS for precision strikes?

(a) By determining the exact location of enemy targets
(b) By guiding soldiers through maps
(c) By transmitting radio signals to bases
(d) By scanning the environment for weather data

Q8. What is the primary use of waypoints in GPS navigation?

(a) To calculate the speed of travel
(b) To mark specific locations for reference or navigation
(c) To determine the weather conditions
(d) To display live traffic updates

Q9. Which GPS feature is most helpful for planning a route with multiple stops?

(a) Waypoints (b) Trackback (c) Satellite imagery (d) Geo fencing

Q10. How does a GPS calculate the time required to reach a destination?

(a) By estimating the traveller's energy level
(b) By using the average speed and distance to the destination
(c) By analysing weather conditions
(d) By measuring elevation changes

Q11. What does the "trackback" feature in GPS do?

(a) It allows users to retrace their route back to the starting point.
(b) It creates alternate routes based on traffic conditions.
(c) It marks the shortest distance between two points.
(d) It provides real-time feedback on nearby waypoints.

Q12. How does GPS enhance artillery targeting in the Army?

(a) It helps calculate the exact position of enemy targets.
(b) It automates calculations of azimuth, elevation and range.
(c) It helps position artillery units quickly and accurately.
(d) All of the above.

Q13. What does the GPS control segment primarily do?

(a) It tracks the user's position.
(b) It manages the satellites and sends updates.
(c) It ensures the satellites' power supply.
(d) It receives signals from GPS devices.

Q14. What is one major benefit of GPS compatibility with smartphones?

(a) It enhances battery life of smartphones.
(b) It improves apps like Google Maps for navigation and fitness tracking.
(c) It eliminates the need for GPS devices.
(d) It provides weather updates only.

Q15. How does GPS technology help with fitness tracking?

(a) It tracks location and elevation for calorie burn and heart rate.
(b) It prevents data loss in fitness apps.
(c) It synchronizes all workout data to the cloud.
(d) It provides automatic weather alerts during exercise.

Short Answers Types Questions

Q1. What are the uses of GPS in the Army?

Q2. What are waypoints and why are they essential for navigation?

Q3. What is the backtrack feature in GPS?

Q4. What are the latest trends in GPS?

Q5. How does GPS assist military operations?

Long Answer Types Questions

Q1. Explain Satellite System of Architecture?

Q2. What are the uses of GPS?

Q3. What are the advantages and limitations of GPS?

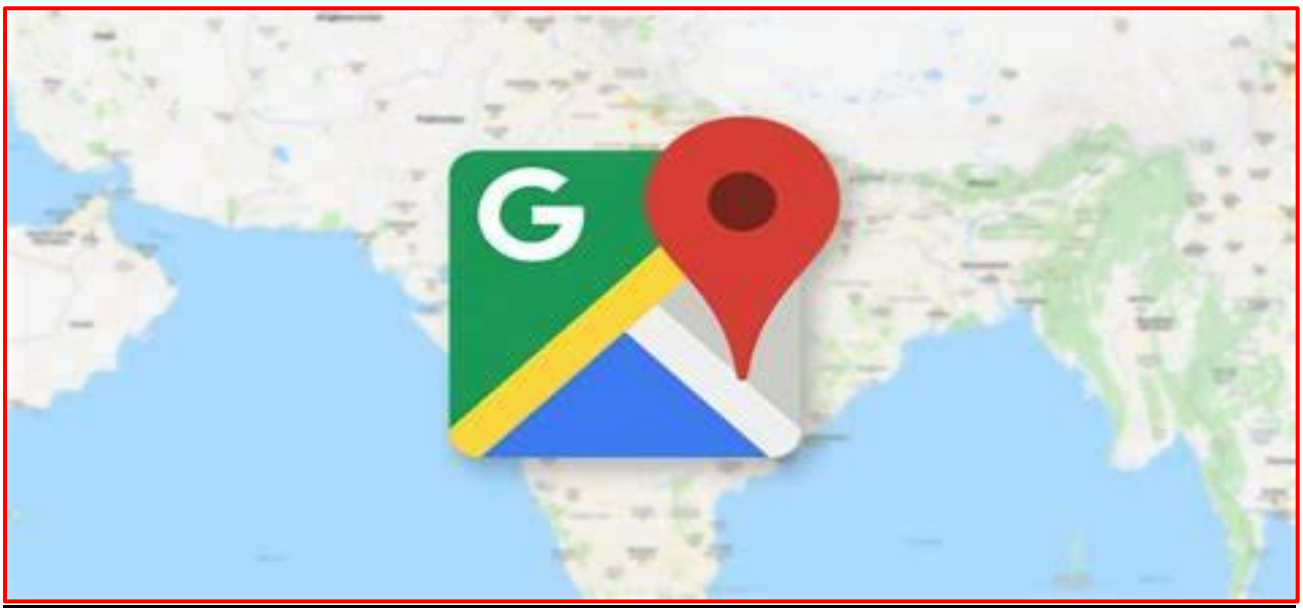
Q4. What are the important features of GPS?

Q5. How GPS aids in navigation including obstacle avoidance and route optimization?

MAP READING

CHAPTER MR-XII: GOOGLE MAPS AND OTHER APPLICATIONS

"Google Maps is more than just directions; it's a window into the world around us."



TEACHING INSTRUCTIONS

Periods : 04 (160 Mins)
Type : Lecture & Practice
Year : 1st(01), 2nd (01) & 3rd (02) Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Smart Phones, Classrooms, OHP, Board Chalks & Marker

Time Plan

- **Introduction : 05 Mins**
- **Part I : 15 Mins**
- **Part II : 20 Mins**
- **Part III : 20 Mins**
- **Practice (3 years) : 80 Mins**
- **Conclusion : 05 Mins**

INTRODUCTION

1. Google Maps is a web-based mapping platform developed by Google, providing users with access to a wide range of geographic and navigational tools. First launched in 2005, it has since become a comprehensive tool for mapping, navigation and real-time updates. It combines advanced technology like satellite imagery, aerial photography and real-time traffic data to deliver a user-friendly experience on smartphones, tablets and desktops. Google Maps are widely used for navigation (driving, walking, cycling and public transit).

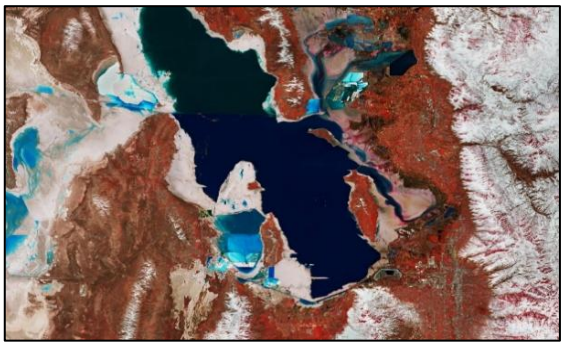
PREVIEW	LEARNING OBJECTIVES
The lecture will be conducted in the following parts:- ➢ **Part I: Introduction and Main features of Google Maps** ➢ **Part II: Applications of Googles Maps** ➢ **Part III - Navigation by Google Map** ➢ **Part IV - Other Applications for Navigation**	➢ **To understand about Google Maps and its various functions**. ➢ **To comprehend the uses of Googles Maps, its features** ➢ **To learn navigation by Google Map**

PART I – INTRODUCTION AND MAIN FEATURES OF GOOGLE MAPS

2. **Google Maps**. is a web-based mapping platform developed by Google, providing users with access to a wide range of geographic and navigational tools.

3. **Main Features of Google Maps**. The main features of google maps are given below:-

(a) **Satellite Imagery**. Google Maps provides detailed and high-resolution satellite images of nearly every corner of the Earth. Google Maps gives out Terrain Mapping Including topographic views of landscapes, helping users understand natural features like mountains, rivers and forests. Which is very useful for geographic studies, urban planning and visual exploration of remote areas.



(b) **Aerial Photography**. Aerial photography gives Bird's-Eye views and offers aerial perspectives that allow users to see landscapes and city layouts from above.

This feature helps in real estate planning, tourism and location scouting. This is often combined with satellite imagery to provide a seamless experience.

(c) **Street Maps**. Streets Maps gives out detailed road networks including streets, highways, lanes and footpaths. It provides landmark-based navigation which displays key landmarks, businesses, parks, schools and points of interest. Search ability allows users to search for specific addresses, nearby locations and custom queries.

(d) **Panoramic Street View**. 360-degree Imagery provides a ground-level view of streets and landmarks, enabling users to "walk" through areas virtually. Interior views of some businesses and public venues (e.g., museums, restaurants) offer interior Street View tours. This feature provides accessibility and enables users to preview destinations, plan routes and explore unfamiliar areas visually.



Real time Traffic Condition reflected on the Google map

(e) **Real-Time Traffic Conditions**. Traffic Flow Visualization displays road congestion with color-coded overlays (green for smooth flow, red for heavy traffic). Alternate route suggestions are given out dynamically adjusts routes based on traffic conditions to save time.

(f) **Navigation and Directions**. Turn-by-turn guidance for driving, walking, cycling, or using public transport. Multiple route options based on travel preferences (fastest, shortest, avoiding tolls). Downloadable maps allow navigation without an internet connection, ideal for remote areas. This feature helps the user navigate in rural areas.

INTERESTING FACTS

- Google Maps has over 1 billion active users each month, making it one of Google's most popular services.
- It can explore maps of the **Moon** and **Mars** through its platform.
- It provides coverage for over 220 countries and territories**,** reaching approximately 99% of the world's population. It is available in more than **80 languages**, making it accessible to people worldwide

PART II: APPLICATIONS OF GOOGLES MAPS

4. **Applications of Google Map**.Some applications of Google maps are given below:

(a) **Travel and Navigation**. Google Maps offers detailed navigation, real-time updates and route suggestions for driving, walking, cycling and public transportation, ensuring smooth travel experiences.



(b) **Voice Navigation**. Turn-by-turn voice instructions in multiple languages make driving and walking navigation hands-free and safer, especially when focusing on the road.

(c) **Tourism and Exploration**. Google Maps helps users discover new places, offering reviews, Street View, 3D imagery and offline maps for trip planning and exploration.

(d) **Urban Planning**. Google Maps supports urban planning by visualizing geographical data, infrastructure and land use, aiding the design and optimization of urban developments.

(e) **Business Search**.Google Maps aids businesses by providing optimized routing, fleet management and seamless integration with Google services for delivery, real estate and logistics.

PART III: NAVIGATION BY GOOGLE MAP

5. **Navigation by Google Map**. Google Maps is an essential tool for transit navigation, offering features that make traveling easier, faster and more efficient. Some of the uses of Google Maps in direction finding are as under:-

(a) **Real-Time Navigation**. Google Maps is an essential tool for finding directions during transit, whether traveling by car, bike, public transport, or on foot. The app updates directions dynamically, helping travellers adjust their routes based on real-time conditions.

(b) **Traffic Monitoring**. One of the most valuable features of Google Maps is its real-time traffic updates, which indicate congestion, accidents, or road closures along the route. It uses live data to provide alternate routes, helping users avoid delays and reach their destination more efficiently.

(c) **Offline Navigation**. Users can download maps for offline use, ensuring uninterrupted navigation in areas with poor or no internet connectivity. This is particularly useful for road trips or traveling in remote locations

(d) **Multi-Stop Routes**. Google Maps allows users to add multiple stops to their journey, making it easier to plan trips that involve visiting several locations. This is especially helpful for delivery drivers or those running errands.

(e) **Location Sharing**. With live location sharing, Google Maps enables users to share their real-time location with others. This feature is useful for coordinating meet-ups or ensuring safety during travel.

6. **Advantages of Google Maps**. The advantages of Google maps are as under:-

(a) **Real-time Traffic Updates**. Provides live traffic information, helping users avoid congested routes.

(b) **Turn-by-turn Navigation**. Offers step-by-step directions for driving, walking, cycling and public transportation.

(c) **Multi-mode Navigation**. Supports multiple modes of travel (car, public transport, walking, cycling).

(d) **Offline Maps**. Users can download maps for offline use in areas with no internet connection.

(e) **Business Listings**. Provides information about local businesses, reviews, hours and contact details.

7. **Disadvantages of Google Maps**. The disadvantages are given below:-

(a) **Dependence on Internet**. Requires an active internet connection for real-time updates and full functionality.

(b) **Battery Drain**. Continuous use of GPS and real-time navigation can consume significant battery power.

(c) **Accuracy Issues**. Can occasionally provide incorrect directions or out-of-date information, especially in remote areas.

(d) **Data Usage**. Using Google Maps frequently can lead to high data consumption, particularly for navigation and updates.

(e) **Over-reliance on GPS**. Users may become too dependent on navigation, potentially lacking basic map-reading skills.

PART IV: OTHER APPLICATIONS FOR NAVIGATION

8. **Other Applications for Navigation**

(a) **NavIC (Navigation with Indian Constellation)**. NavIC (Navigation with Indian Constellation) is an indigenous satellite navigation system developed by India and it serves as an alternative to other global positioning systems (GPS). It is designed to

provide accurate positioning (range of 5 meters) and timing information, especially for India and the surrounding region. NavIC is integrated into India's defence systems, providing secure, accurate navigation for military applications, including troop movements and missile guidance.

(b) **Mappls (Map Location and Place Search)**. Mappls is a comprehensive mapping and navigation app developed by **MapmyIndia**, offering accurate and reliable digital maps for India. The app provides real-time navigation, traffic updates, route planning and location search features for users traveling by car, bike, or on foot.

(c) **Waze**. It is a popular navigation app known for its community-driven features, where users share real-time information about traffic, road conditions, accidents and hazards. Waze was developed by an Israeli company and later acquired by Google in 2013.

9. **Introduction to Google Earth**. Google earth is a virtual mapping service that allows users to explore satellite imagery, maps, terrain and 3D buildings of various locations around the world. It provides a detailed, interactive platform for viewing geographic data and can be used for navigation, education and planning. Google Earth allows users to zoom in on any area, explore street views and even access historical imagery. It's available as a desktop application and online, offering both casual users and professionals a powerful tool for geographic exploration.

HOT

- Use Google Earth to explore and compare satellite images of a specific area from different years to identify significant environmental changes.
- What noticeable changes can you observe in the landscape over the past decade? (e.g., deforestation, urban expansion, changes in water bodies)
- How have these changes impacted the local environment and ecosystem?
- What patterns can be identified in the progression of these changes?

CONCLUSION

10. Google Maps is a versatile and essential tool that has revolutionized navigation and geographic exploration since its inception. With features like satellite imagery, real-time traffic updates and street-level panoramic views, it enhances the travel experience by offering accurate, up-to-date and customizable route guidance for various modes of transport. Its broad applications in tourism, business, urban planning and real-time navigation make it indispensable for users globally, ensuring convenience and efficiency in daily travel and exploration. Other apps like NavIC focusing on regional accuracy and security, Mappls providing reliable mapping for India and Waze offering community-driven traffic updates. Together, they enhance the overall navigation experience across different platforms and regions.

SUMMARY

- Google Maps is a web-based mapping and navigation platform that provides detailed geographic information and real-time services to users worldwide.
- Features of Google Maps are Satellite View, Aerial Photography, Street Maps, Panoramic Street View
- Google Maps provides detailed and high-resolution satellite images of nearly every corner of the Earth.
- Real Time Traffic Condition Google Maps is an essential tool for transit navigation, offering features that make traveling easier, faster and more efficient.
- Google maps also guide travellers during transit by providing Real-Time Navigation, Route Planning, Public Transit Support, integration with Ridesharing Apps
- Google Maps is an essential tool for transit navigation, offering features that make traveling easier, faster and more efficient.
- NavIC is designed to provide accurate positioning (range of 5 meters) and timing information, especially for India and the surrounding region
- Mappls is a comprehensive mapping and navigation app developed by Map my India.

ASSESMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary function of Google Maps?

(a) Social media networking
(b) Navigation and location services
(c) File sharing
(d) Online shopping

Q2. What is the primary function of NavIC (Navigation with Indian Constellation)?

(a) To provide global positioning services
(b) To provide navigation and timing information specifically for India and nearby regions
(c) To offer real-time traffic updates for India
(d) To create digital maps for India

Q3. What is the purpose of Google Maps' "Street View"?

(a) To display 3D images of a location
(b) To view panoramic, ground-level images of streets and locations
(c) To measure distances between two points
(d) To save offline maps

Q4. Which of the following is required to save a Google Map for offline use?

(a) Internet connection
(b) Google Maps Premium subscription
(c) Download the map area in advance
(d) GPS tracking

Q5. Which feature is offered by the Mappls app?

(a) Satellite-based navigation for military applications
(b) Real-time navigation, traffic updates and route planning for India
(c) Community-driven traffic updates
(d) Global positioning services for worldwide use

Q6. Which of the following features helps calculate the fastest route to a destination?

(a) Compass Mode
(b) Directions and Navigation
(c) Street View
(d) Location Sharing

Q7. What is a unique characteristic of Waze compared to other navigation apps?

(a) It is designed only for military applications
(b) It offers real-time community-driven information about traffic and road conditions
(c) It provides navigation only for pedestrians
(d) It was developed by the Indian government

Q8. How can a user share their real-time location on Google Maps?

(a) By enabling Traffic Layer
(b) By using Location Sharing feature
(c) By saving the map offline
(d) By turning on Satellite View

Q9. Which feature of Google Maps helps users understand the natural features of landscapes like mountains and rivers?

(a) Aerial Photography
(b) Terrain Mapping
(c) Real-Time Traffic Conditions
(d) Panoramic Street View

Q10. What type of imagery provides detailed views of Earth's surface from above?

(a) Panoramic Street View (b) Aerial Photography

(c) Satellite Imagery (d) Street Maps

Q11. Which feature provides a ground-level view of streets, allowing users to "walk" through areas virtually?

(a) Street Maps
(b) Panoramic Street View
(c) Real-Time Traffic Conditions
(d) Aerial Photography

Q12. How does Google Maps display real-time traffic conditions?

(a) By showing satellite images
(b) Using color-coded overlays
(c) By listing street names
(d) By showing historical data

Q13. Which of the following navigation modes is NOT supported by Google Maps?

(a) Walking
(b) Cycling
(c) Air travel
(d) Driving

Q14. Which feature allows users to explore new places by searching for nearby restaurants, hotels and parks?

(a) Real-Time Traffic Conditions
(b) Business Search
(c) Tourism and Exploration
(d) Street Maps

Q15. What is the key use of Google Maps in urban planning?

(a) Advertising new businesses
(b) Analysing geographical data and infrastructure
(c) Tracking social media activity
(d) Monitoring stock prices

Short Answers Type Question

Q1. Explain the features of Satellite view and Street View on Google map?

Q2. Which Google Maps feature helps businesses with optimized routing and fleet management?

Q3. What feature of Google Maps is useful for navigation in areas with poor or no internet connectivity?

Q4. What are Real-time updates?

Q5. How does Google Maps help users during real-time navigation?

Long Answers Type Questions

Q1. What are the main features of Google Maps. Explain in detail?

Q2 What are the application of the Google Map?

Q3 Explain Navigation by Google maps?

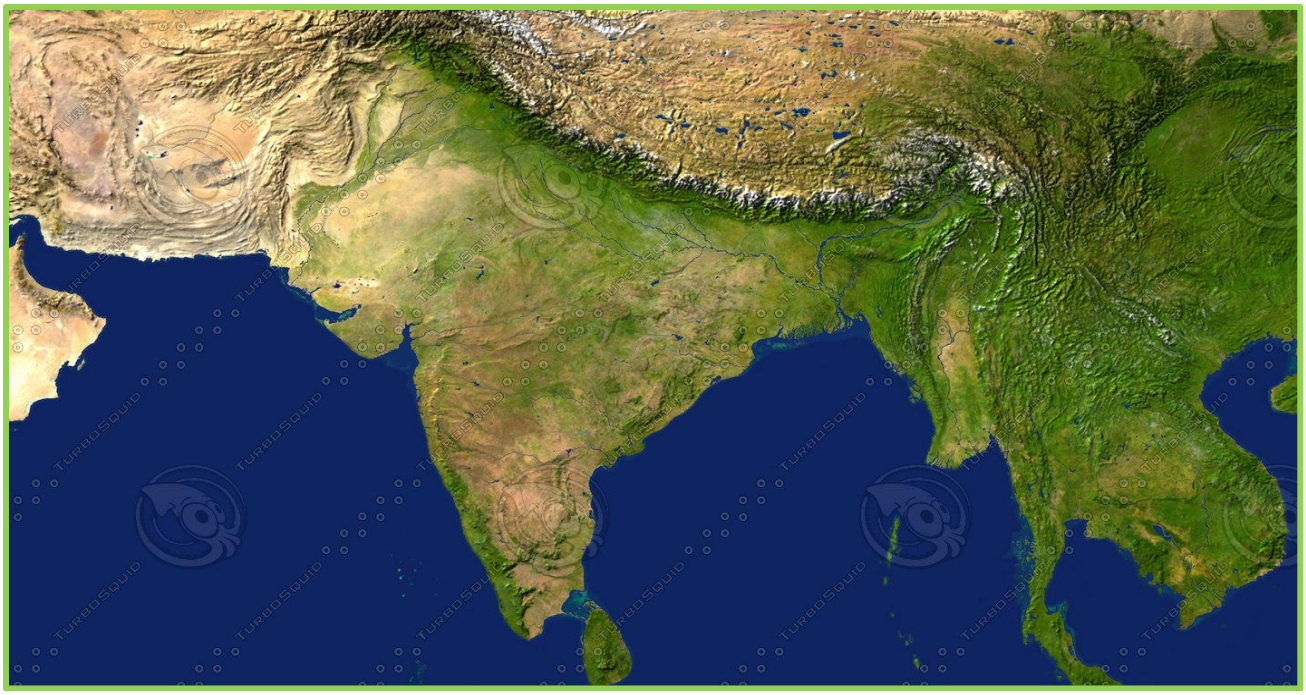
Q4 What is Real-time Navigation?

Q5. Describe Business search application.

MAP READING

CHAPTER MR-XIII: INTRODUCTION TO SATELLITE IMAGERY

"Satellite imagery is not just a tool of observation, but a mirror reflecting humanity's impact on the planet."



Satellite Image of India

TEACHING INSTRUCTIONS

Periods : 01 (40 Mins)
Types : Lecture & Practice
Year : 3rd Year SD/SW
Conducting Officer : Permanent Instructor

Training Aids: Classrooms, Smart Phone, OHP, Board Chalks and Marker

Time Plan

- **Introduction : 02 Mins**
- **Part I : 10 Mins**
- **Part II : 08 Mins**
- **Part III : 10 Mins**
- **Part IV : 08 Mins**
- **Conclusion : 02 Mins**